MTM3502
Theory of Linear Programming

Assoc. Prof. Dr. Hale Gonce Kocken

Contact information

- Email to: hgonce@yildiz.edu.tr
- My web-page: http://avesis.yildiz.edu.tr/hgonce/
- Course web-page: http://avesis.yildiz.edu.tr/hgonce/dokumanlar
- Office Hours and Appointments

The aim of the course

- Model formulation in linear form
- a deep understanding about solution methods of a linear programming problem
- Some special LP problems

Course Content

Textbook and Additional References

- ***H.A.Taha, Operations Research: An Introduction,* Prentice Hall; 9 edition, *Singapore, 2010 (In YTÜ's library)***
- *W.L. Winston, Operations research Applications and Algorithms, Second Edition, Duxbury Press,California, 1991.*
- ***And all OR and LP Books***
- ***Coursera – Linear and Integer Programming***
  Linear and Discrete optimization
- ***Youtube***
- ***Google***

Course Requirements

1. Lectures:

•Students will be expected to attend and participate in the lectures.

2. Problem Sets: There will be a problem set due every week. Students are encouraged to work together to solve them.

3. Exams: Midterm, quiz, final exam.

The date and the rules of the exams will be announced later.

Course Evaluation Criteria

- Midterm Exam : % 35
- Quiz : % 25
- Final Exam : % 40

- Class participation does not effect the final grade.

Lecture 1
Linear Programming (LP) Problem

Two-variable LP Model
Properties of LP Model
Graphical LP Solution (max/min)

2.1 TWO-VARIABLE LP MODEL

Example 2.1-1 (The Reddy Mikks Company)

Reddy Mikks produces both interior and exterior paints from two raw materials, $M1$ and $M2$. The following table provides the basic data of the problem:

	Tons of raw material per ton of		Maximum daily availability (tons)
	Exterior paint	*Interior paint*	
Raw material, $M1$	6	4	24
Raw material, $M2$	1	2	6
Profit per ton ($1000)	5	4	

A market survey indicates that the daily demand for interior paint cannot exceed that for exterior paint by more than 1 ton. Also, the maximum daily demand for interior paint is 2 tons.

Reddy Mikks wants to determine the optimum (best) product mix of interior and exterior paints that maximizes the total daily profit.

The LP model has three basic components.

The proper definition of the decision variables is an essential first step in the development of the model. Once done, the task of constructing the objective function and the constraints becomes more straightforward.

For the Reddy Mikks problem, we need to determine the daily amounts to be produced of exterior and interior paints. Thus the variables of the model are defined as

To construct the objective function, note that the company wants to *maximize* (i.e., increase as much as possible) the total daily profit of both paints. Given that the profits per ton of exterior and interior paints are 5 and 4 (thousand) dollars, respectively, it follows that

Total profit from exterior paint =

Total profit from interior paint =

Letting z represent the total daily profit (in thousands of dollars), the objective of the company is

Next, we construct the constraints that restrict raw material usage and product demand. The raw material restrictions are expressed verbally as

$$\begin{pmatrix}\text{Usage of a raw material}\\ \text{by both paints}\end{pmatrix} \leq \begin{pmatrix}\text{Maximum raw material}\\ \text{availability}\end{pmatrix}$$

The daily usage of raw material $M1$ is 6 tons per ton of exterior paint and 4 tons per ton of interior paint. Thus

Usage of raw material $M1$ by exterior paint =

Usage of raw material $M1$ by interior paint =

Hence Usage of raw material $M1$ by both paints =

In a similar manner, Usage of raw material $M2$ by both paints =

Because the daily availabilities of raw materials $M1$ and $M2$ are limited to 24 and 6 tons, respectively, the associated restrictions are given as

(Raw material $M1$)

(Raw material $M2$)

The first demand restriction stipulates that the excess of the daily production of interior over exterior paint, $x_2 - x_1$, should not exceed 1 ton, which translates to

(Market limit)

The second demand restriction stipulates that the maximum daily demand of interior paint is limited to 2 tons, which translates to

(Demand limit)

An implicit (or "understood-to-be") restriction is that variables x_1 and x_2 cannot assume negative values.

The **nonnegativity restrictions,** account for this requirement.

Any values of x_1 and x_2 that satisfy *all* five constraints constitute a

Otherwise, the solution is

For example, the solution, $x_1 = 3$ tons per day and $x_2 = 1$ ton per day,

On the other hand, the solution $x_1 = 4$ and $x_2 = 1$ is

The goal of the problem is to find the best *feasible* solution, or the that maximizes the total profit.

Properties of the LP Model. In Example 2.1-1, the objective and the constraints are all linear functions. **Linearity** implies that the LP must satisfy three basic properties:

1. **Proportionality:** This property requires the contribution of each decision variable in both the objective function and the constraints to be *directly proportional* to the value of the variable. For example, in the Reddy Mikks model, the quantities $5x_1$ and $4x_2$ give the profits for producing x_1 and x_2 tons of exterior and interior paint, respectively, with the unit profits per ton, 5 and 4, providing the constants of proportionality. If, on the other hand, Reddy Mikks grants some sort of quantity discounts when sales exceed certain amounts, then the profit will no longer be proportional to the production amounts, x_1 and x_2, and the profit function becomes nonlinear.

2. **Additivity:** This property requires the total contribution of all the variables in the objective function and in the constraints to be the direct sum of the individual contributions of each variable. In the Reddy Mikks model, the total profit equals the sum of the two individual profit components. If, however, the two products *compete* for market share in such a way that an increase in sales of one adversely affects the other, then the additivity property is not satisfied and the model is no longer linear.

3. Certainty: All the objective and constraint coefficients of the LP model are deterministic. This means that they are known constants—a rare occurrence in real life, where data are more likely to be represented by probabilistic distributions. In essence, LP coefficients are average-value approximations of the probabilistic distributions. If the standard deviations of these distributions are sufficiently small, then the approximation is acceptable. Large standard deviations can be accounted for directly by using stochastic LP algorithms (Section 19.2.3) or indirectly by applying sensitivity analysis to the optimum solution (Section 3.6).

PROBLEM SET 2.1A

1. For the Reddy Mikks model, construct each of the following constraints and express it with a linear left-hand side and a constant right-hand side:

*(a) The daily demand for interior paint exceeds that of exterior paint by *at least* 1 ton.

(b) The daily usage of raw material *M2* in tons is *at most* 6 and *at least* 3.

*(c) The demand for interior paint cannot be less than the demand for exterior paint.

(d) The minimum quantity that should be produced of both the interior and the exterior paint is 3 tons.

*(e) The proportion of interior paint to the total production of both interior and exterior paints must not exceed .5.

2. Determine the best *feasible* solution among the following (feasible and infeasible) solutions of the Reddy Mikks model:

   (a) $x_1 = 1, x_2 = 4$.

   (b) $x_1 = 2, x_2 = 2$.

   (c) $x_1 = 3, x_2 = 1.5$.

   (d) $x_1 = 2, x_2 = 1$.

   (e) $x_1 = 2, x_2 = -1$.

*3. For the feasible solution $x_1 = 2, x_2 = 2$ of the Reddy Mikks model, determine the unused amounts of raw materials $M1$ and $M2$.

4. Suppose that Reddy Mikks sells its exterior paint to a single wholesaler at a quantity discount. The profit per ton is $5000 if the contractor buys no more than 2 tons daily and $4500 otherwise. Express the objective function mathematically. Is the resulting function linear?

2.2 GRAPHICAL LP SOLUTION

The graphical procedure includes two steps:

1. Determination of the feasible solution space.
2. Determination of the optimum solution from among all the feasible points in the solution space.

The procedure uses two examples to show how maximization and minimization objective functions are handled.

2.2.1 Solution of a Maximization Model

Example 2.2-1

This example solves the Reddy Mikks model of Example 2.1-1.

Step 1. *Determination of the Feasible Solution Space:* → **Figure 2.1 (Next slide)**

Step 2. *Determination of the Optimum Solution:*

The feasible space in Figure 2.1 is delineated by the line segments joining the points *A*, *B*, *C*, *D*, *E*, and *F*. Any point within or on the boundary of the space *ABCDEF* is feasible. Because the feasible space *ABCDEF* consists of an *infinite* number of points, we need a systematic procedure to identify the optimum solution.

Figure 2.1 Feasible space of the Reddy Mikks model

The determination of the optimum solution requires identifying the direction in which the profit function $z = 5x_1 + 4x_2$ increases (recall that we are *maximizing* z). We can do so by assigning *arbitrary* increasing values to z. For example, using $z = 10$ and $z = 15$ would be equivalent to graphing the two lines $5x_1 + 4x_2 = 10$ and $5x_1 + 4x_2 = 15$. Thus, the direction of increase in z is as shown Figure 2.2. The optimum solution occurs at which is the point in the solution space beyond which any further increase will put z outside the boundaries of $ABCDEF$.

The values of x_1 and x_2 associated with the optimum point C are determined by solving the equations associated with lines (1) and (2)—that is,

The solution is $x_1 = 3$ and $x_2 = 1.5$ with $z = 5 \times 3 + 4 \times 1.5 = 21$. This calls for a daily product mix of 3 tons of exterior paint and 1.5 tons of interior paint. The associated daily profit is \$21,000.

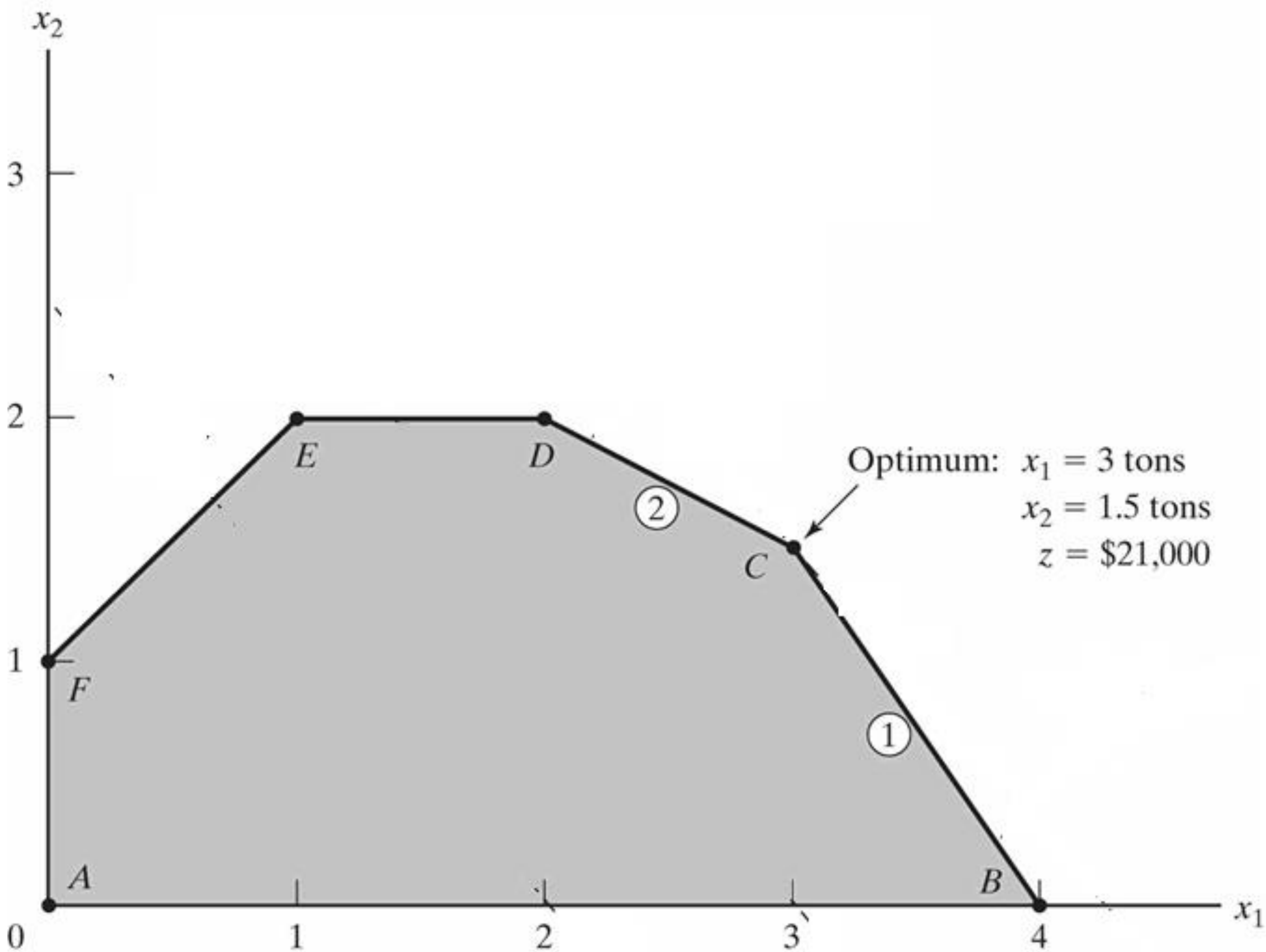


Figure 2.2 Optimum solution of the Reddy Mikks model

An important characteristic of the optimum LP solution is that it is *always* associated with a **corner point** of the solution space (where two lines intersect). This is true even if the objective function happens to be parallel to a constraint. For example, if the objective function is $z = 6x_1 + 4x_2$, which is parallel to constraint 1, we can always say that the optimum occurs at either corner point B or corner point C. Actually any point on the line segment BC will be an *alternative* optimum (see also Example 3.5-2), but the important observation here is that the line segment BC is totally defined by the *corner points* B and C.

The observation that the LP optimum is always associated with a corner point means that the optimum solution can be found simply by enumerating all the corner points as the following table shows:

Corner point	(x_1, x_2)	z
A	(0, 0)	
B	(4, 0)	
C	(3, 1.5)	
D	(2, 2)	
E	(1, 2)	
F	(0, 1)	

Method 1: Enumerating all the corner points

1. Determine the feasible space.
2. List all the corner points.
3. Evaluate the z-values of corner points and select the ‘best’ as optimum.

Method 2: Determining the increasing/decreasing direction of z

1. Assign two arbitrary values to z and plot two corresponding iso-profit/iso-cost lines.
2. Determine the direction.
3. Generate other z-lines by moving parallel to the drawn ones in a direction that improving z.
4. Determine the lost z-line intersecting(touching) the feasible region defines the best z-value and indicate the optimal solution to the LP.

Method 3: Gradient Method

We know from calculus that moving in the direction of gradient ($\vec{c}$) will increase the z-value. And similarly moving in the opposite direction of gradient ($-\vec{c}$) will decrease the objective function value.

Thus, for max. problems, we will generate iso-profit lines in the direction of gradient ($\vec{c}$) for min. problems, iso-cost lines in the direction of (-c).

Determine the lost point that z-lines touch the feasible region.

PROBLEM SET 2.2A

1. Determine the feasible space for each of the following independent constraints, given that $x_1, x_2 \geq 0$.

*(a) $-3x_1 + x_2 \leq 6$.

(b) $x_1 - 2x_2 \geq 5$.

(c) $2x_1 - 3x_2 \leq 12$.

*(d) $x_1 - x_2 \leq 0$.

(e) $-x_1 + x_2 \geq 0$.

2. Identify the direction of increase in z in each of the following cases:

*(a) Maximize $z = x_1 - x_2$.

(b) Maximize $z = -5x_1 - 6x_2$.

(c) Maximize $z = -x_1 + 2x_2$.

*(d) Maximize $z = -3x_1 + x_2$.

3. Determine the solution space and the optimum solution of the Reddy Mikks model for each of the following independent changes:

(a) The maximum daily demand for exterior paint is at most 2.5 tons.

(b) The daily demand for interior paint is at least 2 tons.

(c) The daily demand for interior paint is exactly 1 ton higher than that for exterior paint.

(d) The daily availability of raw material $M1$ is at least 24 tons.

(e) The daily availability of raw material $M1$ is at least 24 tons, and the daily demand for interior paint exceeds that for exterior paint by at least 1 ton.

4. A company that operates 10 hours a day manufactures two products on three processes. The following table summarizes the data of the problem:

Product	Minutes per unit			Unit profit
	Process 1	*Process 2*	*Process 3*	
1	10	6	8	\$2
2	5	20	10	\$3

Determine the optimal mix of the two products.

*5. A company produces two products, *A* and *B*. The sales volume for *A* is at least 80% of the total sales of both *A* and *B*. However, the company cannot sell more than 100 units of *A* per day. Both products use one raw material, of which the maximum daily availability is 240 lb. The usage rates of the raw material are 2 lb per unit of *A* and 4 lb per unit of *B*. The profit units for *A* and *B* are \$20 and \$50, respectively. Determine the optimal product mix for the company.

6. Alumco manufactures aluminum sheets and aluminum bars. The maximum production capacity is estimated at either 800 sheets or 600 bars per day. The maximum daily demand is 550 sheets and 580 bars. The profit per ton is \$40 per sheet and \$35 per bar. Determine the optimal daily production mix.

*7. An individual wishes to invest \$5000 over the next year in two types of investment: Investment *A* yields 5% and investment *B* yields 8%. Market research recommends an allocation of at least 25% in *A* and at most 50% in *B*. Moreover, investment in *A* should be at least half the investment in *B*. How should the fund be allocated to the two investments?

8. The Continuing Education Division at the Ozark Community College offers a total of 30 courses each semester. The courses offered are usually of two types: practical, such as woodworking, word processing, and car maintenance; and humanistic, such as history, music, and fine arts. To satisfy the demands of the community, at least 10 courses of each type must be offered each semester. The division estimates that the revenues of offering practical and humanistic courses are approximately \$1500 and \$1000 per course, respectively.
   (a) Devise an optimal course offering for the college.
   (b) Show that the worth per additional course is \$1500, which is the same as the revenue per practical course. What does this result mean in terms of offering additional courses?

9. ChemLabs uses raw materials *I* and *II* to produce two domestic cleaning solutions, *A* and *B*. The daily availabilities of raw materials *I* and *II* are 150 and 145 units, respectively. One unit of solution *A* consumes .5 unit of raw material *I* and .6 unit of raw material *II*, and one unit of solution *B* uses .5 unit of raw material *I* and .4 unit of raw material *II*. The profits per unit of solutions *A* and *B* are \$8 and \$10, respectively. The daily demand for solution *A* lies between 30 and 150 units, and that for solution *B* between 40 and 200 units. Find the optimal production amounts of *A* and *B*.

10. In the Ma-and-Pa grocery store, shelf space is limited and must be used effectively to increase profit. Two cereal items, Grano and Wheatie, compete for a total shelf space of 60 ft^2. A box of Grano occupies .2 ft^2 and a box of Wheatie needs .4 ft^2. The maximum daily demands of Grano and Wheatie are 200 and 120 boxes, respectively. A box of Grano nets \$1.00 in profit and a box of Wheatie \$1.35. Ma-and-Pa thinks that because the unit profit of Wheatie is 35% higher than that of Grano, Wheatie should be allocated 35% more space than Grano, which amounts to allocating about 57% to Wheatie and 43% to Grano. What do you think?

11. Jack is an aspiring freshman at Ulern University. He realizes that "all work and no play make Jack a dull boy." As a result, Jack wants to apportion his available time of about 10 hours a day between work and play. He estimates that play is twice as much fun as work. He also wants to study at least as much as he plays. However, Jack realizes that if he is going to get all his homework assignments done, he cannot play more than 4 hours a day. How should Jack allocate his time to maximize his pleasure from both work and play?

12. Wild West produces two types of cowboy hats. A type 1 hat requires twice as much labor time as a type 2. If the all available labor time is dedicated to Type 2 alone, the company can produce a total of 400 Type 2 hats a day. The respective market limits for the two types are 150 and 200 hats per day. The profit is \$8 per Type 1 hat and \$5 per Type 2 hat. Determine the number of hats of each type that would maximize profit.

13. Show & Sell can advertise its products on local radio and television (TV). The advertising budget is limited to $10,000 a month. Each minute of radio advertising costs $15 and each minute of TV commercials $300. Show & Sell likes to advertise on radio at least twice as much as on TV. In the meantime, it is not practical to use more than 400 minutes of radio advertising a month. From past experience, advertising on TV is estimated to be 25 times as effective as on radio. Determine the optimum allocation of the budget to radio and TV advertising.

*14. Wyoming Electric Coop owns a steam-turbine power-generating plant. Because Wyoming is rich in coal deposits, the plant generates its steam from coal. This, however, may result in emission that does not meet the Environmental Protection Agency standards. EPA regulations limit sulfur dioxide discharge to 2000 parts per million per ton of coal burned and smoke discharge from the plant stacks to 20 lb per hour. The Coop receives two grades of pulverized coal, $C1$ and $C2$, for use in the steam plant. The two grades are usually mixed together before burning. For simplicity, it can be assumed that the amount of sulfur pollutant discharged (in parts per million) is a weighted average of the proportion of each grade used in the mixture. The following data are based on consumption of 1 ton per hour of each of the two coal grades.

Coal grade	Sulfur discharge in parts per million	Smoke discharge in lb per hour	Steam generated in lb per hour
$C1$	1800	2.1	12,000
$C2$	2100	.9	9,000

(a) Determine the optimal ratio for mixing the two coal grades.

(b) Determine the effect of relaxing the smoke discharge limit by 1 lb on the amount of generated steam per hour.

2.2.2 Solution of a Minimization Model

Example 2.2-2 (Diet Problem)

Ozark Farms uses at least 800 lb of special feed daily. The special feed is a mixture of corn and soybean meal with the following compositions:

	lb per lb of feedstuff		
Feedstuff	*Protein*	*Fiber*	Cost ($/lb)
Corn	.09	.02	.30
Soybean meal	.60	.06	.90

The dietary requirements of the special feed are at least 30% protein and at most 5% fiber. Ozark Farms wishes to determine the daily minimum-cost feed mix.

Because the feed mix consists of corn and soybean meal, the decision variables of the model are defined as

The objective function seeks to minimize the total daily cost (in dollars) of the feed mix and is thus expressed as

The constraints of the model reflect the daily amount needed and the dietary requirements. Because Ozark Farms needs at least 800 lb of feed a day, the associated constraint can be expressed as

As for the protein dietary requirement constraint, the amount of protein included in x_1 lb of corn and x_2 lb of soybean meal is $(.09x_1 + .6x_2)$ lb. This quantity should equal at least 30% of the total feed mix $(x_1 + x_2)$ lb—that is,

In a similar manner, the fiber requirement of at most 5% is constructed as

The constraints are simplified by moving the terms in x_1 and x_2 to the left-hand side of each inequality, leaving only a constant on the right-hand side. The complete model thus becomes

Figure 2.3 Graphical solution of the diet model

PROBLEM SET 2.2B

1. Identify the direction of decrease in z in each of the following cases:
   *(a) Minimize $z = 4x_1 - 2x_2$.
   (b) Minimize $z = -3x_1 + x_2$.
   (c) Minimize $z = -x_1 - 2x_2$.

2. For the diet model, suppose that the daily availability of corn is limited to 450 lb. Identify the new solution space, and determine the new optimum solution.

3. For the diet model, what type of optimum solution would the model yield if the feed mix should not exceed 800 lb a day? Does the solution make sense?

4. John must work at least 20 hours a week to supplement his income while attending school. He has the opportunity to work in two retail stores. In store 1, he can work between 5 and 12 hours a week, and in store 2 he is allowed between 6 and 10 hours. Both stores pay the same hourly wage. In deciding how many hours to work in each store, John wants to base his decision on work stress. Based on interviews with present employees, John estimates that, on an ascending scale of 1 to 10, the stress factors are 8 and 6 at stores 1 and 2, respectively. Because stress mounts by the hour, he assumes that the total stress for each store at the end of the week is proportional to the number of hours he works in the store. How many hours should John work in each store?

*5. OilCo is building a refinery to produce four products: diesel, gasoline, lubricants, and jet fuel. The minimum demand (in bbl/day) for each of these products is 14,000, 30,000, 10,000, and 8,000, respectively. Iran and Dubai are under contract to ship crude to OilCo. Because of the production quotas specified by OPEC (Organization of Petroleum Exporting Countries) the new refinery can receive at least 40% of its crude from Iran and the remaining amount from Dubai. OilCo predicts that the demand and crude oil quotas will remain steady over the next ten years.

The specifications of the two crude oils lead to different product mixes: One barrel of Iran crude yields .2 bbl of diesel, .25 bbl of gasoline, .1 bbl of lubricant, and .15 bbl of jet fuel. The corresponding yields from Dubai crude are .1, .6, .15, and .1, respectively. OilCo needs to determine the minimum capacity of the refinery (in bbl/ day).

6. Day Trader wants to invest a sum of money that would generate an annual yield of at least $10,000. Two stock groups are available: blue chips and high tech, with average annual yields of 10% and 25%, respectively. Though high-tech stocks provide higher yield, they are more risky, and Trader wants to limit the amount invested in these stocks to no more than 60% of the total investment. What is the minimum amount Trader should invest in each stock group to accomplish the investment goal?

Reference: *H.A.Taha, Operations Research: An Introduction,* Prentice Hall; 9th edition, *Singapore, 2010.*

Lecture 2
The Simplex Method (I)

LP model in equation form
Transition from graphical solution to algebraic solution

3.1 LP MODEL IN EQUATION FORM

The development of the simplex method computations is facilitated by imposing two requirements on the constraints of the problem:

1. All the constraints (with the exception of the nonnegativity of the variables) are equations with nonnegative right-hand side.
2. All the variables are nonnegative.

These two requirements are imposed here primarily to standardize and streamline the simplex method calculations. It is important to know that all commercial packages (and TORA) directly accept inequality constraints, nonnegative right-hand side, and unrestricted variables. Any necessary preconditioning of the model is done internally in the software before the simplex method solves the problem.

3.1.1 Converting Inequalities into Equations with Nonnegative Right-Hand Side

In ($\leq$) constraints, the right-hand side can be thought of as representing the limit on the availability of a resource, in which case the left-hand side would represent the usage of this limited resource by the activities (variables) of the model. The difference between the right-hand side and the left-hand side of the ($\leq$) constraint thus yields the *unused* or *slack* amount of the resource.

To convert a ($\leq$)-inequality to an equation, a nonnegative **slack variable** is added to the left-hand side of the constraint. For example, in the Reddy Mikks model (Example 2.1-1), the constraint associated with the use of raw material $M1$ is given as

Defining s_1 as the slack or unused amount of $M1$, the constraint can be converted to the following equation:

Next, a ($\geq$)-constraint sets a lower limit on the activities of the LP model, so that the amount by which the left-hand side exceeds the minimum limit represents a *surplus*. The conversion from ($\geq$) to ($=$) is achieved by subtracting a nonnegative **surplus variable** from the left-hand side of the inequality. For example, in the diet model (Example 2.2-2), the constraint representing the minimum feed requirements is

Defining S_1 as the surplus variable, the constraint can be converted to the following equation

The only remaining requirement is for the right-hand side of the resulting equation to be nonnegative. The condition can always be satisfied by multiplying both sides of the resulting equation by -1, where necessary. For example, the constraint

is equivalent to the equation

Now, multiplying both sides by -1 will render a nonnegative right-hand side, as desired—that is,

PROBLEM SET 3.1A

*1. In the Reddy Mikks model (Example 2.2-1), consider the feasible solution $x_1 = 3$ tons and $x_2 = 1$ ton. Determine the value of the associated slacks for raw materials $M1$ and $M2$.

2. In the diet model (Example 2.2-2), determine the surplus amount of feed consisting of 500 lb of corn and 600 lb of soybean meal.

3. Consider the following inequality

$$10x_1 - 3x_2 \geq -5$$

Show that multiplying both sides of the inequality by -1 and then converting the resulting inequality into an equation is the same as converting it first to an equation and then multiplying both sides by -1.

*4. Two different products, $P1$ and $P2$, can be manufactured by one or both of two different machines, $M1$ and $M2$. The unit processing time of either product on either machine is the same. The daily capacity of machine $M1$ is 200 units (of either $P1$ or $P2$, or a mixture of both) and the daily capacity of machine $M2$ is 250 units. The shop supervisor wants to balance the production schedule of the two machines such that the total number of units produced on one machine is within 5 units of the number produced on the other. The profit per unit of $P1$ is \$10 and that of $P2$ is \$15. Set up the problem as an LP in equation form.

5. Show how the following objective function can be presented in equation form:

$$\text{Minimize } z = \max\{|x_1 - x_2 + 3x_3|, |-x_1 + 3x_2 - x_3|\}$$

$$x_1, x_2, x_3 \geq 0$$

(*Hint:* $|a| \leq b$ is equivalent to $a \leq b$ and $a \geq -b$.)

3.1.2 Dealing with Unrestricted Variables

In Example 2.3-6 we presented a multiperiod production smoothing model in which the workforce at the start of each period is adjusted up or down depending on the demand for that period. Specifically, if $x_i\ (\geq 0)$ is the workforce size in period i, then $x_{i+1}\ (\geq 0)$ the workforce size in period $i+1$ can be expressed as

$$x_{i+1} = x_i + y_{i+1}$$

The variable y_{i+1} must be unrestricted in sign to allow x_{i+1} to increase or decrease relative to x_i depending on whether workers are hired or fired, respectively.

As we will see shortly, the simplex method computations require all the variables be nonnegative. We can always account for this requirement by using the substitution

$$y_{i+1} = y_{i+1}^{-} - y_{i+1}^{+}, \text{ where } y_{i+1}^{-} \geq 0 \text{ and } y_{i+1}^{+} \geq 0$$

To show how this substitution works, suppose that in period 1 the workforce is $x_1 = 20$ workers and that the workforce in period 2 will be increased by 5 to reach 25 workers. In terms of the variables y_2^- and y_2^+, this will be equivalent to $y_2^- = 5$ and $y_2^+ = 0$ or $y_2 = 5 - 0 = 5$. Similarly, if the workforce in period 2 is reduced to 16, then we have $y_2^- = 0$ and $y_2^+ = 4$, or $y_2 = 0 - 4 = -4$. The substitution also allows for the possibility of no change in the workforce by letting both variables assume a zero value.

PROBLEM SET 3.1B

1. McBurger fast-food restaurant sells quarter-pounders and cheeseburgers. A quarter-pounder uses a quarter of a pound of meat, and a cheeseburger uses only .2 lb. The restaurant starts the day with 200 lb of meat but may order more at an additional cost of 25 cents per pound to cover the delivery cost. Any surplus meat at the end of the day is donated to charity. McBurger's profits are 20 cents for a quarter-pounder and 15 cents for a cheeseburger. McBurger does not expect to sell more than 900 sandwiches in any one day. How many of each type sandwich should McBurger plan for the day? Solve the problem using TORA, Solver, or AMPL.

2. Two products are manufactured in a machining center. The productions times per unit of products 1 and 2 are 10 and 12 minutes, respectively. The total regular machine time is 2500 minutes per day. In any one day, the manufacturer can produce between 150 and 200 units of product 1, but no more than 45 units of product 2. Overtime may be used to meet the demand at an additional cost of $.50 per minute. Assuming that the unit profits for products 1 and 2 are $6.00 and $7.50, respectively, formulate the problem as an LP model, then solve with TORA, Solver, or AMPL to determine the optimum production level for each product as well as any overtime needed in the center.

*3. JoShop manufactures three products whose unit profits are \$2, \$5, and \$3, respectively. The company has budgeted 80 hours of labor time and 65 hours of machine time for the production of three products. The labor requirements per unit of products 1, 2, and 3 are 2, 1, and 2 hours, respectively. The corresponding machine-time requirements per unit are 1, 1, and 2 hours. JoShop regards the budgeted labor and machine hours as goals that may be exceeded, if necessary, but at the additional cost of \$15 per labor hour and \$10 per machine hour. Formulate the problem as an LP, and determine its optimum solution using TORA, Solver, or AMPL.

3.2 TRANSITION FROM GRAPHICAL TO ALGEBRAIC SOLUTION

Graphical Method	Algebraic Method
Graph all constraints, including nonnegativity restrictions	Represent the solution space by m equations in n variables and restrict all variables to nonnegative values, $m < n$
Solution space consists of infinity of **feasible** points	The system has infinity of **feasible** solutions
↓	↓
Identify **feasible corner points** of the solution space	Determine the **feasible basic solutions** of the equations
Candidates for the optimum solution are given by a *finite* number of **corner points**	Candidates for the optimum solution are given by a *finite* number of **basic feasible solutions**
↓	↓
Use the objective function to determine the **optimum corner point** from among all the candidates	Use the objective function to determine the **optimum basic feasible solution** from among all the candidates

Figure 3.1 Transition from graphical to algebraic solution

In the graphical method, the solution space is delineated by the half-spaces representing the constraints, and in the simplex method the solution space is represented by m simultaneous linear equations and n nonnegative variables.

We can see visually why the graphical solution space has an infinite number of solution points, but how can we draw a similar conclusion from the algebraic representation of the solution space?

The answer is that in the algebraic representation the number of equations m is always *less than or equal to* the number of variables n.[1] If $m = n$, and the equations are consistent, the system has only one solution; but if $m < n$ (which represents the majority of LPs), then the system of equations, again if consistent, will yield an infinite number of solutions.

If the number of equations m is larger than the number of variables n, then at least $m - n$ equations must be redundant.

To provide a simple illustration, the equation $x = 2$ has $m = n = 1$, and the solution is obviously unique. But, the equation $x + y = 1$ has $m = 1$ and $n = 2$, and it yields an infinite number of solutions (any point on the straight line $x + y = 1$ is a solution).

Having shown how the LP solution space is represented algebraically, the candidates for the optimum (i.e., corner points) are determined from the simultaneous linear equations in the following manner:

Algebraic Determination of Corner Points.

In a set of $m \times n$ equations $(m < n)$, if we set $n - m$ variables equal to zero and then solve the m equations for the remaining m variables, the resulting solution, if unique, is called a **basic solution** and must correspond to a (feasible or infeasible) corner point of the solution space. This means that the *maximum* number of corner points is

$$C_m^n = \frac{n!}{m!(n-m)!}$$

The following example demonstrates the procedure.

Example 3.2-1 Consider the following LP with two variables:

$$\text{Maximize } z = 2x_1 + 3x_2$$

$$\text{subject to } \begin{aligned} 2x_1 + x_2 &\leq 4 \\ x_1 + 2x_2 &\leq 5 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Figure 3.2 provides the graphical solution space for the problem.

Algebraically, the solution space of the LP is represented as:

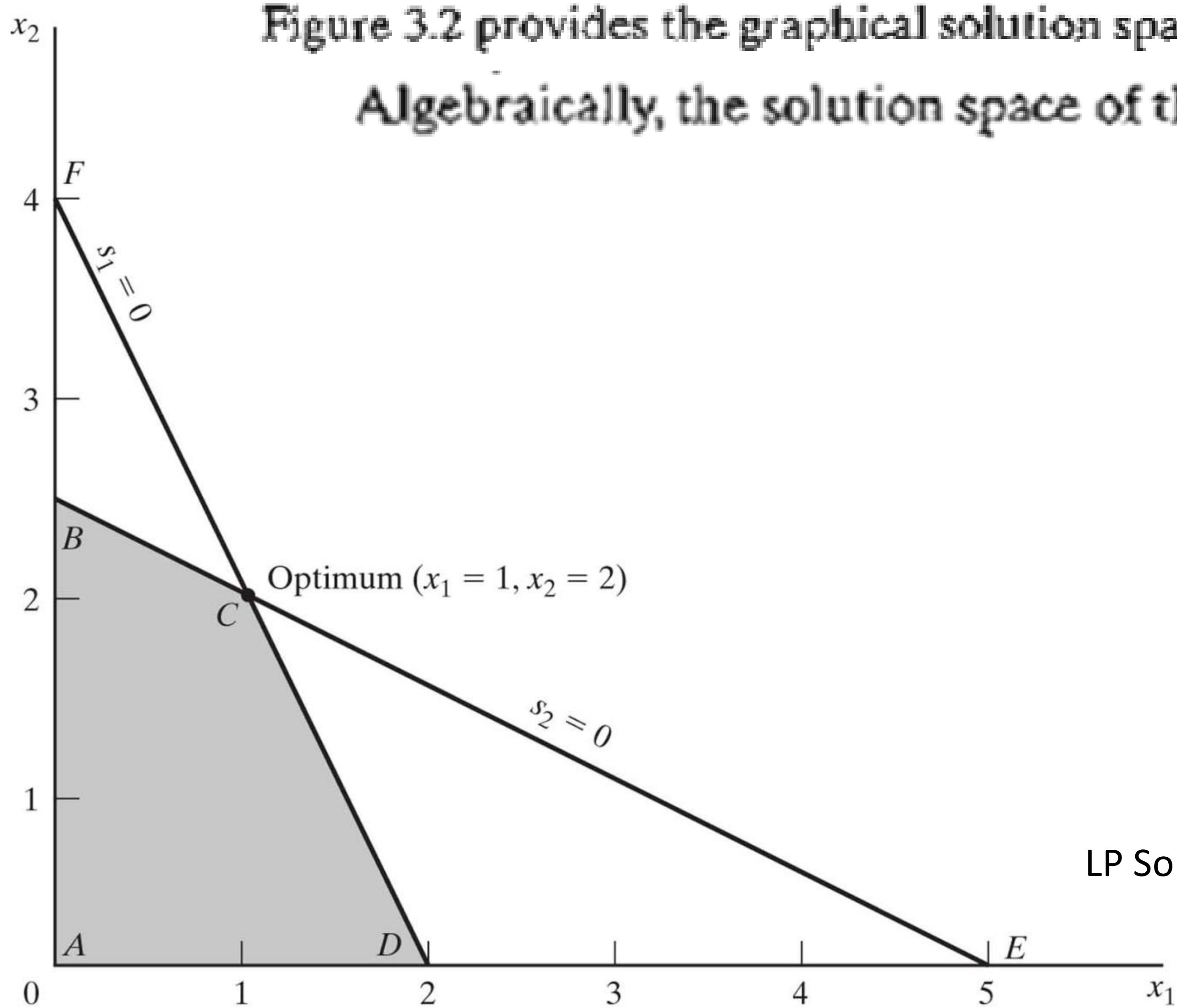


Figure 3.2
LP Solution space of Example 3.2-1

The system has $m = 2$ equations and $n = 4$ variables. Thus, according to the given rule, the corner points can be determined algebraically by setting $n - m = 4 - 2 = 2$ variables equal to zero and then solving for the remaining $m = 2$ variables. For example, if we set $x_1 = 0$ and $x_2 = 0$, the equations provide the unique (basic) solution

This solution corresponds to point A in Figure 3.2 (convince yourself that $s_1 = 4$ and $s_2 = 5$ at point A). Another point can be determined by setting $s_1 = 0$ and $s_2 = 0$ and then solving the two equations

This yields the basic solution ($x_1 = 1, x_2 = 2$), which is point C in Figure 3.2.

1: consider *all* combinations in which $n - m$ variables are set to zero and solve the resulting equations.

2: Once done, the optimum solution is the feasible basic solution (corner point) that yields the best objective value.

***Every corner point of the feasible region (f.r.) is obtained by solving the corresponding linear equation system

In the present example we have $C_2^4 = \frac{4!}{2!2!} = 6$ corner points. Looking at Figure 3.2, we can immediately spot the four corner points A, B, C, and D. Where, then, are the remaining two? In fact, points E and F also are corner points for the problem, but they are *infeasible* because they do not satisfy all the constraints. These infeasible corner points are not candidates for the optimum.

Nonbasic (zero) variables	Basic variables	Basic solution	Associated corner point	Feasible?	Objective value, z

Algebraic method (Enumerating the basic solutions)

Step 1: Convert the inequalities into equations by adding slack variables.

Step 2: Assuming that there are m equations and n variables, set n-m nonbasic variables to zero and evaluate the solution for the remaining m basic variables. Evaluate the objective function value if the basic solution is feasible.

Step 3: Perform step 2 for all combinations of basic variables

Step 4: Identify the optimum solution as the one with the maximum/minimum value of the objective function.

PROBLEM SET 3.2A

1. Consider the following LP:

$$\text{Maximize } z = 2x_1 + 3x_2$$

subject to

$$x_1 + 3x_2 \leq 6$$
$$3x_1 + 2x_2 \leq 6$$
$$x_1, x_2 \geq 0$$

   (a) Express the problem in equation form.

   (b) Determine all the basic solutions of the problem, and classify them as feasible and infeasible.

   *(c) Use direct substitution in the objective function to determine the optimum basic feasible solution.

   (d) Verify graphically that the solution obtained in (c) is the optimum LP solution—hence, conclude that the optimum solution can be determined algebraically by considering the basic feasible solutions only.

   *(e) Show how the *infeasible* basic solutions are represented on the graphical solution space.

2. Determine the optimum solution for each of the following LPs by enumerating all the basic solutions.

(a) Maximize $z = 2x_1 - 4x_2 + 5x_3 - 6x_4$

subject to

$$x_1 + 4x_2 - 2x_3 + 8x_4 \leq 2$$
$$-x_1 + 2x_2 + 3x_3 + 4x_4 \leq 1$$
$$x_1, x_2, x_3, x_4 \geq 0$$

(b) Minimize $z = x_1 + 2x_2 - 3x_3 - 2x_4$

subject to

$$x_1 + 2x_2 - 3x_3 + x_4 = 4$$
$$x_1 + 2x_2 + x_3 + 2x_4 = 4$$
$$x_1, x_2, x_3, x_4 \geq 0$$

*3. Show algebraically that all the basic solutions of the following LP are infeasible.

$$\text{Maximize } z = x_1 + x_2$$

subject to

$$x_1 + 2x_2 \le 6$$

$$2x_1 + x_2 \le 16$$

$$x_1, x_2 \ge 0$$

4. Consider the following LP:

$$\text{Maximize } z = 2x_1 + 3x_2 + 5x_3$$

subject to

$$-6x_1 + 7x_2 - 9x_3 \ge 4$$

$$x_1 + x_2 + 4x_3 = 10$$

$$x_1, x_3 \ge 0$$

$$x_2 \text{ unrestricted}$$

Conversion to the equation form involves using the substitution $x_2 = x_2^- - x_2^+$. Show that a basic solution cannot include both x_2^- and x_2^+ simultaneously.

5. Consider the following LP:

$$\text{Maximize } z = x_1 + 3x_2$$

subject to

$$x_1 + x_2 \leq 2$$

$$-x_1 + x_2 \leq 4$$

$$x_1 \text{ unrestricted}$$

$$x_2 \geq 0$$

(a) Determine all the basic feasible solutions of the problem.

(b) Use direct substitution in the objective function to determine the best basic solution.

(c) Solve the problem graphically, and verify that the solution obtained in (c) is the optimum.

Reference: *H.A.Taha, Operations Research: An Introduction,* Prentice Hall; 9th edition, *Singapore, 2010.*

Lecture 3
The Simplex Method (II)

Iterative nature of the simplex method
Computational details of simplex method
Optimality condition
Feasibility condition

Disadvantages of the algebraic method

- It evaluates a large number of solutions.
- It evaluates infeasible solutions.
- It doesn’t generate progressively better solutions.
- It doesn’t have a mechanism that we we have got the optimum.

So we need a new method that enables the following:

- It does not evaluate any infeasible solutions.
- It generates progressively better solutions.
- The moment it has found the optimum, it should be able to terminate.

3.3 THE SIMPLEX METHOD

Rather than enumerating *all* the basic solutions (corner points) of the LP problem (as we did in Section 3.2), the simplex method investigates only a "select few" of these solutions. Section 3.3.1 describes the *iterative* nature of the method, and Section 3.3.2 provides the computational details of the simplex algorithm.

3.3.1 Iterative Nature of the Simplex Method

Figure 3.3 provides the solution space of the LP of Example 3.2-1. Normally, the simplex method starts at the origin (point A) where $x_1 = x_2 = 0$. At this starting point, the value of the objective function, z, is zero, and the logical question is whether an increase in nonbasic x_1 and/or x_2 above their current zero values can improve (increase) the value of z. We answer this question by investigating the objective function:

The function shows that an increase in either x_1 or x_2 (or both) above their current zero values will *improve* the value of z. The design of the simplex method calls for increasing *one variable at a time*, with the selected variable being the one with the *largest rate of improvement* in z. In the present example, the value of z will increase by 2 for each unit increase in x_1 and by 3 for each unit increase in x_2. This means that the *rate* of improvement in the value of z is 2 for x_1 and 3 for x_2. We thus elect to increase x_2, the variable with the largest rate of improvement.

Example 3.2-1

Maximize $z = 2x_1 + 3x_2$

subject to $2x_1 + x_2 \le 4$

$x_1 + 2x_2 \le 5$

$x_1, x_2 \ge 0$

Maximize $z = 2x_1 + 3x_2$

$$
\begin{aligned}
2x_1 + x_2 + s_1 \quad &= 4 \\
x_1 + 2x_2 \quad + s_2 &= 5 \\
x_1, x_2, s_1, s_2 &\ge 0
\end{aligned}
$$

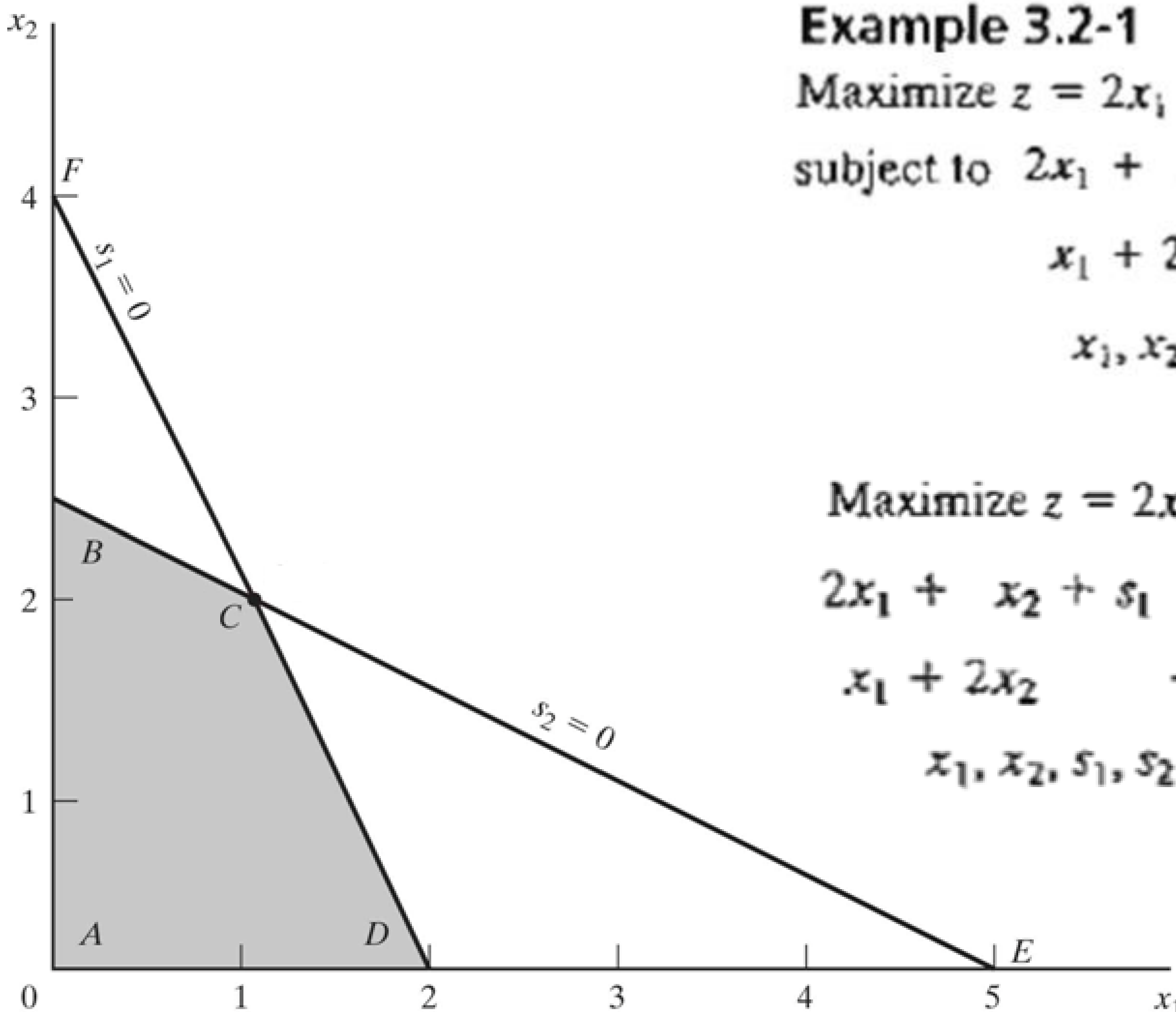


Figure 3.3 Iterative process of the simplex method

Figure 3.3 shows that the value of x_2 must be increased until corner point B is reached (recall that stopping short of reaching corner point B is not optimal because a candidate for the optimum must be a corner point). At point B, the simplex method will then increase the value of x_1 to reach the improved corner point C, which is the optimum. The path of the simplex algorithm is thus defined as $A \rightarrow B \rightarrow C$. Each corner point along the path is associated with an **iteration**. It is important to note that the simplex method moves alongside the **edges** of the solution space, which means that the method cannot cut across the solution space, going from A to C directly.

We need to make the transition from the graphical solution to the algebraic solution by showing how the points A, B, and C are represented by their basic and nonbasic variables. The following table summarizes these representations:

Corner point	Basic variables	Nonbasic (zero) variables
A		
B		
C		

Notice the change pattern in the basic and nonbasic variables as the solution moves along the path $A \rightarrow B \rightarrow C$. From A to B, nonbasic x_2 at A becomes basic at B and basic s_2 at A becomes nonbasic at B. In the terminology of the simplex method, we say that x_2 is the **entering variable** (because it enters the basic solution) and s_2 is the **leaving variable** (because it leaves the basic solution). In a similar manner, at point B, x_1 *enters* (the basic solution) and s_1 *leaves*, thus leading to point C.

PROBLEM SET 3.3A

1. In Figure 3.3, suppose that the objective function is changed to

$$\text{Maximize } z = 8x_1 + 4x_2$$

Identify the path of the simplex method and the basic and nonbasic variables that define this path.

2. Consider the graphical solution of the Reddy Mikks model given in Figure 2.2. Identify the path of the simplex method and the basic and nonbasic variables that define this path.

*3. Consider the three-dimensional LP solution space in Figure 3.4, whose feasible extreme points are $A, B, \ldots,$ and J.

(a) Which of the following pairs of corner points cannot represent *successive* simplex iterations: (A, B), (B, D), (E, H), and (A, I)? Explain the reason.

(b) Suppose that the simplex iterations start at A and that the optimum occurs at H. Indicate whether any of the following paths are *not* legitimate for the simplex algorithm, and state the reason.

(i) $A \rightarrow B \rightarrow G \rightarrow H$.

(ii) $A \rightarrow E \rightarrow I \rightarrow H$.

(iii) $A \rightarrow C \rightarrow E \rightarrow B \rightarrow A \rightarrow D \rightarrow G \rightarrow H$.

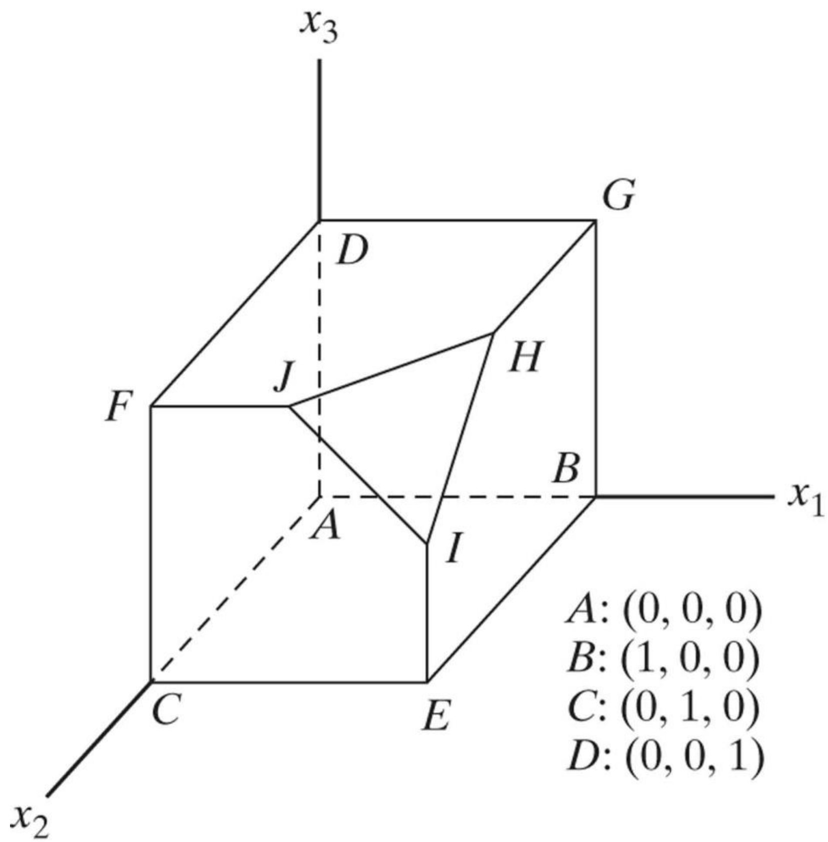


Figure 3.4

4. For the solution space in Figure 3.4, all the constraints are of the type $\leq$ and all the variables x_1, x_2, and x_3 are nonnegative. Suppose that s_1, s_2, s_3, and s_4 (≥ 0) are the slacks associated with constraints represented by the planes *CEIJF*, *BEIHG*, *DFJHG*, and *IJH*, respectively. Identify the basic and nonbasic variables associated with each feasible extreme point of the solution space.

5. Consider the solution space in Figure 3.4, where the simplex algorithm starts at point A. Determine the entering variable in the *first* iteration together with its value and the improvement in z for each of the following objective functions:

*(a) Maximize $z = x_1 - 2x_2 + 3x_3$

(b) Maximize $z = 5x_1 + 2x_2 + 4x_3$

(c) Maximize $z = -2x_1 + 7x_2 + 2x_3$

(d) Maximize $z = x_1 + x_2 + x_3$

Algebraic form of simplex method

Example 3.2-1

Maximize $z = 2x_1 + 3x_2$

subject to
$$2x_1 + x_2 \le 4$$
$$x_1 + 2x_2 \le 5$$
$$x_1, x_2 \ge 0$$

Maximize $z = 2x_1 + 3x_2$
$$2x_1 + x_2 + s_1 = 4$$
$$x_1 + 2x_2 + s_2 = 5$$
$$x_1, x_2, s_1, s_2 \ge 0$$

- We need to begin by considering a b.f.s. The easiest way for obtaining a b.f.s. is fixing x_1=x_2=0 and getting s_1=4 and s_2=5.
- The first step of each iteration in simplex method is to identify the basic variables and also the objective function in terms of the non-basic variables.

<u>İteration 1</u>

3.3.2 Computational Details of the Simplex Algorithm

This section provides the computational details of a simplex iteration, including the rules for determining the entering and leaving variables as well as for stopping the computations when the optimum solution has been reached. The vehicle of explanation is a numerical example.

$$\text{Maximize } z = 5x_1 + 4x_2$$

subject to

$$\begin{aligned} 6x_1 + 4x_2 &\le 24 \\ x_1 + 2x_2 &\le 6 \\ -x_1 + x_2 &\le 1 \\ x_2 &\le 2 \\ x_1, x_2 &\ge 0 \end{aligned}$$

The variables s_1, s_2, s_3, and s_4 are the slacks associated with the respective constraints.

Next, we write the objective equation as

In this manner, the starting simplex tableau can be represented as follows:

Table 0

Basic	z	x_1	x_2	S_1	S_2	S_3	S_4	Solution	
z									z-row
S_1									S_1 -row
S_2									S_2-row
S_3									S_3-row
S_4									S_4-row

The design of the tableau specifies the set of basic and nonbasic variables as well as provides the solution associated with the starting iteration. As explained in Section 3.3.1, the simplex iterations start at the origin $(x_1, x_2) = (0, 0)$ whose associated set of nonbasic and basic variables are defined as

Substituting the nonbasic variables $(x_1, x_2) = (0, 0)$ and noting the special 0-1 arrangement of the coefficients of z and the basic variables (s_1, s_2, s_3, s_4) in the tableau, the following solution is immediately available (without any calculations):

Is the starting solution optimal? The objective function $z = 5x_1 + 4x_2$ shows that the solution can be improved by increasing x_1 or x_2. Using the argument in Section 3.3.1, x_1 with the *most positive* coefficient is selected as the *entering variable*. Equivalently, because the simplex tableau expresses the objective function as $z - 5x_1 - 4x_2 = 0$, the entering variable will correspond to the variable with the *most negative* coefficient in the objective equation. This rule is referred to as the **optimality condition.**

The mechanics of determining the leaving variable from the simplex tableau calls for computing the *nonnegative* **ratios** of the right-hand side of the equations (*Solution* column) to the corresponding constraint coefficients under the entering variable, x_1, as the following table shows.

Basic	**Entering**	**Solution**	**Ratio (or Intercept)**
s_1			
s_2			
s_3			
s_4			
	Conclusion:		

The *minimum nonnegative* ratio automatically identifies the current basic variable s_1 as the leaving variable and assigns the entering variable x_1 the new value of 4.

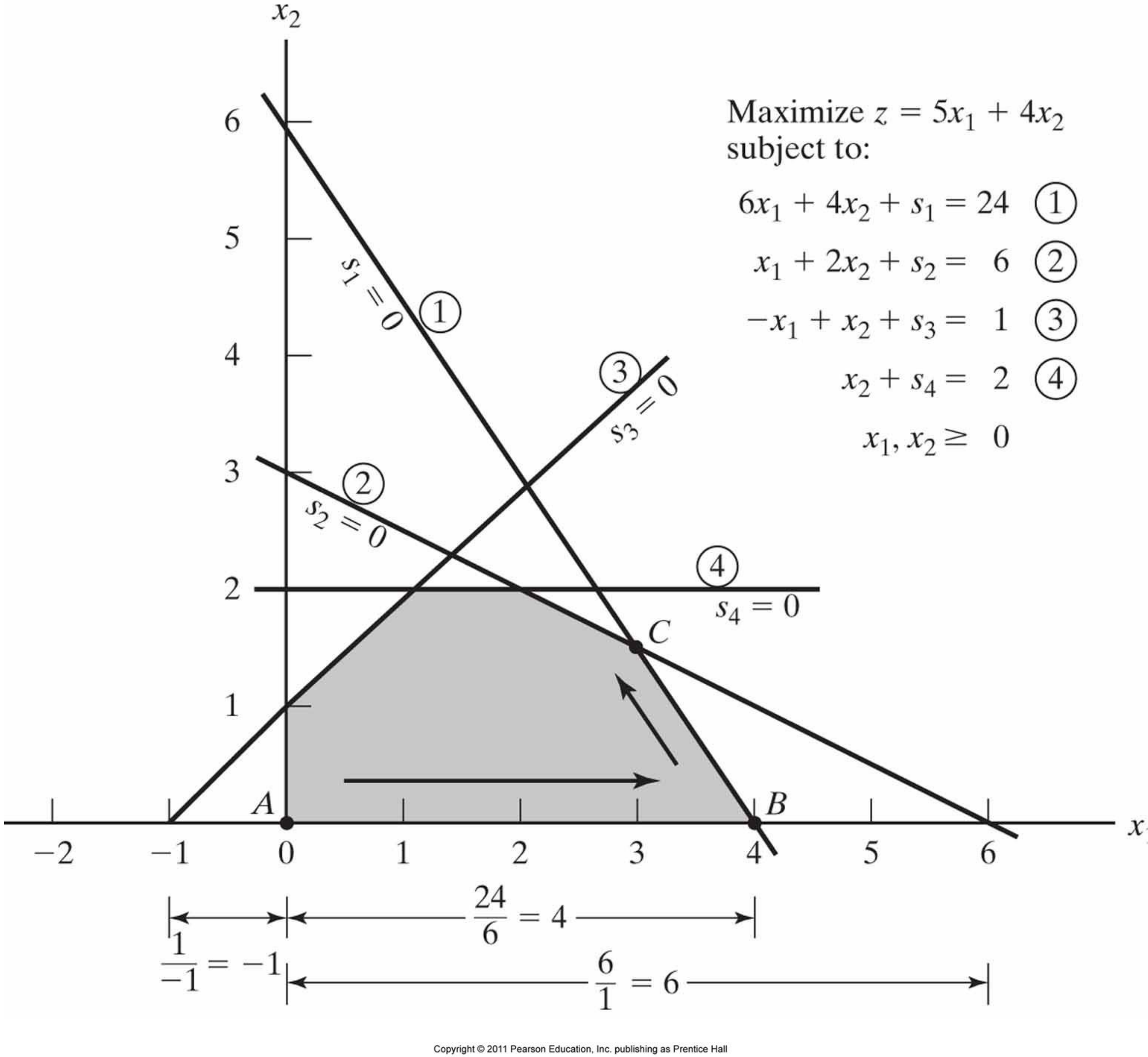


Figure 3.5 Graphical interpretation of the simplex method ratios in the Reddy Mikks model

How do the computed ratios determine the leaving variable and the value of the entering variable? Figure 3.5 shows that the computed ratios are actually the intercepts of the constraints with the entering variable (x_1) axis. We can see that the value of x_1 must be increased to 4 at corner point B, which is the smallest nonnegative intercept with the x_1-axis. An increase beyond B is infeasible. At point B, the current basic variable s_1 associated with constraint 1 assumes a zero value and becomes the *leaving variable*. The rule associated with the ratio computations is referred to as the **feasibility condition** because it guarantees the feasibility of the new solution.

The new solution point B is determined by "swapping" the entering variable x_1 and the leaving variable s_1 in the simplex tableau to produce the following sets of nonbasic and basic variables:

Nonbasic (zero) variables at B: (s_1, x_2)

Basic variables at B: (x_1, s_2, s_3, s_4)

The swapping process is based on the **Gauss-Jordan row operations**. It identifies the entering variable column as the **pivot column** and the leaving variable row as the **pivot row**. The intersection of the pivot column and the pivot row is called the **pivot element**. The following tableau is a restatement of the starting tableau with its pivot row and column highlighted.

Table 0

	Enter ↓									
	Basic	z	x_1	x_2	S_1	S_2	S_3	S_4	Solution	
	z	1	-5	-4	0	0	0	0	0	
Leave ←	S_1	0	6	4	1	0	0	0	24	Pivot row
	S_2	0	1	2	0	1	0	0	6	
	S_3	0	-1	1	0	0	1	0	1	
	S_4	0	0	1	0	0	0	1	2	
			Pivot Col.							

Table 1

Basic	z	x_1	x_2	S_1	S_2	S_3	S_4	Solution
z								
x_1								
S_2								
S_3								
S_4								

The Gauss-Jordan computations needed to produce the new basic solution include two types.

1. *Pivot row*

   a. Replace the leaving variable in the *Basic* column with the entering variable.

   b. New pivot row = Current pivot row ÷ Pivot element

2. *All other rows, including z*

   New Row = (Current row) − (Its pivot column coefficient) ×
   (New pivot row)

These computations are applied to the preceding tableau in the following manner:

1. Replace s_1 in the *Basic* column with x_1:

   New x_1-row = Current s_1-row ÷ 6

   $= \frac{1}{6}(0\ 6\ 4\ 1\ 0\ 0\ 0\ 24)$

   $= \left(0\ 1\ \frac{2}{3}\ \frac{1}{6}\ 0\ 0\ 0\ 4\right)$

2. New z-row = Current z-row − (−5) × New x_1-row

   $= \left(1\ -5\ -4\ 0\ 0\ 0\ 0\ 0\right) - (-5) \times \left(0\ 1\ \frac{2}{3}\ \frac{1}{6}\ 0\ 0\ 0\ 4\right)$

   $= \left(1\ 0\ -\frac{2}{3}\ \frac{5}{6}\ 0\ 0\ 0\ 20\right)$

3. New s_2-row = Current s_2-row − (1) × New x_1-row

   $= (0\ 1\ 2\ 0\ 1\ 0\ 0\ 6) - (1) \times \left(0\ 1\ \frac{2}{3}\ \frac{1}{6}\ 0\ 0\ 0\ 4\right)$

   $= \left(0\ 0\ \frac{4}{3}\ -\frac{1}{6}\ 1\ 0\ 0\ 2\right)$

4. New s_3-row = Current s_3-row − (−1) × New x_1-row

   $= (0\ -1\ 1\ 0\ 0\ 1\ 0\ 1) - (-1) \times \left(0\ 1\ \frac{2}{3}\ \frac{1}{6}\ 0\ 0\ 0\ 4\right)$

   $= \left(0\ 0\ \frac{5}{3}\ \frac{1}{6}\ 0\ 1\ 0\ 5\right)$

5. New s_4-row = Current s_4-row − (0) × New x_1-row

   $= (0\ 0\ 1\ 0\ 0\ 0\ 1\ 2) - (0)\left(0\ 1\ \frac{2}{3}\ \frac{1}{6}\ 0\ 0\ 0\ 4\right)$

   $= (0\ 0\ 1\ 0\ 0\ 0\ 1\ 2)$

The new basic solution is (x_1, s_2, s_3, s_4), and the new tableau becomes

Table 1

Basic	z	x_1	x_2	S_1	S_2	S_3	S_4	Solution
z	1	0	-2/3	5/6	0	0	0	20
x_1	0	1	2/3	1/6	0	0	0	4
S_2	0	0	4/3	-1/6	1	0	0	2
S_3	0	0	5/3	1/6	0	1	0	5
S_4	0	0	1	0	0	0	1	2

Observe that the new tableau has the same properties as the starting tableau. When we set the new nonbasic variables x_2 and s_1 to zero, the *Solution* column automatically yields the new basic solution $(x_1 = 4, s_2 = 2, s_3 = 5, s_4 = 2)$. This "conditioning" of the tableau is the result of the application of the Gauss-Jordan row operations. The corresponding new objective value is $z = 20$, which is consistent with

$$\text{New } z = \text{Old } z + \text{New } x_1\text{-value} \times \text{its objective coefficient}$$

$$= 0 + 4 \times 5 = 20$$

In the last tableau, the *optimality condition* shows that x_2 is the entering variable. The feasibility condition produces the following

Basic	Entering	Solution	Ratio (or Intercept)

Thus, s_2 leaves the basic solution and new value of x_2 is 1.5. The corresponding increase in z is $\frac{2}{3}x_2 = \frac{2}{3} \times 1.5 = 1$, which yields new $z = 20 + 1 = 21$.

Replacing s_2 in the *Basic* column with entering x_2, the following Gauss-Jordan row operations are applied:

1. New pivot x_2-row = Current s_2-row $\div \frac{4}{3}$
2. New z-row = Current z-row $- \left(-\frac{2}{3}\right) \times$ New x_2-row
3. New x_1-row = Current x_1-row $- \left(\frac{2}{3}\right) \times$ New x_2-row
4. New s_3-row = Current s_3-row $- \left(\frac{5}{3}\right) \times$ New x_2-row
5. New s_4-row = Current s_4-row $- (1) \times$ New x_2-row

Table 1

	Basic	z	x_1	x_2	S_1	S_2	S_3	S_4	Solution
	z	1	0	-2/3	5/6	0	0	0	20
	x_1	0	1	2/3	1/6	0	0	0	4
←	S_2	0	0	4/3	-1/6	1	0	0	2
	S_3	0	0	5/3	1/6	0	1	0	5
	S_4	0	0	1	0	0	0	1	2

These computations produce the following tableau:

Table 2

Basic	z	x_1	x_2	S_1	S_2	S_3	S_4	Solution
z								
x_1								
x_2								
S_3								
S_4								

Based on the optimality condition, *none* of the z-row coefficients associated with the nonbasic variables, s_1 and s_2, are negative. Hence, the last tableau is optimal.

The optimum solution can be read from the simplex tableau in the following manner. The optimal values of the variables in the *Basic* column are given in the right-hand-side *Solution* column and can be interpreted as

Decision variable	Optimum value	Recommendation
x_1	3	Produce 3 tons of exterior paint daily
x_2	$\frac{3}{2}$	Produce 1.5 tons of interior paint daily
z	21	Daily profit is $21,000

You can verify that the values $s_1 = s_2 = 0$, $s_3 = \frac{5}{2}$, $s_4 = \frac{1}{2}$ are consistent with the given values of x_1 and x_2 by substituting out the values of x_1 and x_2 in the constraints.

The solution also gives the status of the resources. A resource is designated as **scarce** if the activities (variables) of the model use the resource completely. Otherwise, the resource is **abundant**. This information is secured from the optimum tableau by checking the value of the slack variable associated with the constraint representing the resource. If the slack value is zero, the resource is used completely and, hence, is classified as scarce. Otherwise, a positive slack indicates that the resource is abundant.

The following table classifies the constraints of the model:

Resource	Slack value	Status
Raw material, $M1$	$s_1 = 0$	Scarce
Raw material, $M2$	$s_2 = 0$	Scarce
Market limit	$s_3 = \frac{5}{2}$	Abundant
Demand limit	$s_4 = \frac{1}{2}$	Abundant

Remarks. The simplex tableau offers a wealth of additional information that includes:

1. *Sensitivity analysis*, which deals with determining the conditions that will keep the current solution unchanged.

2. *Post-optimal analysis*, which deals with finding a new optimal solution when the data of the model are changed.

3.3.3 Summary of the Simplex Method

So far we have dealt with the maximization case. In minimization problems, the *optimality condition* calls for selecting the entering variable as the nonbasic variable with the most *positive* objective coefficient in the objective equation, the exact opposite rule of the maximization case. This follows because max z is equivalent to min $(-z)$. As for the *feasibility condition* for selecting the leaving variable, the rule remains unchanged.

Optimality condition. The entering variable in a maximization (minimization) problem is the *nonbasic* variable having the most negative (positive) coefficient in the z-row. Ties are broken arbitrarily. The optimum is reached at the iteration where all the z-row coefficients of the nonbasic variables are nonnegative (nonpositive).

Feasibility condition. For both the maximization and the minimization problems, the leaving variable is the *basic* variable associated with the smallest nonnegative ratio (with *strictly positive* denominator). Ties are broken arbitrarily.

Gauss-Jordan row operations.

1. Pivot row

   a. Replace the leaving variable in the *Basic* column with the entering variable.

   b. New pivot row = Current pivot row ÷ Pivot element

2. All other rows, including z

   New row = (Current row) − (pivot column coefficient) × (New pivot row)

The steps of the simplex method are

Step 1. Determine a starting basic feasible solution.

Step 2. Select an *entering variable* using the optimality condition. Stop if there is no entering variable; the last solution is optimal. Else, go to step 3.

Step 3. Select a *leaving variable* using the feasibility condition.

Step 4. Determine the new basic solution by using the appropriate Gauss-Jordan computations. Go to step 2.

Tabular form of simplex method

Example 1:

$\max z = 6x_1 + 5x_2$

s.t. $x_1 + x_2 \leq 5$

$3x_1 + 2x_2 \leq 12$

$x_1, x_2 \geq 0$

Example 2:

$$\min z = 2x_1 + 7x_2 - 2x_3$$

$$\text{s.t. } x_1 + x_2 + x_3 \le 1$$

$$-4x_1 - 2x_2 + 3x_3 \le 2$$

$$x_1, x_2, x_3 \ge 0$$

PROBLEM SET 3.3B

1. This problem is designed to reinforce your understanding of the simplex feasibility condition. In the first tableau in Example 3.3-1, we used the minimum (nonnegative) ratio test to determine the leaving variable. Such a condition guarantees that none of the new values of the basic variables will become negative (as stipulated by the definition of the LP). To demonstrate this point, force s_2, instead of s_1, to leave the basic solution. Now, look at the resulting simplex tableau, and you will note that s_1 assumes a negative value ($= -12$), meaning that the new solution is infeasible. This situation will never occur if we employ the minimum-ratio feasibility condition.

2. Consider the following set of constraints:

$$
\begin{aligned}
x_1 + 2x_2 + 2x_3 + 4x_4 &\le 40 \\
2x_1 - x_2 + x_3 + 2x_4 &\le 8 \\
4x_1 - 2x_2 + x_3 - x_4 &\le 10 \\
x_1, x_2, x_3, x_4 &\ge 0
\end{aligned}
$$

Solve the problem for each of the following objective functions.

(a) Maximize $z = 2x_1 + x_2 - 3x_3 + 5x_4$.

(b) Maximize $z = 8x_1 + 6x_2 + 3x_3 - 2x_4$.

(c) Maximize $z = 3x_1 - x_2 + 3x_3 + 4x_4$.

(d) Minimize $z = 5x_1 - 4x_2 + 6x_3 - 8x_4$.

*3. Consider the following system of equations:

$$
\begin{array}{rcrcrcrcrcrcrcrcl}
x_1 & + & 2x_2 & - & 3x_3 & + & 5x_4 & + & x_5 & & & & & & & = & 4 \\
5x_1 & - & 2x_2 & & & + & 6x_4 & & & + & x_6 & & & & & = & 8 \\
2x_1 & + & 3x_2 & - & 2x_3 & + & 3x_4 & & & & & + & x_7 & & & = & 3 \\
-x_1 & & & + & x_3 & - & 2x_4 & & & & & & & + & x_8 & = & 0
\end{array}
$$

$$x_1, x_2, \ldots, x_8 \geq 0$$

Let $x_5, x_6, \ldots,$ and x_8 be a given initial basic feasible solution. Suppose that x_1 becomes basic. Which of the given basic variables must become nonbasic at zero level to guarantee that all the variables remain nonnegative, and what is the value of x_1 in the new solution? Repeat this procedure for x_2, x_3, and x_4.

4. Consider the following LP:

Maximize $z = x_1$

subject to

$$
\begin{array}{rcrcrcrcl}
5x_1 & + & x_2 & & & & & = & 4 \\
6x_1 & & & + & x_3 & & & = & 8 \\
3x_1 & & & & & + & x_4 & = & 3
\end{array}
$$

$$x_1, x_2, x_3, x_4 \geq 0$$

(a) Solve the problem *by inspection* (do not use the Gauss-Jordan row operations), and justify the answer in terms of the basic solutions of the simplex method.

(b) Repeat (a) assuming that the objective function calls for minimizing $z = x_1$.

5. Solve the following problem by *inspection*, and justify the method of solution in terms of the basic solutions of the simplex method.

$$\text{Maximize } z = 5x_1 - 6x_2 + 3x_3 - 5x_4 + 12x_5$$

subject to

$$x_1 + 3x_2 + 5x_3 + 6x_4 + 3x_5 \leq 90$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

(*Hint:* A basic solution consists of one variable only.)

6. The following tableau represents a specific simplex iteration. All variables are nonnegative. The tableau is not optimal for either a maximization or a minimization problem. Thus, when a nonbasic variable enters the solution it can either increase or decrease z or leave it unchanged, depending on the parameters of the entering nonbasic variable.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	Solution
z	0	-5	0	4	-1	-10	0	0	620
x_8	0	3	0	-2	-3	-1	5	1	12
x_3	0	1	1	3	1	0	3	0	6
x_1	1	-1	0	0	6	-4	0	0	0

(a) Categorize the variables as basic and nonbasic and provide the current values of all the variables.

*(b) Assuming that the problem is of the maximization type, identify the nonbasic variables that have the potential to improve the value of z. If each such variable enters the basic solution, determine the associated leaving variable, if any, and the associated change in z. Do not use the Gauss-Jordan row operations.

(c) Repeat part (b) assuming that the problem is of the minimization type.

(d) Which nonbasic variable(s) will not cause a change in the value of z when selected to enter the solution?

7. Consider the two-dimensional solution space in Figure 3.6.

(a) Suppose that the objective function is given as

$$\text{Maximize } z = 3x_1 + 6x_2$$

If the simplex iterations start at point A, identify the path to the optimum point E.

(b) Determine the entering variable, the corresponding ratios of the feasibility condition, and the change in the value of z, assuming that the starting iteration occurs at point A and that the objective function is given as

$$\text{Maximize } z = 4x_1 + x_2$$

(c) Repeat (b), assuming that the objective function is

$$\text{Maximize } z = x_1 + 4x_2$$

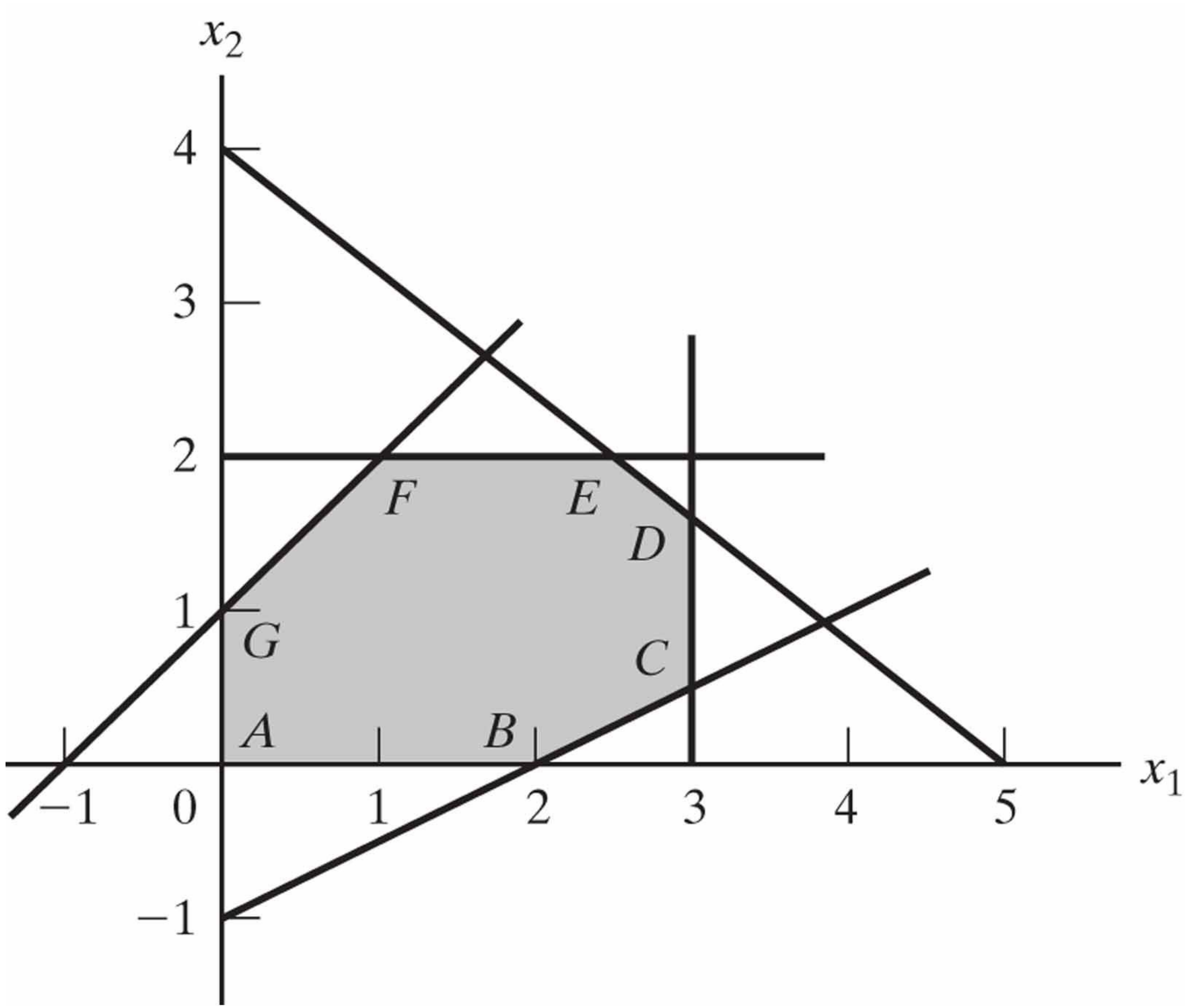


Figure 3.6 Solution Space for Problem 7, Set 3.3b

8. Consider the following LP:

$$\text{Maximize } z = 16x_1 + 15x_2$$

subject to

$$40x_1 + 31x_2 \le 124$$
$$-x_1 + x_2 \le 1$$
$$x_1 \le 3$$
$$x_1, x_2 \ge 0$$

(a) Solve the problem by the simplex method, where the entering variable is the nonbasic variable with the *most* negative z-row coefficient.

(b) Resolve the problem by the simplex algorithm, always selecting the entering variable as the nonbasic variable with the *least* negative z-row coefficient.

(c) Compare the number of iterations in (a) and (b). Does the selection of the entering variable as the nonbasic variable with the *most* negative z-row coefficient lead to a smaller number of iterations? What conclusion can be made regarding the optimality condition?

(d) Suppose that the sense of optimization is changed to minimization by multiplying z by -1. How does this change affect the simplex iterations?

*9. In Example 3.3-1, show how the second best optimal value of z can be determined from the optimal tableau.

10. Can you extend the procedure in Problem 9 to determine the third best optimal value of z?

11. The Gutchi Company manufactures purses, shaving bags, and backpacks. The construction includes leather and synthetics, leather being the scarce raw material. The production process requires two types of skilled labor: sewing and finishing. The following table gives the availability of the resources, their usage by the three products, and the profits per unit.

	Resource requirements per unit			
Resource	*Purse*	*Bag*	*Backpack*	Daily availability
Leather (ft^2)	2	1	3	42 ft^2
Sewing (hr)	2	1	2	40 hr
Finishing (hr)	1	.5	1	45 hr
Selling price ($)	24	22	45	

(a) Formulate the problem as a linear program and find the optimum solution (using TORA, Excel Solver, or AMPL).

(b) From the optimum solution determine the status of each resource.

Reference: *H.A.Taha, Operations Research: An Introduction,* Prentice Hall; 9th edition, *Singapore, 2010.*

Lecture 4
Artificial Starting Solution

Penalty rule for artificial variables
M-Method

3.4 ARTIFICIAL STARTING SOLUTION

As demonstrated in Example 3.3-1, LPs in which all the constraints are ($\leq$) with nonnegative right-hand sides offer a convenient all-slack starting basic feasible solution. Models involving ($=$) and/or ($\geq$) constraints do not.

The procedure for starting "ill-behaved" LPs with ($=$) and ($\geq$) constraints is to use **artificial variables** that play the role of slacks at the first iteration, and then dispose of them legitimately at a later iteration. Two closely related methods are introduced here: the *M*-method and the two-phase method.

3.4.1 *M*-Method

The *M*-method starts with the LP in equation form (Section 3.1). If equation i does not have a slack (or a variable that can play the role of a slack), an artificial variable, R_i, is added to form a starting solution similar to the convenient all-slack basic solution. However, because the artificial variables are not part of the original LP model, they are assigned a very high **penalty** in the objective function, thus forcing them (eventually) to equal zero in the optimum solution. This will always be the case if the problem has a feasible solution. The following rule shows how the penalty is assigned in the cases of maximization and minimization:

Penalty Rule for Artificial Variables.

Given M, a sufficiently large positive value (mathematically, $M \rightarrow \infty$), the objective coefficient of an artificial variable represents an appropriate **penalty** if:

$$\text{Artificial variable objective coefficient} = \begin{cases} -M, & \text{in maximization problems} \\ M, & \text{in minimization problems} \end{cases}$$

Example 3.4-1 Minimize $z = 4x_1 + x_2$

$$\text{subject to} \quad \begin{aligned} 3x_1 + x_2 &= 3 \\ 4x_1 + 3x_2 &\geq 6 \\ x_1 + 2x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Using x_3 as a surplus in the second constraint and x_4 as a slack in the third constraint, the equation form of the problem is given as

The third equation has its slack variable, x_4, but the first and second equations do not. Thus, we add the artificial variables R_1 and R_2 in the first two equations and penalize them in the objective function with $MR_1 + MR_2$ (because we are minimizing). The resulting LP is given as

The associated starting basic solution is now given by $(R_1, R_2, x_4) = (3, 6, 4)$

What value of M should we use? The answer depends on the data of the original LP. Recall that M must be sufficiently large *relative to the original objective coefficients* so it will act as a penalty that forces the artificial variables to zero level in the optimal solution. At the same time, since computers are the main tool for solving LPs, we do not want M to be too large (even though mathematically it should tend to infinity) because potential severe round-off error can result when very large values are manipulated with much smaller values. In the present example, the objective coefficients of x_1 and x_2 are 4 and 1, respectively. It thus appears reasonable to set $M = 100$.

Using $M = 100$, the starting simplex tableau is given as follows (for convenience, the z-column is eliminated because it does not change in all the iterations):

Before proceeding with the simplex method computations, we need to make the z-row consistent with the rest of the tableau. Specifically, in the tableau, $x_1 = x_2 = x_3 = 0$, which yields the starting basic solution $R_1 = 3$, $R_2 = 6$, and $x_4 = 4$. This solution yields $z = 100 \times 3 + 100 \times 6 = 900$ (instead of 0, as the right-hand side of the z-row currently shows). This inconsistency stems from the fact that R_1 and R_2 have nonzero coefficients $(-100, -100)$ in the z-row (compare with the all-slack starting solution in Example 3.3-1, where the z-row coefficients of the slacks are zero).

We can eliminate this inconsistency by substituting out R_1 and R_2 in the z-row using the appropriate constraint equations. In particular, notice the highlighted elements $(= 1)$ in the R_1-row and the R_2-row. Multiplying *each* of R_1-row and R_2-row by 100 and adding the *sum* to the z-row will substitute out R_1 and R_2 in the objective row—that is,

$$\text{New } z\text{-row} = \text{Old } z\text{-row} + (100 \times R_1\text{-row} + 100 \times R_2\text{-row})$$

The modified tableau thus becomes (verify!)

Notice that $z = 900$, which is consistent now with the values of the starting basic feasible solution: $R_1 = 3$, $R_2 = 6$, and $x_4 = 4$.

The last tableau is ready for us to apply the simplex method using the simplex optimality and the feasibility conditions, exactly as we did in Section 3.3.2. Because we are minimizing the objective function, the variable x_1 having the most *positive* coefficient in the z-row ($= 696$) enters the solution. The minimum ratio of the feasibility condition specifies R_1 as the leaving variable (verify!).

Once the entering and the leaving variables have been determined, the new tableau can be computed by using the familiar Gauss-Jordan operations.

Basic	x_1	x_2	x_3	R_1	R_2	x_4	Solution
z	0	167	-100	-232	0	0	204
x_1	1	$\frac{1}{3}$	0	$\frac{1}{3}$	0	0	1
R_2	0	$\frac{5}{3}$	-1	$-\frac{4}{3}$	1	0	2
x_4	0	$\frac{5}{3}$	0	$-\frac{1}{3}$	0	1	3

The last tableau shows that x_2 and R_2 are the entering and leaving variables, respectively. Continuing with the simplex computations, two more iterations are needed to reach the optimum: $x_1 = \frac{2}{5}$, $x_2 = \frac{9}{5}$, $z = \frac{17}{5}$ (verify with TORA!).

Note that the artificial variables R_1 and R_2 leave the basic solution in the first and second iterations, a result that is consistent with the concept of penalizing them in the objective function.

Remarks. The use of the penalty M will not force an artificial variable to zero level in the final simplex iteration if the LP does not have a feasible solution (i.e., the constraints are not consistent). In this case, the final simplex iteration will include at least one artificial variable at a positive level. Section 3.5.4 explains this situation.

Big-M method

If all artificial variables are equal to zero in the optimal solution, then we have found the optimal solution to the original problem.
If any artificial variables are positive in the optimal solution, then the original problem is infeasible.

Example 1:

min $z = -3x_1 + 4x_2$

s.t. $x_1 + x_2 \leq 4$

$2x_1 + 3x_2 \geq 18$

$x_1, x_2 \geq 0$

Example 2:

max $z = 5x_1+4x_2$

s.t. $x_1+2x_2 \geq 6$

$4x_1+2x_2 \leq 8$

$x_1, x_2 \geq 0$

PROBLEM SET 3.4A

1. Use hand computations to complete the simplex iteration of Example 3.4-1 and obtain the optimum solution.

3. In Example 3.4-1, identify the starting tableau for each of the following (independent) cases, and develop the associated z-row after substituting out all the artificial variables:
   *(a) The third constraint is $x_1 + 2x_2 \geq 4$.
   *(b) The second constraint is $4x_1 + 3x_2 \leq 6$.
   (c) The second constraint is $4x_1 + 3x_2 = 6$.
   (d) The objective function is to maximize $z = 4x_1 + x_2$.

5. Consider the following set of constraints:

$$x_1 + x_2 + x_3 = 7$$
$$2x_1 - 5x_2 + x_3 \geq 10$$
$$x_1, x_2, x_3 \geq 0$$

   Solve the problem for each of the following objective functions:
   (a) Maximize $z = 2x_1 + 3x_2 - 5x_3$.
   (b) Minimize $z = 2x_1 + 3x_2 - 5x_3$.
   (c) Maximize $z = x_1 + 2x_2 + x_3$.
   (d) Minimize $z = 4x_1 - 8x_2 + 3x_3$.

4. Consider the following set of constraints:

$$
\begin{aligned}
-2x_1 + 3x_2 &= 3 && (1)\\
4x_1 + 5x_2 &\geq 10 && (2)\\
x_1 + 2x_2 &\leq 5 && (3)\\
6x_1 + 7x_2 &\leq 3 && (4)\\
4x_1 + 8x_2 &\geq 5 && (5)\\
x_1, x_2 &\geq 0
\end{aligned}
$$

For each of the following problems, develop the z-row after substituting out the artificial variables:

(a) Maximize $z = 5x_1 + 6x_2$ subject to (1), (3), and (4).

(b) Maximize $z = 2x_1 - 7x_2$ subject to (1), (2), (4), and (5).

(c) Minimize $z = 3x_1 + 6x_2$ subject to (3), (4), and (5).

(d) Minimize $z = 4x_1 + 6x_2$ subject to (1), (2), and (5).

(e) Minimize $z = 3x_1 + 2x_2$ subject to (1) and (5).

*6. Consider the problem

$$\text{Maximize } z = 2x_1 + 4x_2 + 4x_3 - 3x_4$$

subject to

$$\begin{aligned} x_1 + x_2 + x_3 \quad\quad &= 4 \\ x_1 + 4x_2 \quad\quad + x_4 &= 8 \\ x_1, x_2, x_3, x_4 &\geq 0 \end{aligned}$$

The problem shows that x_3 and x_4 can play the role of slacks for the two equations. They differ from slacks in that they have nonzero coefficients in the objective function. We can use x_3 and x_4 as starting variable, but, as in the case of artificial variables, they must be substituted out in the objective function before the simplex iterations are carried out. Solve the problem with x_3 and x_4 as the starting basic variables and without using any artificial variables.

9. Show how the M-method will indicate that the following problem has no feasible solution.

$$\text{Maximize } z = 2x_1 + 5x_2$$

subject to

$$\begin{aligned} 3x_1 + 2x_2 &\geq 6 \\ 2x_1 + x_2 &\leq 2 \\ x_1, x_2 &\geq 0 \end{aligned}$$

7. Solve the following problem using x_3 and x_4 as starting basic feasible variables. As in Problem 6, do not use any artificial variables.

$$\text{Minimize } z = 3x_1 + 2x_2 + 3x_3$$

subject to

$$\begin{aligned} x_1 + 4x_2 + x_3 \quad\quad &\geq 7 \\ 2x_1 + x_2 \quad\quad + x_4 &\geq 10 \\ x_1, x_2, x_3, x_4 &\geq 0 \end{aligned}$$

8. Consider the problem

$$\text{Maximize } z = x_1 + 5x_2 + 3x_3$$

subject to

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 3 \\ 2x_1 - x_2 \quad\quad &= 4 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

The variable x_3 plays the role of a slack. Thus, no artificial variable is needed in the first constraint. However, in the second constraint, an artificial variable is needed. Use this starting solution (i.e., x_3 in the first constraint and R_2 in the second constraint) to solve this problem.

Reference: *H.A.Taha, Operations Research: An Introduction,* Prentice Hall; 9th edition, *Singapore, 2010.*

Lecture 5
Special Cases in the Simplex Method

Degeneracy
Alternative optima
Unbounded solutions
Nonexisting (or infeasible) solutions

3.5 SPECIAL CASES IN THE SIMPLEX METHOD

This section considers four special cases that arise in the use of the simplex method.

1. Degeneracy
2. Alternative optima
3. Unbounded solutions
4. Nonexisting (or infeasible) solutions

Our interest in studying these special cases is twofold: (1) to present a *theoretical* explanation of these situations and (2) to provide a *practical* interpretation of what these special results could mean in a real-life problem.

3.5.1 Degeneracy

In the application of the feasibility condition of the simplex method, a tie for the minimum ratio may occur and can be broken arbitrarily. When this happens, at least one *basic* variable will be zero in the next iteration and the new solution is said to be **degenerate**.

There is nothing alarming about a degenerate solution, with the exception of a small theoretical inconvenience, called **cycling** or **circling**, which we shall discuss shortly. From the practical standpoint, the condition reveals that the model has at least one *redundant* constraint. To provide more insight into the practical and theoretical impacts of degeneracy, a numeric example is used.

Example 3.5-1 (Degenerate Optimal Solution)

$$\text{Maximize } z = 3x_1 + 9x_2$$

subject to

$$x_1 + 4x_2 \leq 8$$
$$x_1 + 2x_2 \leq 4$$
$$x_1, x_2 \geq 0$$

Given the slack variables x_3 and x_4, the following tableaus provide the simplex iterations of the problem:

Iteration	Basic	x_1	x_2	x_3	x_4	Solution
0	z	-3	-9	0	0	0
x_2 enters	x_3	1	4	1	0	8
x_3 leaves	x_4	1	2	0	1	4
1	z	$-\frac{3}{4}$	0	$\frac{9}{4}$	0	18
x_1 enters	x_2	$\frac{1}{4}$	1	$\frac{1}{4}$	0	2
x_4 leaves	x_4	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	0
2	z	0	0	$\frac{3}{2}$	$\frac{3}{2}$	18
(optimum)	x_2	0	1	$\frac{1}{2}$	$-\frac{1}{2}$	2
	x_1	1	0	-1	2	0

Figure 3.7
LP degeneracy in Example 3.5-1

PROBLEM SET 3.5A

2. Consider the following LP:

$$\text{Maximize } z = 3x_1 + 2x_2$$

subject to

$$4x_1 - x_2 \le 8$$
$$4x_1 + 3x_2 \le 12$$
$$4x_1 + x_2 \le 8$$
$$x_1, x_2 \ge 0$$

(a) Show that the associated simplex iterations are temporarily degenerate (you may use TORA for convenience).

(b) Verify the result by solving the problem graphically (TORA's Graphic module can be used here).

3.5.2 Alternative Optima

When the objective function is parallel to a nonredundant **binding constraint** (i.e., a constraint that is satisfied as an equation at the optimal solution), the objective function can assume the same optimal value at more than one solution point, thus giving rise to alternative optima. The next example shows that there is an *infinite* number of such solutions. It also demonstrates the practical significance of encountering such solutions.

Example 3.5-2 (Infinite Number of Solutions)

$$\text{Maximize } z = 2x_1 + 4x_2$$

subject to

$$x_1 + 2x_2 \le 5$$

$$x_1 + x_2 \le 4$$

$$x_1, x_2 \ge 0$$

Figure 3.9 demonstrates how alternative optima can arise in the LP model when the objective function is parallel to a binding constraint. Any point on the *line segment BC* represents an alternative optimum with the same objective value $z = 10$.

The iterations of the model are given by the following tableaus.

Iteration	Basic	x_1	x_2	x_3	x_4	Solution
0	z	-2	-4	0	0	0
x_2 enters	x_3	1	2	1	0	5
x_3 leaves	x_4	1	1	0	1	4
1 (optimum)	z	0	0	2	0	10
x_1 enters	x_2	$\frac{1}{2}$	1	$\frac{1}{2}$	0	$\frac{5}{2}$
x_4 leaves	x_4	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	$\frac{3}{2}$
2	z	0	0	2	0	10
(alternative optimum)	x_2	0	1	1	-1	1
	x_1	1	0	-1	2	3

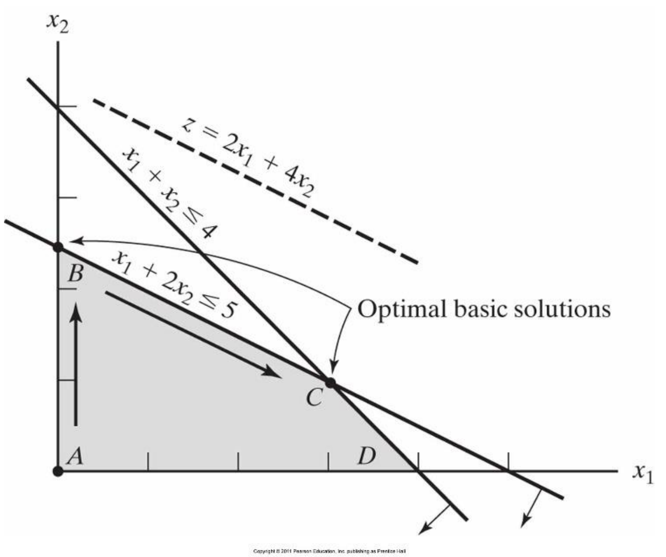


Figure 3.9

LP alternative optima in Example 3.5-2

Iteration 1 gives the optimum solution $x_1 = 0, x_2 = \frac{5}{2}$, and $z = 10$, which coincides with point B in Figure 3.9. How do we know from this tableau that alternative optima exist? Look at the z-equation coefficients of the *non*basic variables in iteration 1. The coefficient of nonbasic x_1 is zero, indicating that x_1 can enter the basic solution without changing the value of z, but causing a change in the values of the variables. Iteration 2 does just that—letting x_1 enter the basic solution and forcing x_4 to leave. The new solution point occurs at $C(x_1 = 3, x_2 = 1, z = 10)$. (TORA's Iterations option allows determining one alternative optimum at a time.)

The simplex method determines only the two corner points B and C. Mathematically, we can determine all the points (x_1, x_2) on the line segment BC as a nonnegative weighted average of points B and C. Thus, given

$$B: x_1 = 0, x_2 = \frac{5}{2}$$

$$C: x_1 = 3, x_2 = 1$$

then all the points on the line segment BC are given by

When $\alpha = 0$, $(\hat{x}_1, \hat{x}_2) = (3,1)$, which is point C. When $\alpha = 1$, $(\hat{x}_1, \hat{x}_2) = \left(0, \frac{5}{2}\right)$, which is point B. For values of α between 0 and 1, $(\hat{x}_1, \hat{x}_2)$ lies between B and C.

Remarks. In practice, alternative optima are useful because we can choose from many solutions without experiencing deterioration in the objective value. For instance, in the present example, the solution at B shows that activity 2 only is at a positive level, whereas at C both activities are positive. If the example represents a product-mix situation, there may be advantages in producing two products rather than one to meet market competition. In this case, the solution at C may be more appealing.

PROBLEM SET 3.5B

*1. For the following LP, identify three alternative optimal basic solutions, and then write a general expression for all the nonbasic alternative optima comprising these three basic solutions.

$$\text{Maximize } z = x_1 + 2x_2 + 3x_3$$

subject to

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &\leq 10 \\ x_1 + x_2 \qquad &\leq 5 \\ x_1 \qquad\qquad &\leq 1 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

Note: Although the problem has more than three alternative basic solution optima, you are only required to identify three of them. You may use TORA for convenience.

2. Solve the following LP:

$$\text{Maximize } z = 2x_1 - x_2 + 3x_3$$

subject to

$$x_1 - x_2 + 5x_3 \leq 10$$
$$2x_1 - x_2 + 3x_3 \leq 40$$
$$x_1, x_2, x_3 \geq 0$$

From the optimal tableau, show that all the alternative optima are not corner points (i.e., nonbasic). Give a two-dimensional graphical demonstration of the type of solution space and objective function that will produce this result. (You may use TORA for convenience.)

3. For the following LP, show that the optimal solution is degenerate and that none of the alternative solutions are corner points (you may use TORA for convenience).

$$\text{Maximize } z = 3x_1 + x_2$$

subject to

$$x_1 + 2x_2 \leq 5$$
$$x_1 + x_2 - x_3 \leq 2$$
$$7x_1 + 3x_2 - 5x_3 \leq 20$$
$$x_1, x_2, x_3 \geq 0$$

3.5.3 Unbounded Solution

In some LP models, the values of the variables may be increased indefinitely without violating any of the constraints—meaning that the solution space is *unbounded* in at least one variable. As a result, the objective value may increase (maximization case) or decrease (minimization case) indefinitely. In this case, both the solution space and the optimum objective value are unbounded.

Unboundedness points to the possibility that the model is poorly constructed. The most likely irregularity in such models is that one or more nonredundant constraints have not been accounted for, and the parameters (constants) of some constraints may not have been estimated correctly.

The following examples show how unboundedness, in both the solution space and the objective value, can be recognized in the simplex tableau.

Example 3.5-3 (Unbounded Objective Value)

Maximize $z = 2x_1 + x_2$

subject to $x_1 - x_2 \leq 10$

$$2x_1 \leq 40$$

$$x_1, x_2 \geq 0$$

Starting Iteration

Basic	x_1	x_2	x_3	x_4	Solution
z	-2	-1	0	0	0
x_3	1	-1	1	0	10
x_4	2	0	0	1	40

In the starting tableau, both x_1 and x_2 have negative z-equation coefficients. Hence either one can improve the solution. Because x_1 has the most negative coefficient, it is normally selected as the entering variable. However, *all* the *constraint* coefficients under x_2 (i.e., the denominators of the ratios of the feasibility condition) are *negative* or *zero*. This means that there is no leaving variable and that x_2 can be increased indefinitely without violating any of the constraints (compare with the graphical interpretation of the minimum ratio in Figure 3.5). Because each unit increase in x_2 will increase z by 1, an infinite increase in x_2 leads to an infinite increase in z. Thus, the problem has no bounded solution. This result can be seen in Figure 3.10. The solution space is unbounded in the direction of x_2, and the value of z can be increased indefinitely.

PROBLEM SET 3.5C

1. *TORA Experiment.* Solve Example 3.5-3 using TORA's Iterations option and show that even though the solution starts with x_1 as the entering variable (per the optimality condition), the simplex algorithm will point eventually to an unbounded solution.

*2. Consider the LP:

$$\text{Maximize } z = 20x_1 + 10x_2 + x_3$$

subject to

$$\begin{aligned} 3x_1 - 3x_2 + 5x_3 &\le 50 \\ x_1 \qquad\quad + x_3 &\le 10 \\ x_1 - x_2 + 4x_3 &\le 20 \\ x_1, x_2, x_3 &\ge 0 \end{aligned}$$

(a) By inspecting the constraints, determine the direction (x_1, x_2, or x_3) in which the solution space is unbounded.

(b) Without further computations, what can you conclude regarding the optimum objective value?

3.5.4 Infeasible Solution

LP models with inconsistent constraints have no feasible solution. This situation can never occur if *all* the constraints are of the type $\leq$ with nonnegative right-hand sides because the slacks provide a feasible solution. For other types of constraints, we use artificial variables. Although the artificial variables are penalized in the objective function to force them to zero at the optimum, this can occur only if the model has a feasible space. Otherwise, at least one artificial variable will be *positive* in the optimum iteration. From the practical standpoint, an infeasible space points to the possibility that the model is not formulated correctly.

Example 3.5-4 (Infeasible Solution Space)

Consider the following LP:

$$\text{Maximize } z = 3x_1 + 2x_2$$

subject to

$$\begin{aligned} 2x_1 + x_2 &\leq 2 \\ 3x_1 + 4x_2 &\geq 12 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Using the penalty $M = 100$ for the artificial variable R, the following tableaux provide the simplex iterations of the model.

Iteration	Basic	x_1	x_2	x_4	x_3	R	Solution
0	z	-303	-402	100	0	0	-1200
x_2 enters	x_3	2	1	0	1	0	2
x_3 leaves	R	3	4	-1	0	1	12
1	z	501	0	100	402	0	-396
(pseudo-optimum)	x_2	2	1	0	1	0	2
	R	-5	0	-1	-4	1	4

Optimum iteration 1 shows that the artificial variable R is *positive* (= 4), which indicates that the problem is infeasible. Figure 3.11 demonstrates the infeasible solution space. By allowing the artificial variable to be positive, the simplex method, in essence, has reversed the direction of the inequality from $3x_1 + 4x_2 \geq 12$ to $3x_1 + 4x_2 \leq 12$ (can you explain how?). The result is what we may call a **pseudo-optimal** solution.

Figure 3.11

Infeasible solution of Example 3.5-4

PROBLEM SET 3.5D

*1. Toolco produces three types of tools, $T1$, $T2$, and $T3$. The tools use two raw materials, $M1$ and $M2$, according to the data in the following table:

	Number of units of raw materials per tool		
Raw material	$T1$	$T2$	$T3$
$M1$	3	5	6
$M2$	5	3	4

The available daily quantities of raw materials $M1$ and $M2$ are 1000 units and 1200 units, respectively. The marketing department informed the production manager that according to their research, the daily demand for all three tools must be at least 500 units. Will the manufacturing department be able to satisfy the demand? If not, what is the most Toolco can provide of the three tools?

2. *TORA Experiment*. Consider the LP model

$$\text{Maximize } z = 3x_1 + 2x_1 + 3x_3$$

subject to

$$2x_1 + x_2 + x_3 \le 2$$
$$3x_1 + 4x_2 + 2x_3 \ge 8$$
$$x_1, x_2, x_3 \ge 0$$

Use TORA's Iterations ⇒ M-Method to show that the optimal solution includes an artificial basic variable, but at zero level. Does the problem have a *feasible* optimal solution?

Reference: *H.A.Taha, Operations Research: An Introduction,* Prentice Hall; 9th edition, *Singapore, 2010.*

Lecture 6
Converting an LP to standart form

Why standart form?
Variables with lower/upper/interval bounds
Free variables

- Why standart form?
  The implementation of simplex method requires the LP problem in standard form which means
  - all variables involved are restricted to be non-negative
  - all constraints are equalities, with constant, non-negative right-hand sides
- Converting may require new variables and rearranging constraints:
  - an inequality can be multiplied by −1 to get non-negative RHS.
  - inequalities can be converted to equalities by adding or subtracting non-negative variables (slack or surplus)
  - Unrestricted variables can be dealt with by writing the variable as the difference of two new non-negative variables

Every LP can be Transformed to Standard Form

Free variables with upper bounds ($x_i \le u_i$)

If a free variable x_i has an upper bound u_i , that is

$x_i \le u_i \Rightarrow 0 \le u_i - x_i \Rightarrow x_i' \ge 0$ where $x_i' = u_i - x_i$.

Example 1: Transform the following LP to standart form.

max $z = x_1 + x_2$

s.t. $-x_1 + 2x_2 \le 8$

$x_1 + 2x_2 \le 10$

$x_1 \le 8 \;,\; x_2 \ge 0$

Example 2:

$$\max z = -x_1 + 2x_2$$

$$\text{s.t. } -3x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 16$$

$$0 \leq x_1 \leq 9 \ , \ x_2 \geq 0$$

Free variables with lower bounds **($l_i \le x_i$)**

If a free variable x_i has lower bound l_i which is not zero, that is

$$x_i \ge l_i \;\Rightarrow\; x_i - l_i \ge 0 \;\Rightarrow\; x_i' \ge 0 \quad \text{where } x_i' = x_i - l_i .$$

Example 3:

max $z= -x_1+3x_2$

s.t. $-2x_1+3x_2 \le 24$

$x_1+2x_2 \le 8$

$x_1 \ge -8 \;,\; x_2 \ge 0$

<u>Variables with interval bounds</u> ($l_i \le x_i \le u_i$)

An interval bound of the form $l_i \le x_i \le u_i$ can be transformed into one non-negativity constraint and one linear inequality constraint in standard form by:

$l_i \le x_i \le u_i \Rightarrow l_i - l_i \le x_i - l_i \le u_i - l_i \Rightarrow 0 \le x_i' \le u_i - l_i$

<u>Example 4:</u>

max $z = -x_1 + 3x_2$

s.t. $-x_1 + 2x_2 \le 8$

$x_1 + 2x_2 \le 12$

$4 \le x_1 \le 10$

$x_2 \ge 0$

Free variables

- Sometimes a variable is given without any bounds. Such variables are called free variables.
- To obtain standard form every free variable must be replaced by the difference of two non-negative variables. That is, if x_i is free, then we get

$$x_i = u_i - v_i$$

with $0 \leq u_i$ and $0 \leq v_i$.

Example 5 (sample exam question):

$$\max\ z = 10x_1 - 5x_2 - 2x_3$$

$$\begin{aligned} s.t. \quad & -2x_1 + 3x_2 + 3x_3 \le 10 \\ & x_1 + x_2 - 5x_3 = 8 \\ & x_1 \ge -3 \\ & -1 \le x_2 \le 1 \\ & x_3 \text{ is free} \end{aligned}$$

a) Obtain the standart form of the given problem.

b) Write its starting *basic feasible solution*.

Reference: *http://www.ifp.illinois.edu/~angelia/ge330fall09_stform4.pdf*, *Date accessed: 15.07.2017*
https://sites.math.washington.edu/~burke/crs/407/lectures/L4-lp_standard_form.pdf, *Date accessed: 15.07.2017*

Lecture 7
Theory of LP

Solving systems with more variables than equations
Matrix representation of a standart LP
Getting a basic solution
Theory of optimality and feasibility conditions
Some basic theorems and their proofs

Solving Systems with More Variables than Equations

Suppose now that $\mathbf{A} \in \mathbb{R}^{m \times n}$ where $m \leq n$. Let $\mathbf{b} \in \mathbb{R}^m$. Then the equation:

$$(3.46) \quad \mathbf{A}\mathbf{x} = \mathbf{b}$$

has more variables than equations and is underdetermined and if $\mathbf{A}$ has full row rank then the system will have an infinite number of solutions. We can formulate an expression to describe this infinite set of solutions.

Sine $\mathbf{A}$ has full row rank, we may choose any m linearly independent columns of $\mathbf{A}$ corresponding to a subset of the variables, say $x_{i_1}, \dots, x_{i_m}$. We can use these to form the matrix

$$(3.47) \quad \mathbf{B} = [\mathbf{A}_{\cdot i_1} \cdots \mathbf{A}_{\cdot i_m}]$$

from the columns $\mathbf{A}_{\cdot i_1}, \dots, \mathbf{A}_{\cdot i_m}$ of $\mathbf{A}$, so that B is invertible. It should be clear at this point that B will be invertible precisely because we've chosen m linearly independent column vectors. We can then use elementary column operations to write the matrix $\mathbf{A}$ as:

$$(3.48) \quad \mathbf{A} = [\mathbf{B}|\mathbf{N}]$$

The matrix $\mathbf{N}$ is composed of the $n - m$ other columns of $\mathbf{A}$ not in $\mathbf{B}$. We can similarly sub-divide the column vector $\mathbf{x}$ and write:

$$(3.49) \quad [\mathbf{B}|\mathbf{N}] \begin{bmatrix} \mathbf{x_B} \\ \mathbf{x_N} \end{bmatrix} = \mathbf{b}$$

where the vector $\mathbf{x_B}$ are the variables corresponding to the columns in $\mathbf{B}$ and the vector $\mathbf{x_N}$ are the variables corresponding to the columns of the matrix $\mathbf{N}$.

DEFINITION 3.46 (Basic Variables). For historical reasons, the variables in the vector $\mathbf{x_B}$ are called the *basic variables* and the variables in the vector $\mathbf{x_N}$ are called the *non-basic* variables.

We can use matrix multiplication to expand the left hand side of this expression as:

$$\mathbf{B}\mathbf{x_B} + \mathbf{N}\mathbf{x_N} = \mathbf{b} \tag{3.50}$$

The fact that $\mathbf{B}$ is composed of all linearly independent columns implies that applying Gauss-Jordan elimination to it will yield an $m \times m$ identity and thus that $\mathbf{B}$ is invertible. We can solve for basic variables $\mathbf{x_B}$ in terms of the non-basic variables:

$$\mathbf{x_B} = \mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{N}\mathbf{x_N} \tag{3.51}$$

We can find an arbitrary solution to the system of linear equations by choosing values for the variables the non-basic variables and solving for the basic variable values using Equation 3.51.

DEFINITION 3.47. (Basic Solution) When we assign $\mathbf{x_N} = 0$, the resulting solution for $\mathbf{x}$ is called a *basic solution* and

$$\mathbf{x_B} = \mathbf{B}^{-1}\mathbf{b} \tag{3.52}$$

EXAMPLE 3.48. Consider the problem:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix} \tag{3.53}$$

Then we can let $x_3 = 0$ and:

$$\mathbf{B} = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} \tag{3.54}$$

We then solve[1]:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{B}^{-1} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} \frac{-19}{3} \\ \frac{20}{3} \end{bmatrix} \tag{3.55}$$

Other basic solutions could be formed by creating $\mathbf{B}$ out of columns 1 and 3 or columns 2 and 3.

EXERCISE 38. Find the two other basic solutions in Example 3.48 corresponding to

$$\mathbf{B} = \begin{bmatrix} 2 & 3 \\ 5 & 6 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 & 3 \\ 4 & 6 \end{bmatrix}$$

In each case, determine what the matrix $\mathbf{N}$ is. [Hint: Find the solutions any way you like. Make sure you record exactly which x_i ($i \in \{1, 2, 3\}$) is equal to zero in each case.]

Matrix representation of a standart LP

- Consider the following standart LP problem:

max $z = c^T x$

s.t. $Ax = b,$

$x \geq 0.$

Getting a Basic Feasible Solution

To get a basic solution rewrite the linear program in terms of the basis where B is the basis of A:

$A=[B\ N]$, $c = [c_B\ c_N]^T$, $x = [x_B\ x_N]^T$, $x_B \in R^m$, $x_N \in R^{n-m}$

3. The Simplex Algorithm

Suppose we have a basic feasible solution $\mathbf{x} = (\mathbf{x_B}, \mathbf{x_N})$. We can divide the cost vector $\mathbf{c}$ into its basic and non-basic parts, so we have $\mathbf{c} = [\mathbf{c_B}|\mathbf{c_N}]^T$. Then the objective function becomes:

$$(5.10) \quad \mathbf{c}^T\mathbf{x} = \mathbf{c}_\mathbf{B}^T\mathbf{x_B} + \mathbf{c}_\mathbf{N}^T\mathbf{x_N}$$

We can substitute Equation 5.8 into Equation 5.10 to obtain:

$$(5.11) \quad \mathbf{c}^T\mathbf{x} = \mathbf{c}_\mathbf{B}^T\left(\mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{N}\mathbf{x_N}\right) + \mathbf{c_N}\mathbf{x_N} = \mathbf{c}_\mathbf{B}^T\mathbf{B}^{-1}\mathbf{b} + \left(\mathbf{c}_\mathbf{N}^T - \mathbf{c}_\mathbf{B}^T\mathbf{B}^{-1}\mathbf{N}\right)\mathbf{x}_N$$

Let $\mathcal{J}$ be the set of indices of non-basic variables. Then we can write Equation 5.11 as:

$$(5.12) \quad z(x_1, \dots, x_n) = \mathbf{c}_\mathbf{B}^T\mathbf{B}^{-1}\mathbf{b} + \sum_{j \in \mathcal{J}} \left(\mathbf{c}_j - \mathbf{c}_\mathbf{B}^T\mathbf{B}^{-1}\mathbf{A}_{\cdot j}\right) x_j$$

Consider now the fact $x_j = 0$ for all $j \in \mathcal{J}$. Further, we can see that:

$$(5.13) \quad \frac{\partial z}{\partial x_j} = \mathbf{c}_j - \mathbf{c}_\mathbf{B}^T\mathbf{B}^{-1}\mathbf{A}_{\cdot j}$$

This means that if $\mathbf{c}_j - \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{A}_{\cdot j} > 0$ and we *increase* x_j from zero to some new value, then we will *increase* the value of the objective function. For historic reasons, we actually consider the value $\mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{A}_{\cdot j} - \mathbf{c}_j$, called the *reduced cost* and denote it as:

$$(5.14) \quad -\frac{\partial z}{\partial x_j} = z_j - c_j = \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{A}_{\cdot j} - \mathbf{c}_j$$

In a maximization problem, we chose non-basic variables x_j with negative reduced cost to become basic because, in this case, $\partial z/\partial x_j$ is *positive*.

Assume we chose x_j, a non-basic variable to become non-zero (because $z_j - c_j < 0$). We wish to know which of the basic variables will become zero as we *increase* x_j away from zero. We must also be very careful that *none* of the variables become negative as we do this.

By Equation 5.8 we know that the only current basic variables will be affected by increasing x_j. Let us focus explicitly on Equation 5.8 where we include only variable x_j (since all other non-basic variables are kept zero). Then we have:

$$(5.15) \quad \mathbf{x}_{\mathbf{B}} = \mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{A}_{\cdot j}x_j$$

Let $\overline{\mathbf{b}} = \mathbf{B}^{-1}\mathbf{b}$ be an $m \times 1$ column vector and let that $\overline{\mathbf{a}_j} = \mathbf{B}^{-1}\mathbf{A}_{\cdot j}$ be another $m \times 1$ column. Then we can write:

$$(5.16) \quad \mathbf{x}_{\mathbf{B}} = \overline{\mathbf{b}} - \overline{\mathbf{a}}_j x_j$$

Let $\overline{\mathbf{b}} = [\overline{b}_1, \dots \overline{b}_m]^T$ and $\overline{\mathbf{a}}_j = [\overline{a}_{j_1}, \dots, \overline{a}_{j_m}]$, then we have:

$$(5.17) \quad \begin{bmatrix} x_{B_1} \\ x_{B_2} \\ \vdots \\ x_{B_m} \end{bmatrix} = \begin{bmatrix} \overline{b}_1 \\ \overline{b}_2 \\ \vdots \\ \overline{b}_m \end{bmatrix} - \begin{bmatrix} \overline{a}_{j_1} \\ \overline{b}_{j_2} \\ \vdots \\ \overline{b}_{j_m} \end{bmatrix} x_j = \begin{bmatrix} \overline{b}_1 - \overline{a}_{j_1} x_j \\ \overline{b}_2 - \overline{a}_{j_2} x_j \\ \vdots \\ \overline{b}_m - \overline{a}_{j_m} x_j \end{bmatrix}$$

We know (a priori) that $\overline{b}_i \geq 0$ for $i = 1, \dots, m$. If $\overline{a}_{j_i} \leq 0$, then as we increase x_j, $\overline{b}_i - \overline{a}_{j_i} \geq 0$ no matter how large we make x_j. On the other hand, if $\overline{a}_{j_i} > 0$, then as we increase x_j we know that $\overline{b}_i - \overline{a}_{j_i} x_j$ will get smaller and eventually hit zero. In order to ensure that *all* variables remain non-negative, we cannot increase x_j beyond a certain point.

For each i ($i = 1, \dots, m$) such that $\overline{a}_{j_i} > 0$, the value of x_j that will make x_{B_i} goto 0 can be found by observing that:

$$(5.18) \quad x_{B_i} = \overline{b}_i - \overline{a}_{j_i} x_j$$

and if $x_{B_i} = 0$, then we can solve:

$$(5.19) \quad 0 = \overline{b}_i - \overline{a}_{j_i} x_j \implies x_j = \frac{\overline{b}_i}{\overline{a}_{j_i}}$$

Thus, the *largest possible value* we can assign x_j and ensure that all variables remain positive is:

$$(5.20) \quad \min \left\{ \frac{\overline{b}_i}{\overline{a}_{j_i}} : i = 1, \dots, m \text{ and } a_{j_i} > 0 \right\}$$

Expression 5.20 is called the *minimum ratio test*. We are interested in which index i is the minimum ratio.

Suppose that in executing the minimum ratio test, we find that $x_j = \overline{b}_k/\overline{a}_{j_k}$. The variable x_j (which was non-basic) becomes basic and the variable $x_{\mathbf{B}_k}$ becomes non-basic. All other basic variables remain basic (and positive). In executing this procedure (of exchanging one basic variable and one non-basic variable) we have moved from one extreme point of X to another.

THEOREM 5.6. *If $z_j - c_j \geq 0$ for all $j \in \mathcal{J}$, then the current basic feasible solution is optimal.*

EXAMPLE 5.9. Consider the Toy Maker Problem (from Example 2.3). The linear programming problem given in Equation 2.8 is:

$$\begin{cases} \max & z(x_1, x_2) = 7x_1 + 6x_2 \\ s.t. & 3x_1 + x_2 \leq 120 \\ & x_1 + 2x_2 \leq 160 \\ & x_1 \leq 35 \\ & x_1 \geq 0 \\ & x_2 \geq 0 \end{cases}$$

We can convert this problem to standard form by introducing the slack variables s_1, s_2 and s_3:

$$\begin{cases} \max \quad z(x_1, x_2) = 7x_1 + 6x_2 \\ s.t. \;\; 3x_1 + x_2 + s_1 = 120 \\ x_1 + 2x_2 + s_2 = 160 \\ x_1 + s_3 = 35 \\ x_1, x_2, s_1, s_2, s_3 \geq 0 \end{cases}$$

which yields the matrices

$$\mathbf{c} = \begin{bmatrix} 7 \\ 6 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{bmatrix} \quad \mathbf{A} = \begin{bmatrix} 3 & 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 120 \\ 160 \\ 35 \end{bmatrix}$$

We can begin with the matrices:

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{N} = \begin{bmatrix} 3 & 1 \\ 1 & 2 \\ 1 & 0 \end{bmatrix}$$

In this case we have:

$$\mathbf{x_B} = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} \quad \mathbf{x_N} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \mathbf{c_B} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{c_N} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$$

and $$\mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 120 \\ 160 \\ 35 \end{bmatrix} \quad \mathbf{B}^{-1}\mathbf{N} = \begin{bmatrix} 3 & 1 \\ 1 & 2 \\ 1 & 0 \end{bmatrix}$$

Therefore:

$$\mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{b} = 0 \quad \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{N} = \begin{bmatrix} 0 & 0 \end{bmatrix} \quad \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{N} - \mathbf{c}_{\mathbf{N}} = \begin{bmatrix} -7 & -6 \end{bmatrix}$$

Using this information, we can compute:

$$\mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{A}_{\cdot 1} - \mathbf{c}_1 = -7$$
$$\mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{A}_{\cdot 2} - \mathbf{c}_2 = -6$$

and therefore:

$$\frac{\partial z}{\partial x_1} = 7 \text{ and } \frac{\partial z}{\partial x_2} = 6$$

Based on this information, we could chose either x_1 or x_2 to enter the basis and the value of the objective function would increase. If we chose x_1 to enter the basis, then we must determine which variable will leave the basis. To do this, we must investigate the elements of $\mathbf{B}^{-1}\mathbf{A}_{\cdot 1}$ and the current basic feasible solution $\mathbf{B}^{-1}\mathbf{b}$. Since each element of $\mathbf{B}^{-1}\mathbf{A}_{\cdot 1}$ is positive, we must perform the minimum ratio test on each element of $\mathbf{B}^{-1}\mathbf{A}_{\cdot 1}$. We know that $\mathbf{B}^{-1}\mathbf{A}_{\cdot 1}$ is just the first column of $\mathbf{B}^{-1}\mathbf{N}$ which is:

$$\mathbf{B}^{-1}\mathbf{A}_{\cdot 1} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

Performing the minimum ratio test, we see have:

$$\min\left\{\frac{120}{3}, \frac{160}{1}, \frac{35}{1}\right\}$$

In this case, we see that index 3 (35/1) is the minimum ratio. Therefore, variable x_1 will enter the basis and variable s_3 will leave the basis. The new basic and non-basic variables will be:

$$\mathbf{x_B} = \begin{bmatrix} s_1 \\ s_2 \\ x_1 \end{bmatrix} \quad \mathbf{x_N} = \begin{bmatrix} s_3 \\ x_2 \end{bmatrix} \quad \mathbf{c_B} = \begin{bmatrix} 0 \\ 0 \\ 7 \end{bmatrix} \quad \mathbf{c_N} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

and the matrices become:

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{N} = \begin{bmatrix} 0 & 1 \\ 0 & 2 \\ 1 & 0 \end{bmatrix}$$

Note we have simply swapped the column corresponding to x_1 with the column corresponding to s_3 in the basis matrix $\mathbf{B}$ and the non-basic matrix $\mathbf{N}$. We will do this repeatedly in the example and we recommend the reader keep track of which variables are being exchanged and why certain columns in $\mathbf{B}$ are being swapped with those in $\mathbf{N}$.

Using the new $\mathbf{B}$ and $\mathbf{N}$ matrices, the derived matrices are then:

$$\mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 15 \\ 125 \\ 35 \end{bmatrix} \quad \mathbf{B}^{-1}\mathbf{N} = \begin{bmatrix} -3 & 1 \\ -1 & 2 \\ 1 & 0 \end{bmatrix}$$

The cost information becomes:

$$\mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{b} = 245 \quad \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{N} = \begin{bmatrix} 7 & 0 \end{bmatrix} \quad \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{N} - \mathbf{c}_{\mathbf{N}} = \begin{bmatrix} 7 & -6 \end{bmatrix}$$

Using the new $\mathbf{B}$ and $\mathbf{N}$ matrices, the derived matrices are then:

$$\mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 15 \\ 125 \\ 35 \end{bmatrix} \quad \mathbf{B}^{-1}\mathbf{N} = \begin{bmatrix} -3 & 1 \\ -1 & 2 \\ 1 & 0 \end{bmatrix}$$

The cost information becomes:

$$\mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{b} = 245 \quad \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{N} = \begin{bmatrix} 7 & 0 \end{bmatrix} \quad \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{N} - \mathbf{c}_{\mathbf{N}} = \begin{bmatrix} 7 & -6 \end{bmatrix}$$

using this information, we can compute:

$$\mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{A}_{\cdot 5} - \mathbf{c}_5 = 7$$
$$\mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{A}_{\cdot 2} - \mathbf{c}_2 = -6$$

Based on this information, we can only choose x_2 to enter the basis to ensure that the value of the objective function increases. We can perform the minimum ration test to figure out which basic variable will leave the basis. We know that $\mathbf{B}^{-1}\mathbf{A}_{\cdot 2}$ is just the second column of $\mathbf{B}^{-1}\mathbf{N}$ which is:

$$\mathbf{B}^{-1}\mathbf{A}_{\cdot 2} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

Performing the minimum ratio test, we see have:

$$\min \left\{ \frac{15}{1}, \frac{125}{2} \right\}$$

In this case, we see that index 1 (15/1) is the minimum ratio. Therefore, variable x_2 will enter the basis and variable s_1 will leave the basis. The new basic and non-basic variables will be: The new basic and non-basic variables will be:

$$\mathbf{x_B} = \begin{bmatrix} x_2 \\ s_2 \\ x_1 \end{bmatrix} \quad \mathbf{x_N} = \begin{bmatrix} s_3 \\ s_1 \end{bmatrix} \quad \mathbf{c_B} = \begin{bmatrix} 6 \\ 0 \\ 7 \end{bmatrix} \quad \mathbf{c_N} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and the matrices become:

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{N} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}$$

The derived matrices are then:

$$\mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 15 \\ 95 \\ 35 \end{bmatrix} \quad \mathbf{B}^{-1}\mathbf{N} = \begin{bmatrix} -3 & 1 \\ 5 & -2 \\ 1 & 0 \end{bmatrix}$$

The cost information becomes:

$$\mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{b} = 335 \quad \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{N} = \begin{bmatrix} -11 & 6 \end{bmatrix} \quad \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{N} - \mathbf{c}_{\mathbf{N}} = \begin{bmatrix} -11 & 6 \end{bmatrix}$$

Based on this information, we can only choose s_3 to (re-enter) the basis to ensure that the value of the objective function increases. We can perform the minimum ration test to figure out which basic variable will leave the basis. We know that $\mathbf{B}^{-1}\mathbf{A}_{\cdot 5}$ is just the fifth column of $\mathbf{B}^{-1}\mathbf{N}$ which is:

$$\mathbf{B}^{-1}\mathbf{A}_{\cdot 5} = \begin{bmatrix} -3 \\ 5 \\ 1 \end{bmatrix}$$

Performing the minimum ratio test, we see have:

$$\min \left\{ \frac{95}{5}, \frac{35}{1} \right\}$$

In this case, we see that index 2 (95/5) is the minimum ratio. Therefore, variable s_3 will enter the basis and variable s_2 will leave the basis. The new basic and non-basic variables will be:

$$\mathbf{x}_{\mathbf{B}} = \begin{bmatrix} x_2 \\ s_3 \\ x_1 \end{bmatrix} \quad \mathbf{x}_{\mathbf{N}} = \begin{bmatrix} s_2 \\ s_1 \end{bmatrix} \quad \mathbf{c}_{\mathbf{B}} = \begin{bmatrix} 6 \\ 0 \\ 7 \end{bmatrix} \quad \mathbf{c}_{\mathbf{N}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

and the matrices become:

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \quad \mathbf{N} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}$$

The derived matrices are then:

$$\mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 72 \\ 19 \\ 16 \end{bmatrix} \quad \mathbf{B}^{-1}\mathbf{N} = \begin{bmatrix} 6/10 & -1/5 \\ 1/5 & -2/5 \\ -1/5 & 2/5 \end{bmatrix}$$

The cost information becomes:

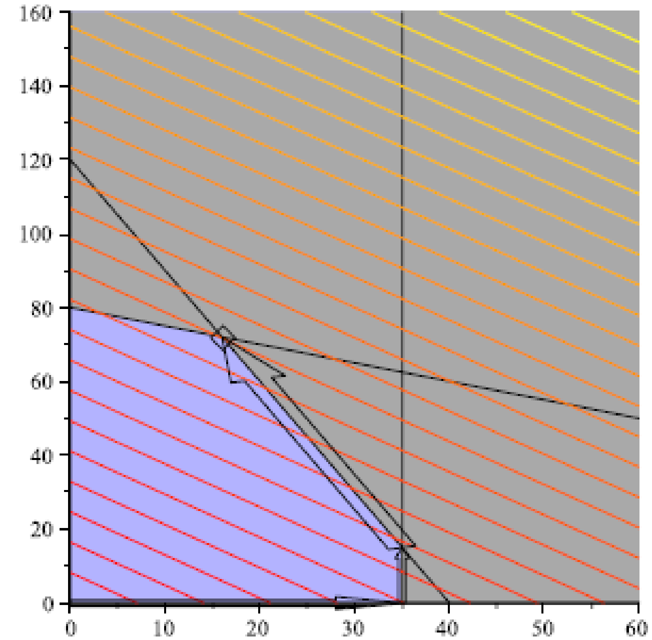


Figure 5.1.

$$\mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{b} = 544 \quad \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{N} = \begin{bmatrix} 11/5 & 8/5 \end{bmatrix} \quad \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{N} - \mathbf{c}_{\mathbf{N}} = \begin{bmatrix} 11/5 & 8/5 \end{bmatrix}$$

Since the reduced costs are now positive, we can conclude that we've obtained an optimal solution because no improvement is possible. The final solution then is:

$$\mathbf{x}_{\mathbf{B}}^* = \begin{bmatrix} x_2 \\ s_3 \\ x_1 \end{bmatrix} = \mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 72 \\ 19 \\ 16 \end{bmatrix}$$

Simply, we have $x_1 = 16$ and $x_2 = 72$ as we obtained in Example 2.3. The path of extreme points we actually took in traversing the boundary of the polyhedral feasible region is shown in Figure 5.1.

4. Simplex Method–Tableau Form

No one executes the simplex algorithm in algebraic form. Instead, several representations (tableau representations) have been developed to lesson the amount of writing that needs to be done and to collect all pertinent information into a single table.

To see how a *Simplex Tableau* is derived, consider Problem P in standard form:

$$P\begin{cases} \max & \mathbf{c}^T\mathbf{x} \\ s.t. & \mathbf{A}\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{cases}$$

We can re-write P in an unusual way by introducing a new variable z and separating $\mathbf{A}$ into its basic and non-basic parts to obtain:

$$(5.22)\qquad \begin{aligned} \max\ & z \\ s.t.\ & z - \mathbf{c}_{\mathbf{B}}^T\mathbf{x}_{\mathbf{B}} - \mathbf{c}_{\mathbf{N}}^T\mathbf{x}_{\mathbf{N}} = 0 \\ & \mathbf{B}\mathbf{x}_{\mathbf{B}} + \mathbf{N}\mathbf{x}_{\mathbf{N}} = \mathbf{b} \\ & \mathbf{x}_{\mathbf{B}}, \mathbf{x}_{\mathbf{N}} \geq 0 \end{aligned}$$

From the second equation, it's clear

$$(5.23)\qquad \mathbf{x}_{\mathbf{B}} + \mathbf{B}^{-1}\mathbf{N}\mathbf{x}_{\mathbf{N}} = \mathbf{B}^{-1}\mathbf{b}$$

We can multiply this equation by $\mathbf{c}_{\mathbf{B}}^{\mathbf{T}}$ to obtain:

$$(5.24)\qquad \mathbf{c}_{\mathbf{B}}^T\mathbf{x}_{\mathbf{B}} + \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{N}\mathbf{x}_{\mathbf{N}} = \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{b}$$

If we add this equation to the equation $z - \mathbf{c}_{\mathbf{B}}^T\mathbf{x}_{\mathbf{B}} - \mathbf{c}_{\mathbf{N}}^T\mathbf{x}_{\mathbf{N}} = 0$ we obtain:

$$(5.25)\qquad z + \mathbf{0}^T\mathbf{x}_{\mathbf{B}} + \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{N}\mathbf{x}_{\mathbf{N}} - \mathbf{c}_{\mathbf{N}}^T\mathbf{x}_{\mathbf{N}} = \mathbf{c}_{\mathbf{B}}^T\mathbf{B}^{-1}\mathbf{b}$$

Here $\mathbf{0}$ is the vector of zeros of appropriate size. This equation can be written as:

$$(5.26) \quad z + \mathbf{0}^T\mathbf{x_B} + \left(\mathbf{c}_\mathbf{B}^T\mathbf{B}^{-1}\mathbf{N} - \mathbf{c}_\mathbf{N}^T\right)\mathbf{x_N} = \mathbf{c}_\mathbf{B}^T\mathbf{B}^{-1}\mathbf{b}$$

We can now represent this set of equations as a large matrix (or tableau):

	z	$\mathbf{x_B}$	$\mathbf{x_N}$	RHS	
z	1	0	$\mathbf{c}_\mathbf{B}^T\mathbf{B}^{-1}\mathbf{N} - \mathbf{c}_\mathbf{N}^T$	$\mathbf{c}_\mathbf{B}^T\mathbf{B}^{-1}\mathbf{b}$	Row 0
$\mathbf{x_B}$	0	1	$\mathbf{B}^{-1}\mathbf{N}$	$\mathbf{B}^{-1}\mathbf{b}$	Rows 1 through m

The augmented matrix shown within the table:

$$(5.27) \quad \left[\begin{array}{ccc|c} 1 & 0 & \mathbf{c}_\mathbf{B}^T\mathbf{B}^{-1}\mathbf{N} - \mathbf{c}_\mathbf{N}^T & \mathbf{c}_\mathbf{B}^T\mathbf{B}^{-1}\mathbf{b} \\ 0 & 1 & \mathbf{B}^{-1}\mathbf{N} & \mathbf{B}^{-1}\mathbf{b} \end{array}\right]$$

is simply the matrix representation of the simultaneous equations described by Equations 5.23 and 5.26. We can see that the first row consists of a row of the first row of the $(m+1)\times(m+1)$ identity matrix, the reduced costs of the non-basic variables and the current objective function values. The remainder of the rows consist of the rest of the $(m+1)\times(m+1)$ identity matrix, the matrix $\mathbf{B}^{-1}\mathbf{N}$ and $\mathbf{B}^{-1}\mathbf{b}$ the current non-zero part of the basic feasible solution.

This matrix representation (or tableau representation) contains all of the information we need to execute the simplex algorithm. An entering variable is chosen from among the columns containing the reduced costs and matrix $\mathbf{B}^{-1}\mathbf{N}$. Naturally, a column with a negative reduced cost is chosen. We then chose a leaving variable by performing the minimum ratio test on the chosen column and the right-hand-side (RHS) column. We pivot on the element at the entering column and leaving row and this transforms the tableau into a new tableau that represents the new basic feasible solution.

Theorem: The set of all feasible solutions to an LP problem is a convex set.

Proof:

Theorem: An LP problem assumes its optimum at an extreme point. If it assumes its optimum at more than one extreme point, then it takes on the same value for every convex combination of these particular points.

Proof:

Appendix

1. Convex Sets

DEFINITION 4.1 (Convex Set). Let $X \subseteq \mathbb{R}^n$. Then the set X is convex if and only if for all pairs $\mathbf{x}_1, \mathbf{x}_2 \in X$ we have $\lambda\mathbf{x}_1 + (1-\lambda)\mathbf{x}_2 \in X$ for all $\lambda \in [0,1]$.

The definition of convexity seems complex, but it is easy to understand. First recall that if $\lambda \in [0,1]$, then the point $\lambda\mathbf{x}_1 + (1-\lambda)\mathbf{x}_2$ is on the line segment connecting $\mathbf{x}_1$ and $\mathbf{x}_2$ in $\mathbb{R}^n$. For example, when $\lambda = 1/2$, then the point $\lambda\mathbf{x}_1 + (1-\lambda)\mathbf{x}_2$ is the midpoint between $\mathbf{x}_1$ and $\mathbf{x}_2$. In fact, for every point $\mathbf{x}$ on the line connecting $\mathbf{x}_1$ and $\mathbf{x}_2$ we can find a value $\lambda \in [0,1]$ so that $\mathbf{x} = \lambda\mathbf{x}_1 + (1-\lambda)\mathbf{x}_2$. Then we can see that, convexity asserts that if $\mathbf{x}_1, \mathbf{x}_2 \in X$, then every point on the line connecting $\mathbf{x}_1$ and $\mathbf{x}_2$ is also in the set X.

6. Extreme Points

DEFINITION 4.27 (Extreme Point of a Convex Set). Let C be a convex set. A point $\mathbf{x}_0 \in C$ is a *extreme point* of C if there are *no points* $\mathbf{x}_1$ and $\mathbf{x}_2$ ($\mathbf{x}_1 \neq \mathbf{x}_0$ or $\mathbf{x}_2 \neq \mathbf{x}_0$) so that $\mathbf{x} = \lambda\mathbf{x}_1 + (1-\lambda)\mathbf{x}_2$ for some $\lambda \in (0,1)$.[2]

An extreme point is simply a point in a convex set C that cannot be expressed as a strict convex combination of any other pair of points in C. We will see that extreme points must be located in specific locations in convex sets.

Definition 4.3 (Convex Combination). Let $\mathbf{x}_1, \dots, \mathbf{x}_m \in \mathbb{R}^n$. If $\lambda_1, \dots, \lambda_m \in [0, 1]$ and

$$\sum_{i=1}^{m} \lambda_i = 1 \quad \text{then}$$

$$\mathbf{x} = \sum_{i=1}^{m} \lambda_i \mathbf{x}_i \tag{4.2}$$

is called a *convex combination* of $\mathbf{x}_1, \dots, \mathbf{x}_m$. If $\lambda_i < 1$ for all $i = 1, \dots, m$, then Equation 4.2 is called a *strict convex combination*.

7. Linear Combinations, Span, Linear Independence

Definition 3.27. Let $\mathbf{x}_1, \dots, \mathbf{x}_m$ be vectors in $\in \mathbb{R}^n$ and let $\alpha_1, \dots, \alpha_m \in \mathbb{R}$ be scalars. Then

$$\alpha_1\mathbf{x}_1 + \cdots + \alpha_m\mathbf{x}_m \tag{3.34}$$

is a *linear combination* of the vectors $\mathbf{x}_1, \dots, \mathbf{x}_m$.

Clearly, any linear combination of vectors in $\mathbb{R}^n$ is also a vector in $\mathbb{R}^n$.

Definition 3.28 (Span). Let $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ be a set of vectors in $\in \mathbb{R}^n$, then the span of $\mathcal{X}$ is the set:

$$\text{span}(\mathcal{X}) = \{\mathbf{y} \in \mathbb{R}^n | \mathbf{y} \text{ is a linear combination of vectors in } \mathcal{X}\} \tag{3.35}$$

Definition 3.29 (Linear Independence). Let $\mathbf{x}_1, \dots, \mathbf{x}_m$ be vectors in $\in \mathbb{R}^n$. The vectors $\mathbf{x}_1, \dots, \mathbf{x}_m$ are *linearly dependent* if there exists $\alpha_1, \dots, \alpha_m \in \mathbb{R}$, not all zero, such that

$$\alpha_1\mathbf{x}_1 + \cdots + \alpha_m\mathbf{x}_m = \mathbf{0} \tag{3.36}$$

If the set of vectors $\mathbf{x}_1, \dots, \mathbf{x}_m$ is not linearly dependent, then they are *linearly independent* and Equation 3.36 holds just in case $\alpha_i = 0$ for all $i = 1, \dots, n$.

8. Basis

Definition 3.35 (Basis). Let $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ be a set of vectors in $\mathbb{R}^n$. The set $\mathcal{X}$ is called a *basis* of $\mathbb{R}^n$ if $\mathcal{X}$ is a linearly independent set of vectors and every vector in $\mathbb{R}^n$ is in the span of $\mathcal{X}$. That is, for any vector $\mathbf{w} \in \mathbb{R}^n$ we can find scalar values $\alpha_1, \dots, \alpha_m$ such that

$$\mathbf{w} = \sum_{i=1}^{m} \alpha_i \mathbf{x}_i \tag{3.37}$$

Theorem 3.37. *If $\mathcal{X}$ is a basis of $\mathbb{R}^n$, then $\mathcal{X}$ contains precisely n vectors.*

Lemma 3.38. *Let $\{\mathbf{x}_1, \dots, \mathbf{x}_{m+1}\}$ be a linearly dependent set of vectors in $\mathbb{R}^n$ and let $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ be a linearly independent set. Further assume that $\mathbf{x}_{m+1} \neq \mathbf{0}$. Assume $\alpha_1, \dots, \alpha_{m+1}$ are a set of scalars, not all zero, so that*

$$\sum_{i=1}^{m+1} \alpha_i \mathbf{x}_i = \mathbf{0} \tag{3.41}$$

For any $j \in \{1, \dots, m\}$ such that $\alpha_j \neq 0$, if we replace $\mathbf{x}_j$ in the set $\mathcal{X}$ with $\mathbf{x}_{m+1}$, then this new set of vectors is linearly independent.

Remark 3.39. This lemma proves an interesting result. If $\mathcal{X}$ is a basis of $\mathbb{R}^m$ and $\mathbf{x}_{m+1}$ is another, non-zero, vector in $\mathbb{R}^m$, we can swap $\mathbf{x}_{m+1}$ for any vector $\mathbf{x}_j$ in $\mathcal{X}$ as long as when we express $\mathbf{x}_{m+1}$ as a linear combination of vectors in $\mathcal{X}$ the coefficient of $\mathbf{x}_j$ is not zero. That is, since $\mathcal{X}$ is a basis of $\mathbb{R}^m$ we can express:

$$\mathbf{x}_{m+1} = \sum_{i=1}^{m} \alpha_i \mathbf{x}_i$$

As long as $\alpha_j \neq 0$, then we can replace $\mathbf{x}_j$ with $\mathbf{x}_{m+1}$ and still have a basis of $\mathbb{R}^m$.

9. Rank

DEFINITION 3.40 (Row Rank). Let $\mathbf{A} \in \mathbb{R}^{m\times n}$. The *row rank* of $\mathbf{A}$ is the size of the largest set of row (vectors) from $\mathbf{A}$ that are linearly independent.

EXERCISE 36. By analogy define the *column rank* of a matrix. [Hint: You don't need a hint.]

THEOREM 3.41. *If $\mathbf{A} \in \mathbb{R}^{m\times n}$ is a matrix, then elementary row operations on A do not change the row rank.*

THEOREM 3.42. *If $\mathbf{A} \in \mathbb{R}^{m\times n}$ is a matrix, then the row rank of $\mathbf{A}$ is equal to the column rank of $\mathbf{A}$. Further,* $\operatorname{rank}(\mathbf{A}) \leq \min\{m, n\}$.

THEOREM 3.43. *If $\mathbf{A} \in \mathbb{R}^{m\times m}$ (i.e., $\mathbf{A}$ is a square matrix) and* $\operatorname{rank}(\mathbf{A}) = m$, *then $\mathbf{A}$ is invertible.*

DEFINITION 3.44. Suppose that $\mathbf{A} \in \mathbb{R}^{m\times n}$ and let $m \leq n$. Then $\mathbf{A}$ has *full row rank* if $\operatorname{rank}(\mathbf{A}) = m$.

Reference: Kevin G Ross, ISM206 Optimization theory and applications, Lecture notes, *https://courses.soe.ucsc.edu/courses/ism206, Date accessed: 16.07.2017*

Christopher Grin, Linear Programming: Penn State Math 484 Lecture Notes, Version 1.8.2.1 *http://www.personal.psu.edu/cxg286/Math484_V1.pdf , Date accessed: 16.07.2017*

Lecture 8
Sensitivity Analysis

Graphical Sensitivity Analysis
Algebraic Sensitivity Analysis

3.6 SENSITIVITY ANALYSIS

In LP, the parameters (input data) of the model can change within certain limits without causing the optimum solution to change. This is referred to as *sensitivity analysis*, and will be the subject matter of this section. Later, in Chapter 4, we will study *post-optimal analysis* which deals with determining the new optimum solution resulting from making targeted changes in the input data.

3.6.1 Graphical Sensitivity Analysis

Example 3.6-1 (Changes in the Right-Hand Side)

JOBCO produces two products on two machines. A unit of product 1 requires 2 hours on machine 1 and 1 hour on machine 2. For product 2, a unit requires 1 hour on machine 1 and 3 hours on machine 2. The revenues per unit of products 1 and 2 are \$30 and \$20, respectively. The total daily processing time available for each machine is 8 hours.

Letting x_1 and x_2 represent the daily number of units of products 1 and 2, respectively, the LP model is given as

$$\text{Maximize } z = 30x_1 + 20x_2$$

subject to

$$\begin{aligned} 2x_1 + x_2 &\leq 8 \quad \text{(Machine 1)} \\ x_1 + 3x_2 &\leq 8 \quad \text{(Machine 2)} \\ x_1, x_2 &\geq 0 \end{aligned}$$

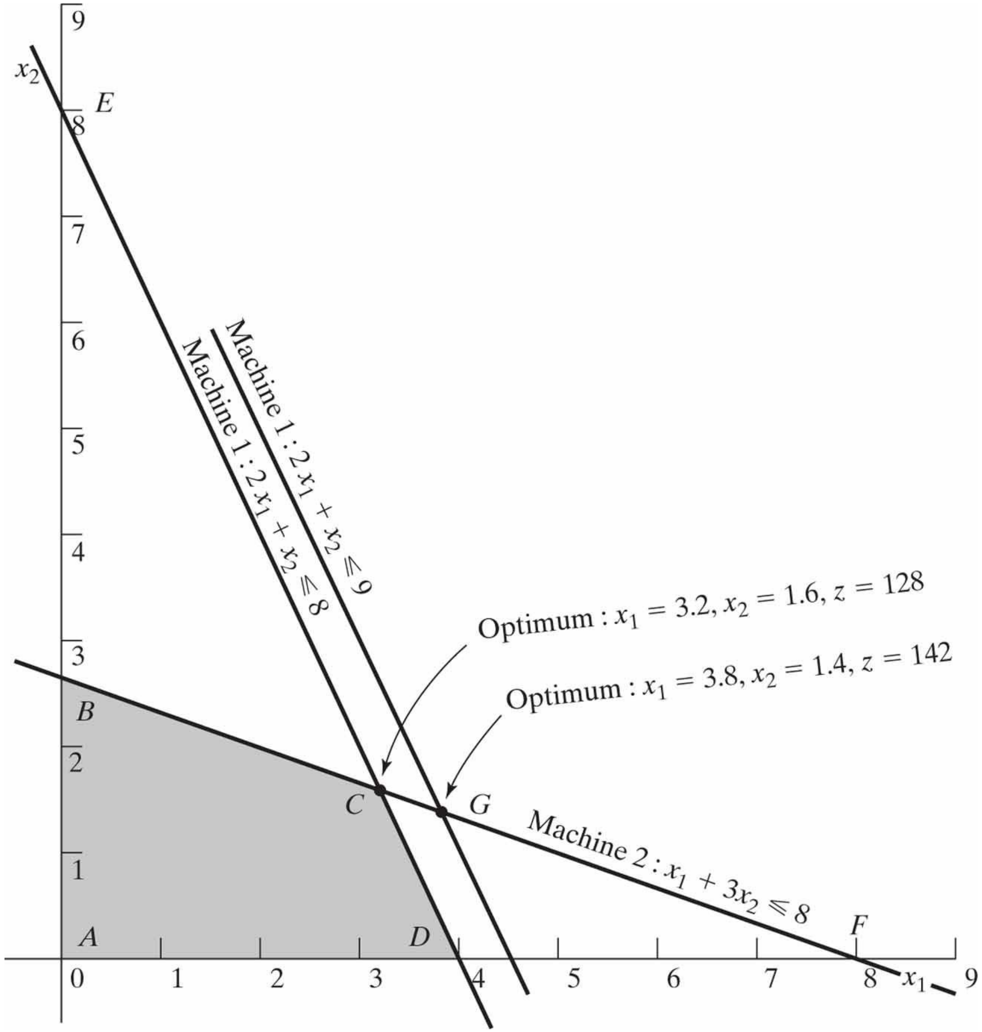


Figure 3.12 Graphical sensitivity of optimal solution to changes in the availability of resources (right-hand side of the constraints)

$$\left(\begin{array}{c} \text{Rate of revenue change} \\ \text{resulting from increasing} \\ \text{machine 1 capacity by 1 hr} \\ \text{(point } C \text{ to point } G\text{)} \end{array} \right) =$$

The computed rate provides a *direct link* between the model input (resources) and its output (total revenue) that represents the **unit worth of a resource** (in \$/hr)—that is, the change in the optimal objective value per unit change in the availability of the resource (machine capacity). This means that a unit increase (decrease) in machine 1 capacity will increase (decrease) revenue by \$14.00. Although *unit worth of a resource* is an apt description of the rate of change of the objective function, the technical name **dual** or **shadow price** is now standard in the LP literature and all software packages and, hence, will be used throughout the book.

$\boxed{\text{the range of applicability of the given dual price}}$

Minimum machine 1 capacity [at $B = (0, 2.67)$] =

Maximum machine 1 capacity [at $F = (8, 0)$] =

We can thus conclude that the dual price of \$14.00/hr will remain valid for the range

$$2.67 \text{ hrs} \leq \text{Machine 1 capacity} \leq 16 \text{ hrs}$$

Changes outside this range will produce a different dual price (worth per unit).

Using similar computations, you can verify that the dual price for machine 2 capacity is \$2.00/hr and it remains valid for changes (increases or decreases) that move its constraint parallel to itself to any point on the line segment DE, which yields the following limits:

$$\text{Minimum machine 2 capacity [at } D = (4, 0)] = 1 \times 4 + 3 \times 0 = 4 \text{ hr}$$

$$\text{Maximum machine 2 capacity [at } E = (8, 0)] = 1 \times 0 + 3 \times 8 = 24 \text{ hr}$$

The conclusion is that the dual price of \$2.00/hr for machine 2 will remain applicable for the range

$$4 \text{ hr} \leq \text{Machine 2 capacity} \leq 24 \text{ hr}$$

The computed limits for machine 1 and 2 are referred to as the **feasibility ranges.**

The dual prices allow making economic decisions about the LP problem, as the following questions demonstrate:

Question 1. If JOBCO can increase the capacity of both machines, which machine should receive higher priority?

The dual prices for machines 1 and 2 are \$14.00/hr and \$2.00/hr. This means that each additional hour of machine 1 will increase revenue by \$14.00, as opposed to only \$2.00 for machine 2. Thus, priority should be given to machine 1.

Question 2. A suggestion is made to increase the capacities of machines 1 and 2 at the additional cost of \$10/hr. Is this advisable?

For machine 1, the additional net revenue per hour is $14.00 - 10.00 = \$4.00$ and for machine 2, the net is $\$2.00 - \$10.00 = -\$8.00$. Hence, only the capacity of machine 1 should be increased.

Question 3. If the capacity of machine 1 is increased from the present 8 hours to 13 hours, how will this increase impact the optimum revenue?

The dual price for machine 1 is \$14.00 and is applicable in the range (2.67, 16) hr. The proposed increase to 13 hours falls within the feasibility range. Hence, the increase in revenue is $\$14.00(13 - 8) = \$70.00$, which means that the total revenue will be increased to (current revenue + change in revenue) $= 128 + 70 = \$198.00$.

Question 4. Suppose that the capacity of machine 1 is increased to 20 hours, how will this increase impact the optimum revenue?

The proposed change is outside the range (2.67, 16) hr for which the dual price of \$14.00 remains applicable. Thus, we can only make an immediate conclusion regarding an increase up to 16 hours. Beyond that, further calculations are needed to find the answer (see Chapter 4). Remember that falling outside the feasibility range does *not* mean that the problem has no solution. It only means that we do not have sufficient information to make an *immediate* decision.

Question 5. We know that the change in the optimum objective value equals (dual price × change in resource) so long as the change in the resource is within the feasibility range. What about the associated optimum values of the variables?

The optimum values of the variables will definitely change. However, the level of information we have from the graphical solution is not sufficient to determine the new values. Section 3.6.2, which treats the sensitivity problem algebraically, provides this detail.

PROBLEM SET 3.6A

1. A company produces two products, A and B. The unit revenues are \$2 and \$3, respectively. Two raw materials, $M1$ and $M2$, used in the manufacture of the two products have respective daily availabilities of 8 and 18 units. One unit of A uses 2 units of $M1$ and 2 units of $M2$, and 1 unit of B uses 3 units of $M1$ and 6 units of $M2$.
   - **(a)** Determine the dual prices of $M1$ and $M2$ and their feasibility ranges.
   - **(b)** Suppose that 4 additional units of $M1$ can be acquired at the cost of 30 cents per unit. Would you recommend the additional purchase?
   - **(c)** What is the most the company should pay per unit of $M2$?
   - **(d)** If $M2$ availability is increased by 5 units, determine the associated optimum revenue.

Example 3.6-2 (Changes in the Objective Coefficients)

Figure 3.13 shows the graphical solution space of the JOBCO problem presented in Example 3.6-1. The optimum occurs at point C ($x_1 = 3.2$, $x_2 = 1.6$, $z = 128$). Changes in revenue units (i.e., objective-function coefficients) will change the slope of z. However, as can be seen from the figure, the optimum solution will remain at point C so long as the objective function lies between lines BF and DE, the two constraints that define the optimum point. This means that there is a range for the coefficients of the objective function that will keep the optimum solution unchanged at C.

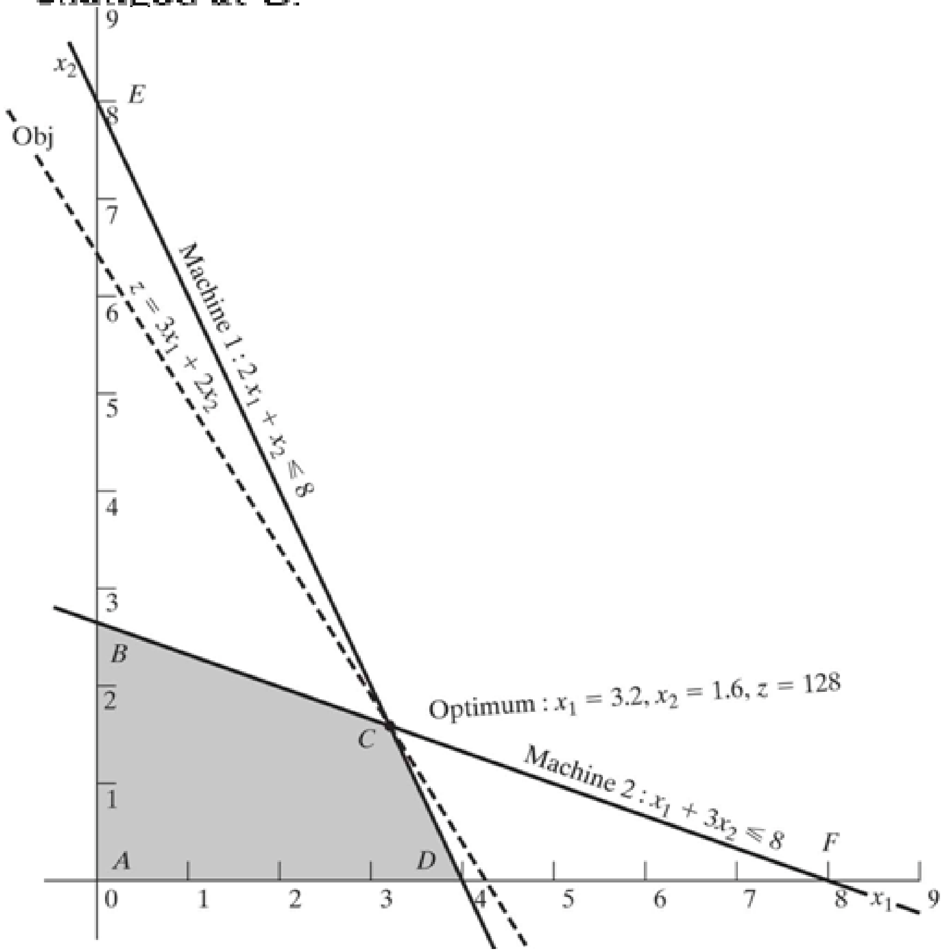


Figure 3.13 Graphical sensitivity of optimal solution to changes in the revenue units (coefficients of the objective function)

We can write the objective function in the general format

$$\text{Maximize } z = c_1x_1 + c_2x_2$$

Imagine now that the line z is pivoted at C and that it can rotate clockwise and counterclockwise. The optimum solution will remain at point C so long as $z = c_1x_1 + c_2x_2$ lies between the two lines $x_1 + 3x_2 = 8$ and $2x_1 + x_2 = 8$. This means that the ratio $\frac{c_1}{c_2}$ can vary between $\frac{1}{3}$ and $\frac{2}{1}$, which yields the following condition:

This information can provide immediate answers regarding the optimum solution as the following questions demonstrate:

Question 1. Suppose that the unit revenues for products 1 and 2 are changed to \$35 and \$25, respectively. Will the current optimum remain the same?

The new objective function is

$$\text{Maximize } z = 35x_1 + 25x_2$$

The solution at C will remain optimal because $\frac{c_1}{c_2} = \frac{35}{25} = 1.4$ remains within the optimality range (.333, 2). When the ratio falls outside this range, additional calculations are needed to find the new optimum (see Chapter 4). Notice that although the values of the variables at the optimum point C remain unchanged, the optimum value of z changes to $35 \times (3.2) + 25 \times (1.6) = \$152.00$.

Question 2. Suppose that the unit revenue of product 2 is fixed at its current value of $c_2 = \$20.00$. What is the associated range for c_1, the unit revenue for product 1 that will keep the optimum unchanged?

Substituting $c_2 = 20$ in the condition $\frac{1}{3} \leq \frac{c_1}{c_2} \leq 2$, we get

Or

This range is referred to as the **optimality range** for c_1, and it implicitly assumes that c_2 is fixed at \$20.00.

We can similarly determine the *optimality range* for c_2 by fixing the value of c_1 at \$30.00. Thus,

$$c_2 \leq 30 \times 3 \text{ and } c_2 \geq \frac{30}{2}$$

Or

$$15 \leq c_2 \leq 90$$

PROBLEM SET 3.6B

1. Consider Problem 1, Set 3.6a.
   - (a) Determine the optimality condition for $\frac{c_A}{c_B}$ that will keep the optimum unchanged.
   - (b) Determine the optimality ranges for c_A and c_B, assuming that the other coefficient is kept constant at its present value.
   - (c) If the unit revenues c_A and c_B are changed simultaneously to \$5 and \$4, respectively, determine the new optimum solution.
   - (d) If the changes in (c) are made one at a time, what can be said about the optimum solution?

3.6.2 Algebraic Sensitivity Analysis—Changes in the Right-Hand Side

Example 3.6-2 (TOYCO Model)

TOYCO assembles three types of toys—trains, trucks, and cars—using three operations. The daily limits on the available times for the three operations are 430, 460, and 420 minutes, respectively, and the revenues per unit of toy train, truck, and car are \$3, \$2, and \$5, respectively. The assembly times per train at the three operations are 1, 3, and 1 minutes, respectively. The corresponding times per train and per car are (2,0,4) and (1,2,0) minutes (a zero time indicates that the operation is not used).

Letting x_1, x_2, and x_3 represent the daily number of units assembled of trains, trucks, and cars, respectively, the associated LP model is given as:

$$\text{Maximize } z = 3x_1 + 2x_2 + 5x_3$$

subject to

$$x_1 + 2x_2 + x_3 \leq 430 \text{ (Operation 1)}$$
$$3x_1 + 2x_3 \leq 460 \text{ (Operation 2)}$$
$$x_1 + 4x_2 \leq 420 \text{ (Operation 3)}$$
$$x_1, x_2, x_3 \geq 0$$

Using x_4, x_5, and x_6 as the slack variables for the constraints of operations 1, 2, and 3, respectively, the optimum tableau is

Basic	x_1	x_2	x_3	x_4	x_5	x_6	Solution
z	4	0	0	1	2	0	1350
x_2	$-\frac{1}{4}$	1	0	$\frac{1}{2}$	$-\frac{1}{4}$	0	100
x_3	$\frac{3}{2}$	0	1	0	$\frac{1}{2}$	0	230
x_6	2	0	0	-2	1	1	20

Determination of Dual Prices. The constraints of the model after adding the slack variables x_4, x_5, and x_6 can be written as follows:

$$\begin{aligned} x_1 + 2x_2 + x_3 + x_4 &= 430 \quad \text{(Operation 1)} \\ 3x_1 + 2x_3 + x_5 &= 460 \quad \text{(Operation 2)} \\ x_1 + 4x_2 + x_6 &= 420 \quad \text{(Operation 3)} \end{aligned}$$

or

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 430 - x_4 \quad \text{(Operation 1)} \\ 3x_1 + 2x_3 &= 460 - x_5 \quad \text{(Operation 2)} \\ x_1 + 4x_2 &= 420 - x_6 \quad \text{(Operation 3)} \end{aligned}$$

With this representation, the slack variables have the same units (minutes) as the operation times. Thus, we can say that a one-minute *decrease* in the slack variable is equivalent to a one-minute *increase* in the operation time.

We can use the information above to determine the *dual prices* from the z-equation in the optimal tableau:

$$z + 4x_1 + x_4 + 2x_5 + 0x_6 = 1350$$

This equation can be written as

Given that a *decrease* in the value of a slack variable is equivalent to an *increase* in its operation time, we get

This equation reveals that (1) a one-minute increase in operation 1 time increases z by \$1, (2) a one-minute increase in operation 2 time increases z by \$2, and (3) a one-minute increase in operation 3 time does not change z.

To summarize, the z-row in the optimal tableau:

Basic	x_1	x_2	x_3	x_4	x_5	x_6	Solution
z	4	0	0	1	2	0	1350

yields directly the dual prices, as the following table shows:

Resource	Slack variable	Optimal z-equation coefficient of slack variable	Dual price
Operation 1	x_4	1	\$1/min
Operation 2	x_5	2	\$2/min
Operation 3	x_6	0	\$0/min

The zero dual price for operation 3 means that there is no economic advantage in allocating more production time to this operation. The result makes sense because the resource is already abundant, as is evident by the fact that the slack variable associated with Operation 3 is positive (= 20) in the optimum solution. As for each of Operations 1 and 2, a one minute increase will improve revenue by \$1 and \$2, respectively. The dual prices also indicate that, when allocating additional resources, Operation 2 may be given higher priority because its dual price is twice as much as that of Operation 1.

Determination of the Feasibility Ranges. Having determined the dual prices, we show next how the *feasibility ranges* in which they remain valid are determined. Let D_1, D_2, and D_3 be the changes (positive or negative) in the daily manufacturing time allocated to operations 1, 2, and 3, respectively. The model can be written as follows:

$$\text{Maximize } z = 3x_1 + 2x_2 + 5x_3$$

subject to

$$\begin{aligned}
x_1 + 2x_2 + x_3 &\le 430 + D_1 \quad \text{(Operation 1)} \\
3x_1 + 2x_3 &\le 460 + D_2 \quad \text{(Operation 2)} \\
x_1 + 4x_2 &\le 420 + D_3 \quad \text{(Operation 3)} \\
x_1, x_2, x_3 &\ge 0
\end{aligned}$$

The procedure is based on recomputing the optimum simplex tableau with the modified right-hand side and then deriving the conditions that will keep the solution feasible—that is, the right-hand side of the optimum tableau remains nonnegative. To show how the right-hand side is recomputed, we start by modifying the *Solution* column of the starting tableau using the new right-hand sides: $430 + D_1$, $460 + D_2$, and $420 + D_3$. The starting tableau will thus appear as

							Solution			
Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	D_1	D_2	D_3
z	−3	−2	−5	0	0	0	0			
x_4	1	2	1	1	0	0	430			
x_5	3	0	2	0	1	0	460			
x_6	1	4	0	0	0	1	420			

							Solution			
Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	D_1	D_2	D_3
z	4	0	0	1	2	0	1350			
x_2	$-\frac{1}{4}$	1	0	$\frac{1}{2}$	$-\frac{1}{4}$	0	100			
x_3	$\frac{3}{2}$	0	1	0	$\frac{1}{2}$	0	230			
x_6	2	0	0	-2	1	1	20			

The new optimum tableau provides the following optimal solution:

Interestingly, as shown earlier, the new z-value confirms that the dual prices for operations 1, 2, and 3 are 1, 2, and 0, respectively.

The current solution remains feasible so long as all the variables are nonnegative, which leads to the following **feasibility conditions**:

$$x_2 = 100 + \tfrac{1}{2}D_1 - \tfrac{1}{4}D_2 \geq 0$$
$$x_3 = 230 + \tfrac{1}{2}D_2 \geq 0$$
$$x_6 = 20 - 2D_1 + D_2 + D_3 \geq 0$$

Any simultaneous changes D_1, D_2, and D_3 that satisfy these inequalities will keep the solution feasible. If all the conditions are satisfied, then the new optimum solution can be found through direct substitution of D_1, D_2, and D_3 in the equations given above.

To illustrate the use of these conditions, suppose that the manufacturing time available for operations 1, 2, and 3 are 480, 440, and 410 minutes respectively. Then, $D_1 = 480 - 430 = 50$, $D_2 = 440 - 460 = -20$, and $D_3 = 410 - 420 = -10$. Substituting in the feasibility conditions, we get

$$x_2 = 100 + \tfrac{1}{2}(50) - \tfrac{1}{4}(-20) = 130 > 0 \quad \text{(feasible)}$$
$$x_3 = 230 + \tfrac{1}{2}(-20) = 220 > 0 \quad \text{(feasible)}$$
$$x_6 = 20 - 2(50) + (-20) + (-10) = -110 < 0 \quad \text{(infeasible)}$$

The calculations show that $x_6 < 0$, hence the current solution does not remain feasible. Additional calculations will be needed to find the new solution. These calculations are discussed in Chapter 4 as part of the post-optimal analysis.

Alternatively, if the changes in the resources are such that $D_1 = -30$, $D_2 = -12$, and $D_3 = 10$, then

$$\begin{aligned}
x_2 &= 100 + \tfrac{1}{2}(-30) - \tfrac{1}{4}(-12) = 88 > 0 && \text{(feasible)} \\
x_3 &= 230 + \tfrac{1}{2}(-12) = 224 > 0 && \text{(feasible)} \\
x_6 &= 20 - 2(-30) + (-12) + (10) = 78 > 0 && \text{(feasible)}
\end{aligned}$$

The new feasible solution is $x_1 = 88$, $x_3 = 224$, and $x_6 = 68$ with $z = 3(0) + 2(88) + 5(224) = \$1296$. Notice that the optimum objective value can also be computed as $z = 1350 + 1(-30) + 2(-12) = \$1296$.

The given conditions can be specialized to produce the individual *feasibility ranges* that result from changing the resources *one at a time* (as defined in Section 3.6.1).

Case 1. Change in operation 1 time from 460 *to* 460 + D_1 *minutes.* This change is equivalent to setting $D_2 = D_3 = 0$ in the simultaneous conditions, which yields

Case 2. Change in operation 2 time from 430 *to* 430 + D_2 *minutes.* This change is equivalent to setting $D_1 = D_3 = 0$ in the simultaneous conditions, which yields

$$\left.\begin{array}{l} x_2 = 100 - \frac{1}{4}D_2 \geq 0 \Rightarrow D_2 \leq 400 \\ x_3 = 230 + \frac{1}{2}D_2 \geq 0 \Rightarrow D_2 \geq -460 \\ x_6 = 20 + D_2 \geq 0 \quad \Rightarrow D_2 \geq -20 \end{array}\right\} \Rightarrow -20 \leq D_2 \leq 400$$

Case 3. Change in operation 3 time from 420 *to* 420 + D_3 *minutes.* This change is equivalent to setting $D_1 = D_2 = 0$ in the simultaneous conditions, which yields

$$\left.\begin{array}{l} x_2 = 100 > 0 \\ x_3 = 230 > 0 \\ x_6 = 20 + D_3 \geq 0 \end{array}\right\} \Rightarrow -20 \leq D_3 < \infty$$

We can now summarize the dual prices and their feasibility ranges for the TOYCO model as follows:[3]

			Resource amount (minutes)		
Resource	Dual price	Feasibility range	*Minimum*	*Current*	*Maximum*
Operation 1	1	$-200 \le D_1 \le 10$	230	430	440
Operation 2	2	$-20 \le D_2 \le 400$	440	440	860
Operation 3	0	$-20 \le D_3 < \infty$	400	420	∞

It is important to notice that the dual prices will remain applicable for any *simultaneous* changes that keep the solution feasible, even if the changes violate the individual ranges. For example, the changes $D_1 = 30$, $D_2 = -12$, and $D_3 = 100$, will keep the solution feasible even though $D_1 = 30$ violates the feasibility range $-200 \le D_1 \le 10$, as the following computations show:

$$x_2 = 100 + \tfrac{1}{2}(30) - \tfrac{1}{4}(-12) = 118 > 0 \qquad \text{(feasible)}$$

$$x_3 = 230 + \tfrac{1}{2}(-12) = 224 > 0 \qquad \text{(feasible)}$$

$$x_6 = 20 - 2(30) + (-12) + (100) = 48 > 0 \qquad \text{(feasible)}$$

This means that the dual prices will remain applicable, and we can compute the new optimum objective value from the dual prices as $z = 1350 + 1(30) + 2(-12) + 0(100) = \$1356$

The results above can be summarized as follows:

1. The dual prices remain valid so long as the changes D_i, $i = 1, 2, \ldots, m$, in the right-hand sides of the constraints satisfy all the feasibility conditions when the changes are simultaneous or fall within the feasibility ranges when the changes are made individually.

2. For other situations where the dual prices are not valid because the simultaneous feasibility conditions are not satisfied or because the individual feasibility ranges are violated, the recourse is to either re-solve the problem with the new values of D_i or apply the post-optimal analysis presented in Chapter 4.

PROBLEM SET 3.6C[4]

1. In the TOYCO model, suppose that the changes D_1, D_2, and D_3 are made *simultaneously* in the three operations.
   (a) If the availabilities of operations 1, 2, and 3 are changed to 438, 500, and 410 minutes, respectively, use the simultaneous conditions to show that the current basic solution remains feasible, and determine the change in the optimal revenue by using the optimal dual prices.
   (b) If the availabilities of the three operations are changed to 460, 440, and 380 minutes, respectively, use the simultaneous conditions to show that the current basic solution becomes infeasible.

3. A company produces three products, A, B, and C. The sales volume for A is at least 50% of the total sales of all three products. However, the company cannot sell more than 75 units of A per day. The three products use one raw material, of which the maximum daily availability is 240 lb. The usage rates of the raw material are 2 lb per unit of A, 4 lb per unit of B, and 3 lb per unit of C. The unit prices for A, B, and C are \$20, \$50, and \$35, respectively.
   (a) Determine the optimal product mix for the company.
   (b) Determine the dual price of the raw material resource and its allowable range. If available raw material is increased by 120 lb, determine the optimal solution and the change in total revenue using the dual price.
   (c) Use the dual price to determine the effect of changing the maximum demand for product A by ± 10 units.

3.6.3 Algebraic Sensitivity Analysis—Objective Function

Definition of Reduced Cost. To facilitate the explanation of the objective function sensitivity analysis, first we need to define *reduced costs*. In the TOYCO model (Example 3.6-2), the objective z-equation in the optimal tableau is

$$z + 4x_1 + x_4 + 2x_5 = 1350$$

or

$$z = 1350 - 4x_1 - x_4 - 2x_5$$

The optimal solution does not recommend the production of toy trains ($x_1 = 0$). This recommendation is confirmed by the information in the z-equation because each unit increase in x_1 above its current zero level will decrease the value of z by \$4 — namely, $z = 1350 - 4 \times (1) - 1 \times (0) - 2 \times (0) = \$1346$.

We can think of the coefficient of x_1 in the z-equation ($= 4$) as a unit *cost* because it causes a reduction in the revenue z. But where does this "cost" come from? We know that x_1 has a unit revenue of \$3 in the original model. We also know that each toy train consumes resources (operations time), which in turn incur cost. Thus, the "attractiveness" of x_1 from the standpoint of optimization depends on the relative values of the revenue per unit and the cost of the resources consumed by one unit. This relationship is formalized in the LP literature by defining the reduced cost as

$$\boxed{\begin{pmatrix}\text{Reduced cost}\\ \text{per unit}\end{pmatrix} = \begin{pmatrix}\text{Cost of consumed}\\ \text{resources per unit}\end{pmatrix} - (\text{Revenue per unit})}$$

With the given definition of *reduced cost* we can now see that an unprofitable variable (such as x_1) can be made profitable in two ways:

1. By increasing the unit revenue.
2. By decreasing the unit cost of consumed resources.

Determination of the Optimality Ranges. We now turn our attention to determining the conditions that will keep an optimal solution unchanged. The presentation is based on the definition of *reduced cost*.

In the TOYCO model, let d_1, d_2, and d_3 represent the change in unit revenues for toy trucks, trains, and cars, respectively. The objective function then becomes

As we did for the right-hand side sensitivity analysis in Section 3.6.2, we will first deal with the general situation in which all the coefficients of the objective function are changed *simultaneously* and then specialize the results to the one-at-a-time case.

With the simultaneous changes, the z-row in the starting tableau appears as:

Basic	x_1	x_2	x_3	x_4	x_5	x_6	Solution
z	$-3 - d_1$	$-2 - d_2$	$-5 - d_3$	0	0	0	0

When we generate the simplex tableaus using the same sequence of entering and leaving variables in the original model (before the changes d_j are introduced), the optimal iteration will appear as follows (convince yourself that this is indeed the case by carrying out the simplex row operations):

Basic	x_1	x_2	x_3	x_4	x_5	x_6	Solution
z	$4 - \frac{1}{4}d_2 + \frac{3}{2}d_3 - d_1$	0	0	$1 + \frac{1}{2}d_2$	$2 - \frac{1}{4}d_2 + \frac{1}{2}d_3$	0	$1350 + 100d_2 + 230d_3$
x_2	$-\frac{1}{4}$	1	0	$\frac{1}{2}$	$-\frac{1}{4}$	0	100
x_3	$\frac{3}{2}$	0	1	0	$\frac{1}{2}$	0	230
x_6	$-\frac{1}{4}$	0	0	-2	1	1	20

The new optimal tableau is exactly the same as in the *original* optimal tableau except that the *reduced costs* (z-equation coefficients) have changed. This means that *changes in the objective-function coefficients can affect the optimality of the problem only*.

You really do not need to carry out the row operation to compute the new reduced costs. An examination of the new z-row shows that the coefficients of d_j are taken directly from the constraint coefficients of the optimum tableau. A convenient way for computing the new reduced cost is to add a new top row and a new leftmost column to the optimum tableau, as shown by the shaded areas below. The entries in the top row are the change d_j associated with each variable. For the leftmost column, the entries are 1 in the z-row and the associated d_j in the row of each basic variable. Keep in mind that $d_j = 0$ for the slack variables.

Basic	x_1	x_2	x_3	x_4	x_5	x_6	Solution
z	4	0	0	1	2	0	1350
x_2	$-\frac{1}{4}$	1	0	$\frac{1}{2}$	$-\frac{1}{4}$	0	100
x_3	$\frac{3}{2}$	0	1	0	$\frac{1}{2}$	0	230
x_6	2	0	0	-2	1	1	20

Now, to compute the new reduced cost for any variable (or the value of z), multiply the elements of its column by the corresponding elements in the leftmost column, add them up, and subtract the top-row element from the sum. For example, for x_1, we have

	d_1	
Left column	x_1	(x_1-column × left-column)
1	4	4×1
d_2	$-\frac{1}{4}$	$-\frac{1}{4}d_2$
d_3	$\frac{3}{2}$	$\frac{3}{2}d_3$
0	2	2×0
Reduced cost for $x_1 = 4 - \frac{1}{4}d_2 + \frac{3}{2}d_3 - d_1$		

Note that the application of these computations to the *basic* variables will always produce a zero reduced cost, a proven theoretical result. Also, applying the same rule to the *Solution* column produces $z = 1350 + 100d_2 + 230d_3$.

Because we are dealing with a maximization problem, the current solution remains optimal so long as the new reduced costs (z-equation coefficients) remain nonnegative for all the nonbasic variables. We thus have the following **optimality conditions** corresponding to nonbasic x_1, x_4, and x_5:

$$4 - \tfrac{1}{4}d_2 + \tfrac{3}{2}d_3 - d_1 \geq 0$$

$$1 + \tfrac{1}{2}d_2 \geq 0$$

$$2 - \tfrac{1}{4}d_2 + \tfrac{1}{2}d_3 \geq 0$$

These conditions must be satisfied *simultaneously* to maintain the optimality of the current optimum.

To illustrate the use of these conditions, suppose that the objective function of TOYCO is changed from

$$\text{Maximize } z = 3x_1 + 2x_2 + 5x_3$$

to

$$\text{Maximize } z = 2x_1 + x_2 + 6x_3$$

Then, $d_1 = 2 - 3 = -\$1$, $d_2 = 1 - 2 = -\$1$, and $d_3 = 6 - 5 = \$1$. Substitution in the given conditions yields

$$4 - \tfrac{1}{4}d_2 + \tfrac{3}{2}d_3 - d_1 = 4 - \tfrac{1}{4}(-1) + \tfrac{3}{2}(1) - (-1) = 6.75 > 0 \text{ (satisfied)}$$

$$1 + \tfrac{1}{2}d_2 = 1 + \tfrac{1}{2}(-1) = .5 > 0 \qquad \text{(satisfied)}$$

$$2 - \tfrac{1}{4}d_2 + \tfrac{1}{2}d_3 = 2 - \tfrac{1}{4}(-1) + \tfrac{1}{2}(1) = 2.75 > 0 \qquad \text{(satisfied)}$$

The results show that the proposed changes will keep the current solution ($x_1 = 0$, $x_2 = 100$, $x_3 = 230$) optimal. Hence no further calculations are needed, except that the objective value will change to $z = 1350 + 100d_2 + 230d_3 = 1350 + 100 \times -1 + 230 \times 1 = \$1480$. If any of the conditions is not satisfied, a new solution must be determined (see Chapter 4).

The discussion so far has dealt with the maximization case. The only difference in the minimization case is that the reduced costs (z-equations coefficients) must be ≤ 0 to maintain optimality.

The general optimality conditions can be used to determine the special case where the changes d_j occur *one at a time* instead of simultaneously. This analysis is equivalent to considering the following three cases:

1. Maximize $z = (3 + d_1)x_1 + 2x_2 + 5x_3$
2. Maximize $z = 3x_1 + (2 + d_2)x_2 + 5x_3$
3. Maximize $z = 3x_1 + 2x_2 + (5 + d_3)x_3$

The individual conditions can be accounted for as special cases of the simultaneous case.[5]

Case 1. Set $d_2 = d_3 = 0$ in the simultaneous conditions, which gives

Case 2. Set $d_1 = d_3 = 0$ in the simultaneous conditions, which gives

Case 3. Set $d_1 = d_2 = 0$ in the simultaneous conditions, which gives

$$\left.\begin{array}{l} 4 + \frac{3}{2}d_3 \geq 0 \Rightarrow d_3 \geq -\frac{8}{3} \\ 2 + \frac{1}{2}d_3 \geq 0 \Rightarrow d_3 \geq -4 \end{array}\right\} \Rightarrow -\frac{8}{3} \leq d_3 < \infty$$

It is important to notice that the changes d_1, d_2, and d_3 may be within their allowable individual ranges without satisfying the simultaneous conditions, and vice versa. For example, consider

$$\text{Maximize } z = 6x_1 + 8x_2 + 3x_3$$

Here $d_1 = 6 - 3 = \$3$, $d_2 = 8 - 2 = \$6$, and $d_3 = 3 - 5 = -\$2$, which are all within the permissible individual ranges $(-\infty < d_1 \leq 4, -2 \leq d_2 \leq 8$, and $-\frac{8}{3} \leq d_3 < \infty)$. However, the corresponding simultaneous conditions yield

$$4 - \tfrac{1}{4}d_2 + \tfrac{3}{2}d_3 - d_1 = 4 - \tfrac{1}{4}(6) + \tfrac{3}{2}(-2) - 3 = -3.5 < 0 \quad \text{(not satisfied)}$$

$$1 + \tfrac{1}{2}d_2 = 1 + \tfrac{1}{2}(6) = 4 > 0 \quad \text{(satisfied)}$$

$$2 - \tfrac{1}{4}d_2 + \tfrac{1}{2}d_3 = 2 - \tfrac{1}{4}(6) + \tfrac{1}{2}(-2) = -.5 < 0 \quad \text{(not satisfied)}$$

The results above can be summarized as follows:

1. The optimal values of the variables remain unchanged so long as the changes d_j, $j = 1, 2, \ldots, n$, in the objective function coefficients satisfy all the optimality conditions when the changes are simultaneous or fall within the optimality ranges when a change is made individually.
2. For other situations where the simultaneous optimality conditions are not satisfied or the individual feasibility ranges are violated, the recourse is to either re-solve the problem with the new values of d_j or apply the post-optimal analysis presented in Chapter 4.

PROBLEM SET 3.6D[6]

1. In the TOYCO model, determine if the current solution will change in each of the following cases:
   (i) $z = 2x_1 + x_2 + 4x_3$
   (ii) $z = 3x_1 + 6x_2 + x_3$
   (iii) $z = 8x_1 + 3x_2 + 9x_3$

*2. B&K grocery store sells three types of soft drinks: the brand names A1 Cola and A2 Cola and the cheaper store brand BK Cola. The price per can for A1, A2, and BK are 80, 70, and 60 cents, respectively. On the average, the store sells no more than 500 cans of all colas a day. Although A1 is a recognized brand name, customers tend to buy more A2 and BK because they are cheaper. It is estimated that at least 100 cans of A1 are sold daily and that A2 and BK combined outsell A1 by a margin of at least 4:2.
   (a) Show that the optimum solution does not call for selling the A3 brand.
   (b) By how much should the price per can of A3 be increased to be sold by B&K?
   (c) To be competitive with other stores, B&K decided to lower the price on all three types of cola by 5 cents per can. Recompute the reduced costs to determine if this promotion will change the current optimum solution.

Reference: *H.A.Taha, Operations Research: An Introduction,* Prentice Hall; 9th edition, *Singapore, 2010.*

Lecture 9
Duality

Duality in equation form
Primal dual relationships
Inverse Matrix
Optimal Dual Solution

4.1 DEFINITION OF THE DUAL PROBLEM

The **dual** problem is an LP defined directly and systematically from the **primal** (or original) LP model. The two problems are so closely related that the optimal solution of one problem automatically provides the optimal solution to the other.

In most LP treatments, the dual is defined for various forms of the primal depending on the sense of optimization (maximization or minimization), types of constraints ($\leq$, $\geq$, or $=$), and orientation of the variables (nonnegative or unrestricted). This type of treatment is somewhat confusing, and for this reason we offer a *single* definition that automatically subsumes *all* forms of the primal.

Our definition of the dual problem requires expressing the primal problem in the *equation form* presented in Section 3.1 (all the constraints are equations with nonnegative right-hand side and all the variables are nonnegative).

To show how the dual problem is constructed, define the primal in *equation form* as follows:

$$\text{Maximize or minimize } z = \sum_{j=1}^{n} c_j x_j$$

subject to

$$\sum_{j=1}^{n} a_{ij} x_j = b_i,\ i = 1, 2, \ldots, m$$

$$x_j \geq 0,\ j = 1, 2, \ldots, n$$

The variables x_j, $j = 1, 2, \ldots, n$, include the surplus, slack, and artificial variables, if any.

Table 4.1 shows how the dual problem is constructed from the primal. Effectively, we have

1. A dual variable is defined for each primal (constraint) equation.
2. A dual constraint is defined for each primal variable.
3. The constraint (column) coefficients of a primal variable define the left-hand-side coefficients of the dual constraint and its objective coefficient define the right-hand side.
4. The objective coefficients of the dual equal the right-hand side of the primal constraint equations.

TABLE 4.1 Construction of the Dual from the Primal

	Primal variables						
	x_1	x_2	$\cdots$	x_j	$\cdots$	x_n	
Dual variables	c_1	c_2	$\cdots$	c_j	$\cdots$	c_n	Right-hand side
y_1	a_{11}	a_{12}	$\cdots$	a_{1j}	$\cdots$	a_{1n}	b_1
y_2	a_{21}	a_{22}	$\cdots$	a_{2j}	$\cdots$	a_{2n}	b_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
y_m	a_{m1}	a_{m2}	$\cdots$	a_{mj}	$\cdots$	a_{mn}	b_m
				↑ jth dual constraint			↑ Dual objective coefficients

The rules for determining the sense of optimization (maximization or minimization), the type of the constraint ($\leq$, $\geq$, or $=$), and the sign of the dual variables are summarized in Table 4.2. Note that the sense of optimization in the dual is always opposite to that of the primal. An easy way to remember the constraint type in the dual (i.e., $\leq$ or $\geq$) is that if the dual objective is *minimization* (i.e., pointing *down*), then the constraints are all of the type $\geq$ (i.e., pointing *up*). The opposite is true when the dual objective is maximization.

TABLE 4.2 Rules for Constructing the Dual Problem

Primal problem objective[a]	Dual problem		
	Objective	*Constraints type*	*Variables sign*
Maximization	Minimization	$\geq$	Unrestricted
Minimization	Maximization	$\leq$	Unrestricted

[a]All primal constraints are equations with nonnegative right-hand side, and all the variables are nonnegative.

Example 4.1-1

Primal
Maximize $z = 5x_1 + 12x_2 + 4x_3$ subject to $x_1 + 2x_2 + x_3 \le 10$ $2x_1 - x_2 + 3x_3 = 8$ $x_1, x_2, x_3 \ge 0$

Example 4.1-2

Primal
Minimize $z = 15x_1 + 12x_2$ subject to $x_1 + 2x_2 \geq 3$ $2x_1 - 4x_2 \leq 5$ $x_1, x_2 \geq 0$

Example 4.1-3

Primal
Maximize $z = 5x_1 + 6x_2$ subject to $x_1 + 2x_2 = 5$ $-x_1 + 5x_2 \geq 3$ $4x_1 + 7x_2 \leq 8$ x_1 unrestricted, $x_2 \geq 0$

The first and second constraints are replaced by an equation. The general rule in this case is that an unrestricted primal variable always corresponds to an equality dual constraint. Conversely, a primal equation produces an unrestricted dual variable, as the first primal constraint demonstrates.

Summary of the Rules for Constructing the Dual. The general conclusion from the preceding examples is that the variables and constraints in the primal and dual problems are defined by the rules in Table 4.3. It is a good exercise to verify that these explicit rules are subsumed by the general rules in Table 4.2.

TABLE 4.3 Rules for Constructing the Dual Problem

Maximization problem		Minimization problem
Constraints		*Variables*
$\geq$	$\Leftrightarrow$	≤ 0
$\leq$	$\Leftrightarrow$	≥ 0
$=$	$\Leftrightarrow$	Unrestricted
Variables		*Constraints*
≥ 0	$\Leftrightarrow$	$\geq$
≤ 0	$\Leftrightarrow$	$\leq$
Unrestricted	$\Leftrightarrow$	$=$

Note that the table does not use the designation primal and dual. What matters here is the sense of optimization. If the primal is maximization, then the dual is minimization, and vice versa.

PROBLEM SET 4.1A

1. In Example 4.1-1, derive the associated dual problem if the sense of optimization in the primal problem is changed to minimization.

*2. In Example 4.1-2, derive the associated dual problem given that the primal problem is augmented with a third constraint, $3x_1 + x_2 = 4$.

3. In Example 4.1-3, show that even if the sense of optimization in the primal is changed to minimization, an unrestricted primal variable always corresponds to an equality dual constraint.

4. Write the dual for each of the following primal problems:

   (a) Maximize $z = -5x_1 + 2x_2$

   subject to

$$-x_1 + x_2 \leq -2$$

$$2x_1 + 3x_2 \leq 5$$

$$x_1, x_2 \geq 0$$

   (b) Minimize $z = 6x_1 + 3x_2$

   subject to

$$6x_1 - 3x_2 + x_3 \geq 2$$

$$3x_1 + 4x_2 + x_3 \geq 5$$

$$x_1, x_2, x_3 \geq 0$$

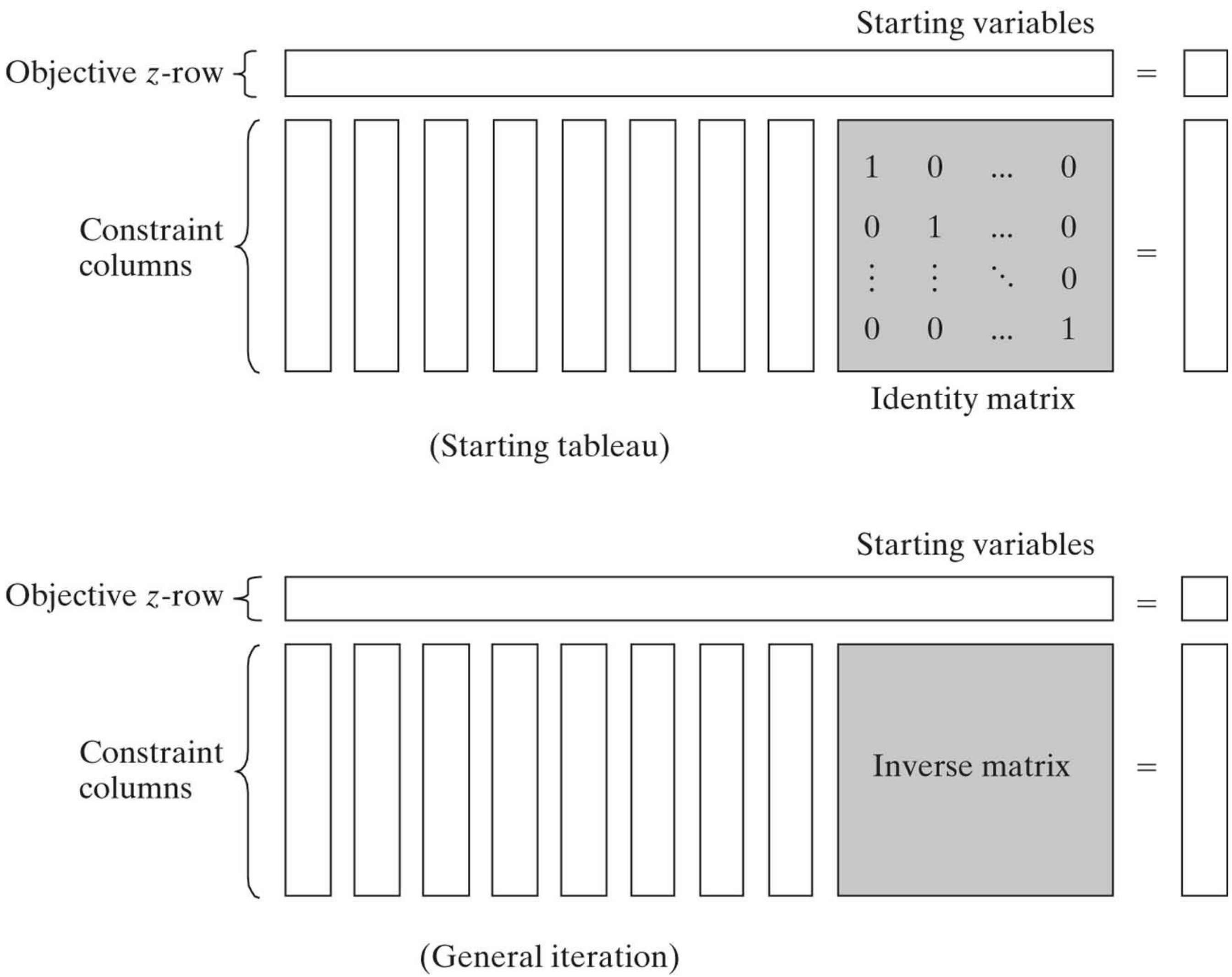
4.2 PRIMAL-DUAL RELATIONSHIPS

Changes made in the original LP model will change the elements of the current optimal tableau, which in turn may affect the optimality and/or the feasibility of the current solution. This section introduces a number of primal-dual relationships that can be used to recompute the elements of the optimal simplex tableau. These relationships will form the basis for the economic interpretation of the LP model as well as for post-optimality analysis.

4.2.2 Simplex Tableau Layout

Figure 4.1 gives a schematic representation of the *starting* and *general* simplex tableaus. In the starting tableau, the constraint coefficients under the starting variables form an **identity matrix** (all main-diagonal elements equal 1 and all off-diagonal elements equal zero). With this arrangement, subsequent iterations of the simplex tableau generated by the Gauss-Jordan row operations (see Chapter 3) will modify the elements of the identity matrix to produce what is known as the **inverse matrix**. As we will see in the remainder of this chapter, the inverse matrix is key to computing all the elements of the associated simplex tableau.

Figure 4.1 Schematic representation of the starting and general simplex tableaus



4.2.3 Optimal Dual Solution

The primal and dual solutions are so closely related that the optimal solution of either problem directly yields (with little additional computation) the optimal solution to the other. Thus, in an LP model in which the number of variables is considerably smaller than the number of constraints, computational savings may be realized by solving the dual, from which the primal solution is determined automatically. This result follows because the amount of simplex computation depends largely (though not totally) on the number of constraints (see Problem 2, Set 4.2c).

This section provides two methods for determining the dual values. Note that the dual of the dual is itself the primal, which means that the dual solution can also be used to yield the optimal primal solution automatically.

Method 1.

$$\begin{pmatrix}\text{Optimal value of}\\ \textit{dual} \text{ variable } y_i\end{pmatrix} = \begin{pmatrix}\text{Optimal primal } z\text{-coefficient of } \textit{starting} \text{ variable } x_i\\ +\\ \textit{Original} \text{ objective coefficient of } x_i\end{pmatrix}$$

Method 2.

$$\begin{pmatrix}\text{Optimal values}\\ \text{of } \textit{dual} \text{ variables}\end{pmatrix} = \begin{pmatrix}\text{Row vector of}\\ \text{original objective coefficients}\\ \text{of optimal } \textit{primal} \text{ basic variables}\end{pmatrix} \times \begin{pmatrix}\text{Optimal } \textit{primal}\\ \text{inverse}\end{pmatrix}$$

The elements of the row vector must appear in the same order in which the basic variables are listed in the *Basic* column of the simplex tableau.

Example 4.2-1

Consider the following LP:

$$\text{Maximize } z = 5x_1 + 12x_2 + 4x_3$$

subject to

$$\begin{aligned} x_1 + 2x_2 + x_3 &\le 10 \\ 2x_1 - x_2 + 3x_3 &= 8 \\ x_1, x_2, x_3 &\ge 0 \end{aligned}$$

To prepare the problem for solution by the simplex method, we add a slack x_4 in the first constraint and an artificial R in the second. The resulting primal and the associated dual problems are thus defined as follows:

Primal	Dual
Maximize $z = 5x_1 + 12x_2 + 4x_3 - MR$ subject to $x_1 + 2x_2 + x_3 + x_4 = 10$ $2x_1 - x_2 + 3x_3 + R = 8$ $x_1, x_2, x_3, x_4, R \ge 0$	Minimize $w = 10y_1 + 8y_2$ subject to $y_1 + 2y_2 \ge 5$ $2y_1 - y_2 \ge 12$ $y_1 + 3y_2 \ge 4$ $y_1 \ge 0$ $y_2 \ge -M$ ($\Rightarrow y_2$ unrestricted)

TABLE 4.4 Optimal Tableau of the Primal of Example 4.2-1

Basic	x_1	x_2	x_3	x_4	R	Solution
z	0	0	$\frac{3}{5}$	$\frac{29}{5}$	$-\frac{2}{5} + M$	$54\frac{4}{5}$
x_2	0	1	$-\frac{1}{5}$	$\frac{2}{5}$	$-\frac{1}{5}$	$\frac{12}{5}$
x_1	1	0	$\frac{7}{5}$	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{26}{5}$

Table 4.4 provides the optimal primal tableau.

We now show how the optimal dual values are determined using the two methods described at the start of this section.

Method 1. In Table 4.4, the starting primal variables x_4 and R uniquely correspond to the dual variables y_1 and y_2, respectively. Thus, we determine the optimum dual solution as follows:

Starting primal basic variables	
z-equation coefficients	
Original objective coefficient	
Dual variables	
Optimal dual values	

Method 2. The optimal inverse matrix, highlighted under the starting variables x_4 and R, is given in Table 4.4 as

First, we note that the optimal primal variables are listed in the tableau in *row order* as x_2 and then x_1. This means that the elements of the original objective coefficients for the two variables must appear in the same order—namely,

$$\begin{aligned}(\text{Original objective coefficients}) &= (\text{Coefficient of } x_2, \text{coefficient of } x_1)\\ &= (12, 5)\end{aligned}$$

Thus, the optimal dual values are computed as

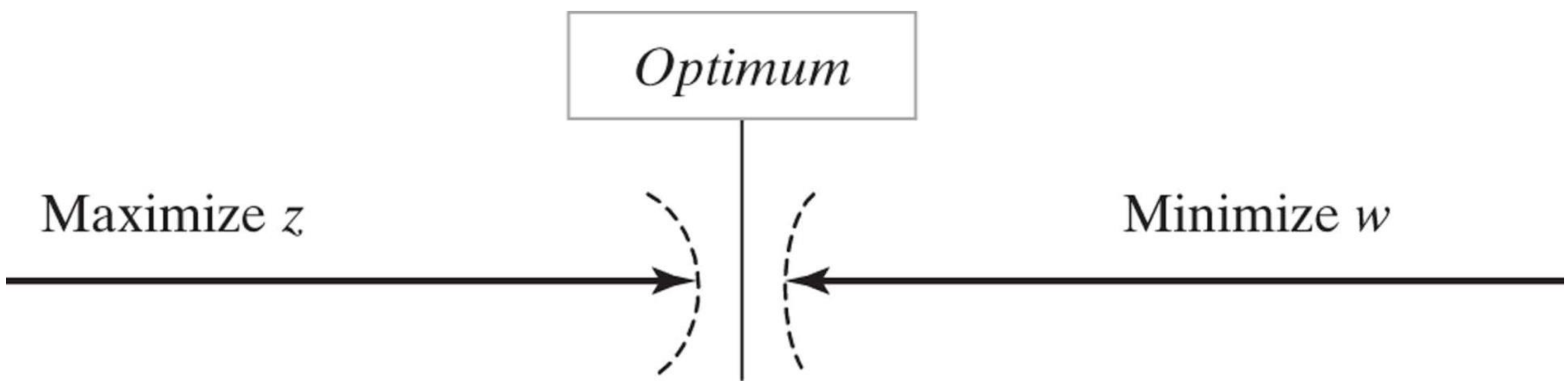
Primal-dual objective values. Having shown how the optimal dual values are determined, next we present the relationship between the primal and dual objective values. For any pair of *feasible* primal and dual solutions,

$$\begin{pmatrix}\text{Objective value in the} \\ \textit{maximization}\text{ problem}\end{pmatrix} \leq \begin{pmatrix}\text{Objective value in the} \\ \textit{minimization}\text{ problem}\end{pmatrix}$$

At the optimum, the relationship holds as a strict equation. The relationship does not specify which problem is primal and which is dual. Only the sense of optimization (maximization or minimization) is important in this case.

The optimum cannot occur with z strictly less than w (i.e., $z < w$) because, no matter how close z and w are, there is always room for improvement, which contradicts optimality as Figure 4.2 demonstrates.

Figure 4.2 Relationship between maximum z and minimum w



Example 4.2-2

In Example 4.2-1, $\left(x_1 = 0, x_2 = 0, x_3 = \frac{8}{3}\right)$ and $(y_1 = 6, y_2 = 0)$ are feasible primal and dual solutions. The associated values of the objective functions are

$$z = 5x_1 + 12x_2 + 4x_3 = 5(0) + 12(0) + 4\left(\tfrac{8}{3}\right) = 10\tfrac{2}{3}$$

$$w = 10y_1 + 8y_2 = 10(6) + 8(0) = 60$$

Thus, $z\left(= 10\frac{2}{3}\right)$ for the maximization problem (primal) is less than w $(= 60)$ for the minimization problem (dual). The optimum value of $z\left(= 54\frac{4}{5}\right)$ falls within the range $\left(10\frac{2}{3}, 60\right)$.

PROBLEM SET 4.2C

1. Find the optimal value of the objective function for the following problem by inspecting only its dual. (Do not solve the dual by the simplex method.)

$$\text{Minimize } z = 10x_1 + 4x_2 + 5x_3$$

subject to

$$5x_1 - 7x_2 + 3x_3 \geq 50$$

$$x_1, x_2, x_3 \geq 0$$

2. Solve the dual of the following problem, then find its optimal solution from the solution of the dual. Does the solution of the dual offer computational advantages over solving the primal directly?

$$\text{Minimize } z = 5x_1 + 6x_2 + 3x_3$$

subject to

$$\begin{aligned}
5x_1 + 5x_2 + 3x_3 &\geq 50 \\
x_1 + x_2 - x_3 &\geq 20 \\
7x_1 + 6x_2 - 9x_3 &\geq 30 \\
5x_1 + 5x_2 + 5x_3 &\geq 35 \\
2x_1 + 4x_2 - 15x_3 &\geq 10 \\
12x_1 + 10x_2 &\geq 90 \\
x_2 - 10x_3 &\geq 20 \\
x_1, x_2, x_3 &\geq 0
\end{aligned}$$

Reference: *H.A.Taha, Operations Research: An Introduction,* Prentice Hall; 9th edition, *Singapore, 2010.*

Lecture 10
Dual Simplex Algorithm

Dual feasibility condition
Dual optimality condition

4.4 ADDITIONAL SIMPLEX ALGORITHMS

In the simplex algorithm presented in Chapter 3 the problem starts at a (basic) feasible solution. Successive iterations continue to be feasible until the optimal is reached at the last iteration. The algorithm is sometimes referred to as the **primal simplex** method.

This section presents two additional algorithms: The **dual simplex** and the **generalized simplex**. In the dual simplex, the LP starts at a better than optimal *infeasible* (basic) solution. Successive iterations remain infeasible and (better than) optimal until feasibility is restored at the last iteration. The generalized simplex combines both the primal and dual simplex methods in one algorithm. It deals with problems that start both nonoptimal and infeasible. In this algorithm, successive iterations are associated with basic feasible or infeasible (basic) solutions. At the final iteration, the solution becomes optimal and feasible (assuming that one exists).

All three algorithms, the primal, the dual, and the generalized, are used in the course of post-optimal analysis calculations, as will be shown in Section 4.5.

4.4.1 Dual Simplex Algorithm

The crux of the dual simplex method is to start with a better than optimal and infeasible basic solution. The optimality and feasibility conditions are designed to preserve the optimality of the basic solutions while moving the solution iterations toward feasibility.

Dual feasibility condition. The leaving variable, x_r, is the basic variable having the most negative value (ties are broken arbitrarily). If all the basic variables are nonnegative, the algorithm ends.

Dual optimality condition. Given that x_r is the leaving variable, let $\bar{c}_j$ be the reduced cost of nonbasic variable x_j and α_{rj} the constraint coefficient in the x_r-row and x_j-column of the tableau. The entering variable is the nonbasic variable with $\alpha_{rj} < 0$ that corresponds to

$$\min_{\text{Nonbasic } x_j} \left\{ \left| \frac{\bar{c}_j}{\alpha_{rj}} \right|, \alpha_{rj} < 0 \right\}$$

(Ties are broken arbitrarily.) If $\alpha_{rj} \geq 0$ for all nonbasic x_j, the problem has no feasible solution.

To start the LP optimal and infeasible, two requirements must be met:

1. The objective function must satisfy the optimality condition of the regular simplex method (Chapter 3).

2. All the constraints must be of the type ($\leq$).

The second condition requires converting any ($\geq$) to ($\leq$) simply by multiplying both sides of the inequality ($\geq$) by -1. If the LP includes ($=$) constraints, the equation can be replaced by two inequalities. For example,

$$x_1 + x_2 = 1$$

is equivalent to

or

After converting all the constraints to ($\leq$), the starting solution is infeasible if at least one of the right-hand sides of the inequalities is strictly negative.

Example 4.4-1 Minimize $z = 3x_1 + 2x_2 + x_3$

subject to

$$\begin{aligned} 3x_1 + \quad x_2 + x_3 &\geq 3. \\ -3x_1 + 3x_2 + x_3 &\geq 6 \\ x_1 + \quad x_2 + x_3 &\leq 3 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

In the present example, the first two inequalities are multiplied by -1 to convert them to ($\leq$) constraints. The starting tableau is thus given as:

The tableau is optimal because all the reduced costs in the z-row are ≤ 0 ($\bar{c}_1 = -3, \bar{c}_2 = -2, \bar{c}_3 = -1, \bar{c}_4 = 0, \bar{c}_5 = 0, \bar{c}_6 = 0$). It is also infeasible because at least one of the basic variables is negative ($x_4 = -3, x_5 = -6, x_6 = 3$).

According to the dual feasibility condition, x_5 ($= -6$) is the leaving variable. The next table shows how the dual optimality condition is used to determine the entering variable.

The ratios show that x_2 is the entering variable. Notice that a nonbasic variable x_j is a candidate for entering the basic solution only if its α_{rj} is strictly negative. This is the reason x_1 is excluded in the table above.

The next tableau is obtained by using the familiar row operations, which give

The preceding tableau shows that x_4 leaves and x_3 enters, thus yielding the following tableau, which is both optimal and feasible:

Notice how the dual simplex works. In all the iterations, optimality is maintained (all reduced costs are ≤ 0). At the same time, each new iteration moves the solution toward feasibility. At iteration 3, feasibility is restored for the first time and the process ends with the optimal feasible solution given as $x_1 = 0$, $x_2 = \frac{3}{2}$, $x_2 = \frac{3}{2}$, and $z = \frac{9}{2}$.

PROBLEM SET 4.4A

2. Generate the dual simplex iterations for the following problems (using TORA for convenience), and trace the path of the algorithm on the graphical solution space.

(a) Minimize $z = 2x_1 + 3x_2$

subject to

$$2x_1 + 2x_2 \leq 30$$
$$x_1 + 2x_2 \geq 10$$
$$x_1, x_2 \geq 0$$

(d) Minimize $z = 2x_1 + 3x_2$

subject to

$$2x_1 + x_2 \geq 3$$
$$x_1 + x_2 = 2$$
$$x_1, x_2 \geq 0$$

Reference: *H.A.Taha, Operations Research: An Introduction,* Prentice Hall; 9th edition, *Singapore, 2010.*

Lecture 11
Transportation Model and Algorithm

Balancing the transportation model
Determination of the starting solution
Iterative computations of the transportation algorithm

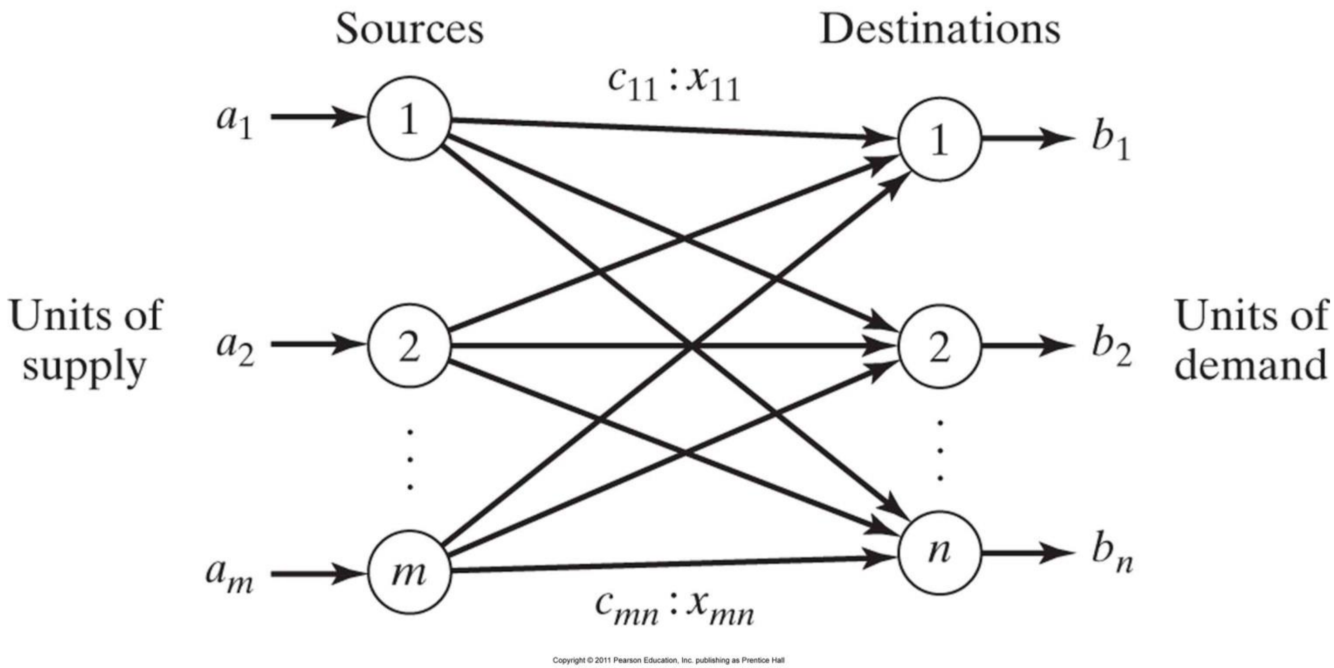
Transportation Model

The transportation model is a special class of linear programs that deals with shipping a commodity from *sources* (e.g., factories) to *destinations* (e.g., warehouses). The objective is to determine the shipping schedule that minimizes the total shipping cost while satisfying supply and demand limits. The application of the transportation model can be extended to other areas of operation, including inventory control, employment scheduling, and personnel assignment.

5.1 DEFINITION OF THE TRANSPORTATION MODEL

The general problem is represented by the network in Figure 5.1. There are m sources and n destinations, each represented by a **node**. The **arcs** represent the routes linking the sources and the destinations. Arc (i, j) joining source i to destination j carries two pieces of information: the transportation cost per unit, c_{ij}, and the amount shipped, x_{ij}. The amount of supply at source i is a_i and the amount of demand at destination j is b_j. The objective of the model is to determine the unknowns x_{ij} that will minimize the total transportation cost while satisfying all the supply and demand restrictions.

Figure 5.1 Representation of the transportation model with nodes and arcs



Example 5.1-1

MG Auto has three plants in Los Angeles, Detroit, and New Orleans, and two major distribution centers in Denver and Miami. The capacities of the three plants during the next quarter are 1000, 1500, and 1200 cars. The quarterly demands at the two distribution centers are 2300 and 1400 cars. The mileage chart between the plants and the distribution centers is given in Table 5.1.

The trucking company in charge of transporting the cars charges 8 cents per mile per car. The transportation costs per car on the different routes, rounded to the closest dollar, are given in Table 5.2.

TABLE 5.1 Mileage Chart

	Denver	Miami
Los Angeles	1000	2690
Detroit	1250	1350
New Orleans	1275	850

TABLE 5.2 Transportation Cost per Car

	Denver (1)	Miami (2)
Los Angeles (1)	\$80	\$215
Detroit (2)	\$100	\$108
New Orleans (3)	\$102	\$68

The LP model of the problem is given as

$$\text{Minimize } z = 80x_{11} + 215x_{12} + 100x_{21} + 108x_{22} + 102x_{31} + 68x_{32}$$

subject to

$$\begin{array}{lr}
x_{11} + x_{12} = 1000 & \text{(Los Angeles)} \\
x_{21} + x_{22} = 1500 & \text{(Detroit)} \\
+ x_{31} + x_{32} = 1200 & \text{(New Oreleans)} \\
x_{11} + x_{21} + x_{31} = 2300 & \text{(Denver)} \\
x_{12} + x_{22} + x_{32} = 1400 & \text{(Miami)} \\
x_{ij} \geq 0, i = 1, 2, 3, j = 1, 2 &
\end{array}$$

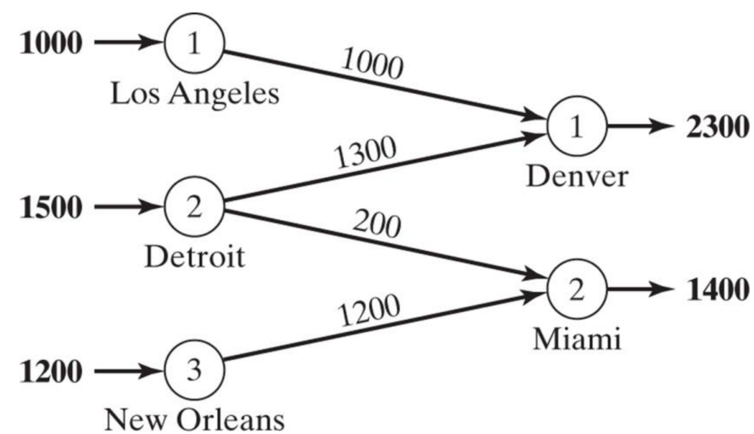
These constraints are all equations because the total supply from the three sources (= 1000 + 1500 + 1200 = 3700 cars) equals the total demand at the two destinations (= 2300 + 1400 = 3700 cars).

The LP model can be solved by the simplex method. However, with the special structure of the constraints we can solve the problem more conveniently using the **transportation tableau** shown in Table 5.3.

TABLE 5.3 MG Transportation Model

	Denver	Miami	Supply
Los Angeles	80 x_{11}	215 x_{12}	**1000**
Detroit	100 x_{21}	108 x_{22}	**1500**
New Orleans	102 x_{31}	68 x_{32}	**1200**
Demand	**2300**	**1400**	

Figure 5.2 Optimal solution of MG Auto model



The optimal solution in Figure 5.2 (obtained by TORA[1]) calls for shipping 1000 cars from Los Angeles to Denver, 1300 from Detroit to Denver, 200 from Detroit to Miami, and 1200 from New Orleans to Miami. The associated minimum transportation cost is computed as $1000 \times \$80 + 1300 \times \$100 + 200 \times \$108 + 1200 \times \$68 = \$313{,}200$.

Balancing the Transportation Model. The transportation algorithm is based on the assumption that the model is balanced, meaning that the total demand equals the total supply. If the model is unbalanced, we can always add a dummy source or a dummy destination to restore balance.

Example 5.1-2

In the MG model, suppose that the Detroit plant capacity is 1300 cars (instead of 1500). The total supply (= 3500 cars) is less than the total demand (= 3700 cars), meaning that part of the demand at Denver and Miami will not be satisfied.

Because the demand exceeds the supply, a dummy source (plant) with a capacity of 200 cars (= 3700 − 3500) is added to balance the transportation model. The unit transportation costs from the dummy plant to the two destinations are zero because the plant does not exist.

Table 5.4 gives the balanced model together with its optimum solution. The solution shows that the dummy plant ships 200 cars to Miami, which means that Miami will be 200 cars short of satisfying its demand of 1400 cars.

We can make sure that a specific destination does not experience shortage by assigning a very high unit transportation cost from the dummy source to that destination. For example, a penalty of $1000 in the dummy-Miami cell will prevent shortage at Miami. Of course, we cannot use this "trick" with all the destinations, because shortage must occur somewhere in the system.

TABLE 5.4 MG Model with Dummy Plant

	Denver	Miami	Supply
Los Angeles	80 **1000**	215	**1000**
Detroit	100 **1300**	108	**1300**
New Orleans	102	68 **1200**	**1200**
Dummy Plant	0	0 **200**	**200**
Demand	**2300**	**1400**	

TABLE 5.5 MG Model with Dummy Destination

	Denver	Miami	Dummy	
Los Angeles	80 **1000**	215	0	**1000**
Detroit	100 **900**	108 **200**	0 **400**	**1500**
New Orleans	102	68 **1200**	0	**1200**
Demand	**1900**	**1400**	**400**	

The case where the supply exceeds the demand can be demonstrated by assuming that the demand at Denver is 1900 cars only. In this case, we need to add a dummy distribution center to "receive" the surplus supply. Again, the unit transportation costs to the dummy distribution center are zero, unless we require a factory to "ship out" completely. In this case, we must assign a high unit transportation cost from the designated factory to the dummy destination.

Table 5.5 gives the new model and its optimal solution (obtained by TORA). The solution shows that the Detroit plant will have a surplus of 400 cars.

PROBLEM SET 5.1A[2]

1. True or False?
   (a) To balance a transportation model, it may be necessary to add both a dummy source and a dummy destination.
   (b) The amounts shipped to a dummy destination represent surplus at the shipping source.
   (c) The amounts shipped from a dummy source represent shortages at the receiving destinations.
2. In each of the following cases, determine whether a dummy source or a dummy destination must be added to balance the model.
   (a) Supply: $a_1 = 10, a_2 = 5, a_3 = 4, a_4 = 6$
   Demand: $b_1 = 10, b_2 = 5, b_3 = 7, b_4 = 9$
   (b) Supply: $a_1 = 30, a_2 = 44$
   Demand: $b_1 = 25, b_2 = 30, b_3 = 10$
3. In Table 5.4 of Example 5.1-2, where a dummy plant is added, what does the solution mean when the dummy plant "ships" 150 cars to Denver and 50 cars to Miami?
*4. In Table 5.5 of Example 5.1-2, where a dummy destination is added, suppose that the Detroit plant must ship out *all* its production. How can this restriction be implemented in the model?

*6. Three electric power plants with capacities of 25, 40, and 30 million kWh supply electricity to three cities. The maximum demands at the three cities are estimated at 30, 35, and 25 million kWh. The price per million kWh at the three cities is given in Table 5.6.

During the month of August, there is a 20% increase in demand at each of the three cities, which can be met by purchasing electricity from another network at a premium rate of $1000 per million kWh. The network is not linked to city 3, however. The utility company wishes to determine the most economical plan for the distribution and purchase of additional energy.

(a) Formulate the problem as a transportation model.

(b) Determine an optimal distribution plan for the utility company.

(c) Determine the cost of the additional power purchased by each of the three cities.

7. Solve Problem 6, assuming that there is a 10% power transmission loss through the network.

TABLE 5.6 Price/Million kWh for Problem 6

		City	
	1	2	3
Plant 1	$600	$700	$400
Plant 2	$320	$300	$350
Plant 3	$500	$480	$450

5.3 THE TRANSPORTATION ALGORITHM

The transportation algorithm follows the *exact steps* of the simplex method (Chapter 3). However, instead of using the regular simplex tableau, we take advantage of the special structure of the transportation model to organize the computations in a more convenient form.

Summary of the Transportation Algorithm. The steps of the transportation algorithm are exact parallels of the simplex algorithm.

Step 1. Determine a *starting* basic feasible solution, and go to step 2.

Step 2. Use the optimality condition of the simplex method to determine the *entering variable* from among all the nonbasic variables. If the optimality condition is satisfied, stop. Otherwise, go to step 3.

Step 3. Use the feasibility condition of the simplex method to determine the *leaving variable* from among all the current basic variables, and find the new basic solution. Return to step 2.

Example 5.3-1 (SunRay Transport)

SunRay Transport Company ships truckloads of grain from three silos to four mills. The supply (in truckloads) and the demand (also in truckloads) together with the unit transportation costs per truckload on the different routes are summarized in the transportation model in Table 5.16. The unit transportation costs, c_{ij}, (shown in the northeast corner of each box) are in hundreds of dollars. The model seeks the minimum-cost shipping schedule x_{ij} between silo i and mill j $(i = 1, 2, 3; j = 1, 2, 3, 4)$.

TABLE 5.16 SunRay Transportation Model

	Mill 1	2	3	4	Supply
Silo 1	10 x_{11}	2 x_{12}	20 x_{13}	11 x_{14}	**15**
Silo 2	12 x_{21}	7 x_{22}	9 x_{23}	20 x_{24}	**25**
Silo 3	4 x_{31}	14 x_{32}	16 x_{33}	18 x_{34}	**10**
Demand	**5**	**15**	**15**	**15**	

5.3.1 Determination of the Starting Solution

A general transportation model with m sources and n destinations has $m + n$ constraint equations, one for each source and each destination. However, because the transportation model is always balanced (sum of the supply = sum of the demand), one of these equations is redundant. Thus, the model has $m + n - 1$ independent constraint equations, which means that the starting basic solution consists of $m + n - 1$ basic variables. Thus, in Example 5.3-1, the starting solution has $3 + 4 - 1 = 6$ basic variables.

The special structure of the transportation problem allows securing a nonartificial starting basic solution using one of three methods:[5]

1. Northwest-corner method
2. Least-cost method
3. Vogel approximation method

The three methods differ in the "quality" of the starting basic solution they produce, in the sense that a better starting solution yields a smaller objective value. In general, though not always, the Vogel method yields the best starting basic solution, and the northwest-corner method yields the worst. The tradeoff is that the northwest-corner method involves the least amount of computations.

Northwest-Corner Method. The method starts at the northwest-corner cell (route) of the tableau (variable x_{11}).

Step 1. Allocate as much as possible to the selected cell, and adjust the associated amounts of supply and demand by subtracting the allocated amount.

Step 2. Cross out the row or column with zero supply or demand to indicate that no further assignments can be made in that row or column. If both a row and a column net to zero simultaneously, *cross out one only*, and leave a zero supply (demand) in the uncrossed-out row (column).

Step 3. If *exactly one* row or column is left uncrossed out, stop. Otherwise, move to the cell to the right if a column has just been crossed out or below if a row has been crossed out. Go to step 1.

TABLE 5.17 North-West Corner Starting Solution

	1	2	3	4	Supply
1	10	2	20	11	**15**
2	12	7	9	20	**25**
3	4	14	16	18	**10**
Demand	**5**	**15**	**15**	**15**	

Example 5.3-2

The application of the procedure to the model of Example 5.3-1 gives the starting basic solution in Table 5.17. The arrows show the order in which the allocated amounts are generated.

The starting basic solution is

$$x_{11} = 5, x_{12} = 10$$

$$x_{22} = 5, x_{23} = 15, x_{24} = 5$$

$$x_{34} = 10$$

The associated cost of the schedule is

$$z = 5 \times 10 + 10 \times 2 + 5 \times 7 + 15 \times 9 + 5 \times 20 + 10 \times 18 = \$520$$

Least-Cost Method. The least-cost method finds a better starting solution by concentrating on the cheapest routes. The method assigns as much as possible to the cell with the smallest unit cost (ties are broken arbitrarily). Next, the satisfied row or column is crossed out and the amounts of supply and demand are adjusted accordingly. If both a row and a column are satisfied simultaneously, *only one is crossed out*, the same as in the northwest-corner method. Next, look for the uncrossed-out cell with the smallest unit cost and repeat the process until exactly one row or column is left uncrossed out.

TABLE 5.18 Least-Cost Starting Solution

	1	2	3	4	Supply
1	10	2	20	11	**15**
2	12	7	9	20	**25**
3	4	14	16	18	**10**
Demand	**5**	**15**	**15**	**15**	

Example 5.3-3

The least-cost method is applied to Example 5.3-1 in the following manner:

1. Cell (1, 2) has the least unit cost in the tableau (= \$2). The most that can be shipped through (1, 2) is $x_{12} = 15$ truckloads, which happens to satisfy both row 1 and column 2 simultaneously. We arbitrarily cross out column 2 and adjust the supply in row 1 to 0.
2. Cell (3, 1) has the smallest uncrossed-out unit cost (= \$4). Assign $x_{31} = 5$, and cross out column 1 because it is satisfied, and adjust the demand of row 3 to $10 - 5 = 5$ truckloads.
3. Continuing in the same manner, we successively assign 15 truckloads to cell (2, 3), 0 truckloads to cell (1, 4), 5 truckloads to cell (3, 4), and 10 truckloads to cell (2, 4) (verify!).

The resulting starting solution is summarized in Table 5.18. The arrows show the order in which the allocations are made. The starting solution (consisting of 6 basic variables) is $x_{12} = 15$, $x_{14} = 0$, $x_{23} = 15$, $x_{24} = 10$, $x_{31} = 5$, $x_{34} = 5$. The associated objective value is

$$z = 15 \times 2 + 0 \times 11 + 15 \times 9 + 10 \times 20 + 5 \times 4 + 5 \times 18 = \$475$$

The quality of the least-cost starting solution is better than that of the northwest-corner method (Example 5.3-2) because it yields a smaller value of z (\$475 versus \$520 in the northwest-corner method).

Vogel Approximation Method (VAM). VAM is an improved version of the least-cost method that generally, but not always, produces better starting solutions.

Step 1. For each row (column), determine a penalty measure by subtracting the *smallest* unit cost element in the row (column) from the *next smallest* unit cost element in the same row (column).

Step 2. Identify the row or column with the largest penalty. Break ties arbitrarily. Allocate as much as possible to the variable with the least unit cost in the selected row or column. Adjust the supply and demand, and cross out the satisfied row *or* column. If a row and a column are satisfied simultaneously, only one of the two is crossed out, and the remaining row (column) is assigned zero supply (demand).

Step 3. **(a)** If exactly one row or column with zero supply or demand remains uncrossed out, stop.

(b) If one row (column) with *positive* supply (demand) remains uncrossed out, determine the basic variables in the row (column) by the least-cost method. Stop.

(c) If all the uncrossed out rows and columns have (remaining) zero supply and demand, determine the *zero* basic variables by the least-cost method. Stop.

(d) Otherwise, go to step 1.

Example 5.3-4

VAM is applied to Example 5.3-1. Table 5.19 computes the first set of penalties.

Because row 3 has the largest penalty (= 10) and cell (3, 1) has the smallest unit cost in that row, the amount 5 is assigned to x_{31}. Column 1 is now satisfied and must be crossed out. Next, new penalties are recomputed as in Table 5.20.

Table 5.20 shows that row 1 has the highest penalty (= 9). Hence, we assign the maximum amount possible to cell (1, 2), which yields $x_{12} = 15$ and simultaneously satisfies both row 1 and column 2. We arbitrarily cross out column 2 and adjust the supply in row 1 to zero.

Continuing in the same manner, row 2 will produce the highest penalty (= 11), and we assign $x_{23} = 15$, which crosses out column 3 and leaves 10 units in row 2. Only column 4 is left, and it has a positive supply of 15 units. Applying the least-cost method to that column, we successively assign $x_{14} = 0$, $x_{34} = 5$, and $x_{24} = 10$ (verify!). The associated objective value for this solution is

$$z = 15 \times 2 + 0 \times 11 + 15 \times 9 + 10 \times 20 + 5 \times 4 + 5 \times 18 = \$475$$

This solution happens to have the same objective value as in the least-cost method.

TABLE 5.19 Row and Column Penalties in VAM

	1	2	3	4	Row penalty
1	10	2	20	11	**15**
2	12	7	9	20	**25**
3	4	14	16	18	**10**
Column penalty	**5**	**15**	**15**	**15**	

PROBLEM SET 5.3A

1. Compare the starting solutions obtained by the northwest-corner, least-cost, and Vogel methods for each of the following models:

*(a)

0	2	1	6
2	1	5	7
2	4	3	7
5	5	10	

(b)

1	2	6	7
0	4	2	12
3	1	5	11
10	10	10	

(c)

5	1	8	12
2	4	0	14
3	6	7	4
9	10	11	

5.3.2 Iterative Computations of the Transportation Algorithm

After determining the starting solution (using any of the three methods in Section 5.3.1), we use the following algorithm to determine the optimum solution:

Step 1. Use the simplex *optimality condition* to determine the *entering variable* as the current nonbasic variable that can improve the solution. If the optimality condition is satisfied, stop. Otherwise, go to step 2.

Step 2. Determine the *leaving variable* using the simplex *feasibility condition*. Change the basis, and return to step 1.

The optimality and feasibility conditions do not involve the familiar row operations used in the simplex method. Instead, the special structure of the transportation model allows simpler computations.

Example 5.3-5

Solve the transportation model of Example 5.3-1, starting with the northwest-corner solution.

Table 5.21 gives the northwest-corner starting solution as determined in Table 5.17, Example 5.3-2.

TABLE 5.21 Starting Iteration

	1	2	3	4	Supply
1	10 **5**	2 **10**	20	11	**15**
2	12	7 **5**	9 **15**	20 **5**	**25**
3	4	14	16	18 **10**	**10**
Demand	**5**	**15**	**15**	**15**	

The determination of the entering variable from among the current nonbasic variables (those that are not part of the starting basic solution) is done by computing the nonbasic coefficients in the z-row, using the **method of multipliers** (which, as we show in Section 5.3.4, is rooted in LP duality theory).

In the method of multipliers, we associate the multipliers u_i and v_j with row i and column j of the transportation tableau. For each current *basic* variable x_{ij}, these multipliers are shown in Section 5.3.4 to satisfy the following equations:

$$u_i + v_j = c_{ij}, \text{ for each } basic\ x_{ij}$$

As Table 5.21 shows, the starting solution has 6 basic variables, which leads to 6 equations in 7 unknowns. To solve these equations, the method of multipliers calls for arbitrarily setting any $u_i = 0$, and then solving for the remaining variables as shown below.

Basic variable	(u,v) Equation	Solution

To summarize, we have

$$u_1 = 0, u_2 = 5, u_3 = 3$$
$$v_1 = 10, v_2 = 2, v_3 = 4, v_4 = 15$$

Next, we use u_i and v_j to evaluate the nonbasic variables by computing

$$u_i + v_j - c_{ij}, \text{ for each } nonbasic\ x_{ij}$$

The results of these evaluations are shown in the following table:

Nonbasic variable	$u_i + v_j - c_{ij}$

The preceding information, together with the fact that $u_i + v_j - c_{ij} = 0$ for each basic x_{ij}, is actually equivalent to computing the z-row of the simplex tableau, as the following summary shows.

Basic	x_{11}	x_{12}	x_{13}	x_{14}	x_{21}	x_{22}	x_{23}	x_{24}	x_{31}	x_{32}	x_{33}	x_{34}
z	0	0	−16	4	3	0	0	0	9	−9	−9	0

Because the transportation model seeks to *minimize* cost, the entering variable is the one having the *most positive* coefficient in the z-row. Thus, x_{31} is the entering variable.

The preceding computations are usually done directly on the transportation tableau as shown in Table 5.22, meaning that it is not necessary really to write the (u, v)-equations explicitly. Instead, we start by setting $u_1 = 0$.[6] Then we can compute the v-values of all the columns that have *basic* variables in row 1—namely, v_1 and v_2. Next, we compute u_2 based on the (u, v)-equation of basic x_{22}. Now, given u_2, we can compute v_3 and v_4. Finally, we determine u_3 using the basic equation of x_{33}. Once all the u's and v's have been determined, we can evaluate the nonbasic variables by computing $u_i + v_j - c_{ij}$ for each nonbasic x_{ij}. These evaluations are shown in Table 5.22 in the boxed southeast corner of each cell.

TABLE 5.22 Iteration 1 Calculations

	$v_1 = 10$	$v_2 = 2$	$v_3 = 4$	$v_4 = 15$	Supply
$u_1 = 0$	10 **5**	2 **10**	20 $\boxed{-16}$	11 $\boxed{4}$	**15**
$u_2 = 5$	12 $\boxed{3}$	7 **5**	9 **15**	20 **5**	**25**
$u_3 = 3$	4 $\boxed{9}$	14 $\boxed{-9}$	16 $\boxed{-9}$	18 **10**	**10**
Demand	**5**	**15**	**15**	**15**	

Having identified x_{31} as the entering variable, we need to determine the leaving variable. Remember that if x_{31} enters the solution to become basic, one of the current basic variables must leave as nonbasic (at zero level).

The selection of x_{31} as the entering variable means that we want to ship through this route because it reduces the total shipping cost. What is the most that we can ship through the new route? Observe in Table 5.22 that if route (3, 1) ships θ units (i.e., $x_{31} = \theta$), then the maximum value of θ is determined based on two conditions.

1. Supply limits and demand requirements remain satisfied.
2. Shipments through all routes remain nonnegative.

These two conditions determine the maximum value of θ and the leaving variable in the following manner. First, construct a *closed loop* that starts and ends at the entering variable cell, (3, 1). The loop consists of *connected horizontal* and *vertical* segments only (no diagonals are allowed).[7] Except for the entering variable cell, each corner of the closed loop must coincide with a basic variable. Table 5.23 shows the loop for x_{31}. Exactly one loop exists for a given entering variable.

Next, we assign the amount θ to the entering variable cell (3, 1). For the supply and demand limits to remain satisfied, we must alternate between subtracting and adding the amount θ at the successive *corners* of the loop as shown in Table 5.23 (it is immaterial whether the loop is traced in a clockwise or counterclockwise direction). For $\theta \geq 0$, the new values of the variables then remain nonnegative if

TABLE 5.23 Determination of Closed Loop for x_{31}

	$v_1 = 10$	$v_2 = 2$	$v_3 = 4$	$v_4 = 15$	Supply
$u_1 = 0$	10 **5**	2 **10**	20 $\boxed{-16}$	11 $\boxed{4}$	**15**
$u_2 = 5$	12 $\boxed{3}$	7 **5**	9 **15**	20 **5**	**25**
$u_3 = 3$	4 $\boxed{9}$	14 $\boxed{-9}$	16 $\boxed{-9}$	18 **10**	**10**
Demand	**5**	**15**	**15**	**15**	

The corresponding maximum value of θ is 5, which occurs when both x_{11} and x_{22} reach zero level. Because only one current basic variable must leave the basic solution, we can choose either x_{11} or x_{22} as the leaving variable. We arbitrarily choose x_{11} to leave the solution.

The selection of x_{31} (= 5) as the entering variable and x_{11} as the leaving variable requires adjusting the values of the basic variables at the corners of the closed loop as Table 5.24 shows. Because each unit shipped through route (3, 1) reduces the shipping cost by \$9 ($= u_3 + v_1 - c_{31}$), the total cost associated with the new schedule is \$9 × 5 = \$45 less than in the previous schedule. Thus, the new cost is \$520 − \$45 = \$475.

TABLE 5.24 Iteration 2 Calculations

					Supply
	10	2	20	11	**15**
	12	7	9	20	**25**
	4	14	16	18	**10**
Demand	**5**	**15**	**15**	**15**	

Given the new basic solution, we repeat the computation of the multipliers u and v, as Table 5.24 shows. The entering variable is x_{14}. The closed loop shows that $x_{14} = 10$ and that the leaving variable is x_{24}.

TABLE 5.25 Iteration 3 Calculations (Optimal)

					Supply
	10	2	20	11	**15**
	12	7	9	20	**25**
	4	14	16	18	**10**
Demand	**5**	**15**	**15**	**15**	

The new solution, shown in Table 5.25, costs $\$4 \times 10 = \$40$ less than the preceding one, thus yielding the new cost $\$475 - \$40 = \$435$. The new $u_i + v_j - c_{ij}$ are now negative for all nonbasic x_{ij}. Thus, the solution in Table 5.25 is optimal.

From silo	To mill	Number of truckloads	From silo	To mill	Number of truckloads
1	2	5	2	3	15
1	4	10	3	1	5
2	2	10	3	4	5

Optimal cost = \$435

PROBLEM SET 5.3B

1. Consider the transportation models in Table 5.26.
   (a) Use the northwest-corner method to find the starting solution.
   (b) Develop the iterations that lead to the optimum solution.

TABLE 5.26 Transportation Models for Problem 1

(i)

\$0	\$2	\$1	**6**
\$2	\$1	\$5	**9**
\$2	\$4	\$3	**5**
5	**5**	**10**	

(ii)

\$10	\$4	\$2	**8**
\$2	\$3	\$4	**5**
\$1	\$2	\$0	**6**
7	**6**	**6**	

(iii)

—	\$3	\$5	**4**
\$7	\$4	\$9	**7**
\$1	\$8	\$6	**19**
5	**6**	**19**	

TABLE 5.27 Data for Problem 2

\$5	\$1	\$7	**10**
\$6	\$4	\$6	**80**
\$3	\$2	\$5	**15**
75	**20**	**50**	

Develop the iterations that lead to the optimum solution.

Reference: *H.A.Taha, Operations Research: An Introduction,* Prentice Hall; 9th edition, *Singapore, 2010.*